Algebra SEP Notes

Ivan Aidun

April—December, 2024

Contents

README

Using these notes

Do not read these notes in order. The original purpose of these notes was to have a reference document while I taught the Algebra SEP, because I often found that students from different backgrounds needed help filling in different gaps for themselves. For this purpose, Dummit and Foote was too lengthy and circuitous, while Evan Dummit's old notes were too terse. I aspired to make the notes very concise, but include enough of the background as exercises to make it a profitable way for someone to try to catch up. I obviously failed to make these notes at all short, because I kept discovering more and more things that students (reasonably) found to be confusing, or poorly explained in D&F, or required additional context to make sense.

I now think the best way to use these notes is: if there's something that you don't understand, that you find confusing, that you want some survival guidance in, you can **flip directly to that section of the notes**. In each section, I have written my side of an imagined conversation with you, the reader, containing some of the things I have found personally helpful to keep in mind when I have practiced this material. I think of the purpose of the text as being a teacher when no teacher is otherwise available, to guide you in what to think about as you practice either in the provided exercises, or in past qual problems (several of which are also included in the notes).

Each section also contains exercises interspersed with exposition. I think of learning math by analogy to music: doing exercises is like practicing your instrument. Some exercises are like practicing your scales and your arpeggios. They aren't supposed to be very difficult, nor novel, nor necessarily interesting, but rather their purpose is to impart some amount of fluency and ease with common techniques. Many of the exercises I have written here are of this sort, for example Exercises 1 and 2 in Linear Algebra 1 are like this. Some exercises are like practicing a difficult passage from the piece you're actually preparing. The best qual problems are like this, and I have tried to place such problems as front-and-center applications in the appropriate sections.

I repeat: *do not read these notes in order.*

Linear Algebra 1

Conventions

In these notes, most vector spaces have been chosen to be real or complex vector spaces for the sake of concreteness. Anything stated over $\mathbb{R}$ can be done over any field, and anything stated over $\mathbb{C}$ can be done over any algebraically closed field. The one exception to this would be the definition of an inner product, which presupposes that your vector space is either real or complex (otherwise you have no inner products at all).

I freely exchange between referring to a matrix and to a linear transformation, except in cases where a single linear transformation may have several different matrices (due to a change of basis occurring).

Elementary techniques and constructions

Here are two pieces of philosophy that can be helpful when thinking about linear algebra:

- An invertible linear transformation is essentially a change of basis, and a change of basis is essentially a change of perspective. Many matrix manipulations involve "changing your perspective" several times until you arrive at a basis that is easy to read information from.
- Most (but not all) true statements about $n \times n$ matrices for n large are true for 2×2 matrices, and most (but not all) false statements about matrices have a 2×2 counterexample. So, always look for 2×2 examples/counterexamples first, since they are much easier to manipulate.

It is also good to keep in mind a list of criteria equivalent to a matrix being invertible.

Proposition. For a square matrix A, the following are equivalent.

- A is invertible,
- A has linearly independent columns (or rows),
- A has rank $(= \dim \operatorname{im} A)$ n,
- A has trivial kernel,
- 0 is not an eigenvalue of A,
- $\det A \neq 0$.

Row Reduction

The most computationally important matrix manipulations are the **Elementary Row (or Column) Operations**. They come in three flavors, creatively named Type I, Type II, and Type III row operations. They are as follows:

Type I: swap two rows,

Type II: multiply a row by a nonzero scalar,

Type III: add a multiple of one row to another.

(Note, by combining type II and type III operations, we can replace one row by an arbitrary linear combination of that row with other rows.)

Exercise 1. The different kinds of elementary row operations correspond to multiplying on the left by certain matrices. What kinds of matrices give you each of the types of row operation?

These operations are employed in an algorithm called **row reduction** or **Gauss-Jordan reduction** (not the same Jordan as Jordan Normal Form, weirdly). The algorithm goes like

1. Choose a row with a nonzero entry. Swap it to the top, and multiply by an appropriate constant so the leftmost entry is 1. This entry is called the *pivot*.

2. Use Type III operations to make all the entries above and below the pivot 0.

3. Recurse onto the submatrix obtained by eliminating the first row and column, and return to step 1.

At the end of the algorithm, you get a block matrix that looks like

$$\begin{pmatrix} I_k & A \\ 0 & 0 \end{pmatrix}$$

for some matrix A. In the same vein, you could also column reduce a matrix, or both row *and* column reduce a matrix.

Here's how to interpret what the row reduction algorithm does: given a linear transformation $A\colon V \to W$ between finite-dimensional vector spaces, and a basis $\{v_1, \ldots, v_m\}$ for V and a basis $\{w_1, \ldots, w_n\}$ for W, we can form the **matrix of the transformation** (with respect to these two bases) as the matrix whose (i, j) entry is the coefficient of w_i in the expansion of $A(v_j)$. The row reduction algorithm changes the basis of W until $w_i = A(v_i)$ for as many i as possible. Conversely, column reduction changes the basis on V until $\operatorname{span}(A(v_i) - w_i)$ is as small as possible. (Notice that if $V = W$, we are keeping track of *two possibly different* bases for V throughout this algorithm.)

Both forms make it easy to read off information about both the kernel and the image, but the row reduced form is particularly conducive to information about the image, and column reduced form to information about the kernel, because of the aforementioned properties. For the next problem, try both parts using row reduction, then both parts using column reduction.

Exercise 2. The matrix

$$A = \begin{pmatrix} 1 & -5 & 5 & 2 \\ -1 & 5 & -4 & -5 \\ -1 & 5 & -3 & -8 \end{pmatrix}$$

defines a linear transformation $\mathbb{R}^4 \to \mathbb{R}^3$ given by $T(v) = Av$.

(a) Compute a basis for the kernel of this transformation.

(b) Compute a basis for the image of this transformation.

Occasionally, row reduction problems, or problems made much easier to think about with row reduction, have actually appeared on the qual.

Exercise 3. (August 2019 Problem 4) This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an E**-matrix** if M has 1's along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an E-matrix. Express the matrix A below as a product of E-matrices.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Exercise 4. (January 2020 Problem 4) Let V be a finite-dimensional vector space over a field. Let $A\colon V \to V$ be a linear transformation.

(a) Show that there exists $B\colon V \to V$ such that $ABA = A$.

(b) Show that there exists $C\colon V \to V$ such that $ACA = A$ and $CAC = C$.

(**NB** a matrix C like in part (b) is known as a "pseudo-inverse" to A.)

Direct sums and complements

As in many areas of algebra, given two vector spaces V, W, you can form a new vector space whose underlying set is $V \times W$. In the case of vector spaces (and, we will later see, abelian groups and modules) this construction is referred to as the **direct sum** $V \oplus W$ rather than the direct product, because in many ways it feels more like adding the vector spaces than like multiplying them. For example, $\dim V \oplus W = \dim V + \dim W$.

There is also, as in other areas, the concept of a vector space being an *internal* direct sum. Given a vector space V and subspaces U_1, U_2, we say V is the **internal direct sum** of U_1 and U_2 if the following two things hold:

(i) every vector $v \in V$ can be written as $v = u_1 + u_2$ for $u_1 \in U_1$, $u_2 \in U_2$, and

(ii) this representation is unique.

The condition (i) is often written as $U_1 + U_2 = V$, or perhaps as $\langle U_1, U_2 \rangle = V$ or $\operatorname{span}(U_1, U_2) = V$.

Exercise 5. Let V be a vector space of dimension n, U_1, U_2 subspaces, and let $\mathcal{B}_1 = \{u_{11}, \dots, u_{1k}\}$ be a basis for U_1 and $\mathcal{B}_2 = \{u_{21}, \dots, u_{2\ell}\}$ be a basis for U_2.

(a) **(The direct sum test)** Prove that V is the internal direct sum of U_1 and U_2 if and only if $\dim U_1 + \dim U_2 = n$ and $U_1 \cap U_2 = \{0\}$.

(b) Prove that $U_1 + U_2 = V$ if and only if $\mathcal{B}_1 \cup \mathcal{B}_2$ contains a basis for V.

(c) Prove that V is the internal direct sum of U_1 and U_2 if and only if $\mathcal{B}_1 \cup \mathcal{B}_2$ *is* a basis for V.

(d) Notice that $V \oplus W$ has subspaces $\{(v, 0)\} \cong V$ and $\{(0, w)\} \cong W$. Show that $V \oplus W$ is the internal direct product of these two subspaces.

If U is a subspace of V, another subspace U' is called a **complement** to U if V is the internal direct sum of U and U'. In general, the complement is not unique, **Exercise 6.** find two different complements to

$$\operatorname{span}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

inside $\mathbb{R}^2$. However, in the next set of notes, the section on inner products will discuss how inner products can make one complement “better” than all the rest.

Given linear maps $A \colon V \to U$ and $B \colon W \to U$, there is a (unique) linear map $A \oplus B \colon V \oplus W \to U$ given by $(A \oplus B)(v, w) = A(v) + B(w)$. I mention this (1) because I reference it in Exercise 9 and (2) because in future notes the theme of constructions of objects being associated to constructions of homomorphisms will be explored much more in-depth.

The rank-nullity theorem

The rank-nullity theorem says that if $A \colon V \to W$ is a linear transformation where $\dim V = n$, then $\operatorname{rank} A + \operatorname{null} A = n$. Here, $\operatorname{null} A = \dim \ker A$, and as mentioned before $\operatorname{rank} A = \dim \operatorname{im} A$. The rank-nullity theorem is the fundamental theorem of elementary linear algebra. In a sense, the Isomorphism Theorems in other parts of algebra can be thought of as the best approximations to the rank-nullity theorem.

Eigen-stuff

This section talks about some basic facts about eigenvalues, eigenvectors, and some associated constructions. Many of these facts make frequent appearance in Qual problems, so you should be comfortable with them.

An **eigenvector** for a square matrix A is a *nonzero* vector v so that A acts on v by scaling it. In an equation, $Av = \lambda v$, for some scalar λ. The number λ is called an **eigenvalue** for A. Eigenvalues turn out to be the most important features of linear transformations of finite-dimensional vector spaces.

(It's important to insist that eigenvectors be nonzero, because $A\,0 = \lambda \cdot 0$ is true for literally every λ. Whenever a problem involves eigenvectors, be on alert that your purported eigenvectors are nonzero.)

Exercise 7. Prove that the eigenvalues of an upper (or lower) triangular matrix are just the diagonal entries.

Exercise 8. *Without* using its characteristic polynomial, find the eigenvectors and eigenvalues of

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

(Set up the eigenvector problem as a linear system with a variable parameter, and solve that system.)

Exercise 9. Suppose V, W are finite-dimensional vector spaces, and $A\colon V \to V$ and $B\colon W \to W$ are linear transformations. Describe the eigenvalues and eigenvectors of $A \oplus B\colon V \oplus W \to V \oplus W$.

Exercise 10. Two matrices A and A' are **similar** if there exists an invertible matrix B such that $A' = BAB^{-1}$.

(a) Prove that similar matrices have the same eigenvalues.

(b) Use this to show that if B is invertible, then AB has the same eigenvalues as BA. (This holds in general, but I couldn't think of a slick proof for the general case. If you know one, tell me!)

Comment. In the context of group theory, two group elements g and g' are said to be **conjugate** if there exists an element h so that $g' = hgh^{-1}$. Thus, if A, A' are invertible, and we're in the context of talking about the general linear group GL_n, saying A and A' are similar (in the linear algebra sense) is the same as saying they are conjugate (in the group theory sense.) Unfortunately, sometimes people also use the term "conjugate" to mean other things when talking about matrices (such as being complex conjugates). Also, if you're talking about a group smaller than GL_n, like SL_n, two matrices may be similar in the linear algebra sense, but may fail to be conjugate in the smaller group.
Exercise 11. Find an example. (There are examples in $\mathrm{SL}_2(\mathbb{R})$.)

It's a confusing world, so be careful out there.

The nicest possible matrices are diagonal matrices, because they are very easy to work with. It turns out if a matrix has "enough" eigenvectors, then it is similar to a diagonal matrix. We call such matrices **diagonalizable**.

Exercise 12.

(a) Prove that if v_1 and v_2 are eigenvectors for A corresponding to eigenvalues $\lambda_1 \neq \lambda_2$, then v_1 and v_2 are linearly independent. Imagine in your mind the induction argument that shows that any collection of eigenvectors belonging to different eigenvalues is linearly independent.

(b) Suppose an $n \times n$ matrix A has n linearly independent eigenvectors $v_1, \ldots, v_n$. Let T be the matrix whose columns are the v_i. Prove that $T^{-1}AT$ is a diagonal matrix. (Note that T is invertible since it has linearly independent columns by assumption.)

(c) Write the matrix
$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$
as $T\Lambda T^{-1}$ for a diagonal matrix Λ and invertible matrix T.

(d) Give an example of a 2×2 matrix which cannot be diagonalized because it only has 1 eigenvalue, and a 1-dimensional eigenspace for that eigenvalue.

Determinants

There are three different definitions of the determinant (that I know of):

1. The determinant of the matrix A is the product of the eigenvalues of A (with multiplicity).
2. The determinant is the unique function $(\mathbb{R}^n)^n \to \mathbb{R}$ (i.e. it takes in n vectors, each of length n, interpreted as the columns of your matrix) that is
   (i) linear in each of the inputs individually;
   (ii) **alternating**, that is, if you switch any two inputs, you multiply the output by -1;
   (iii) and sends the standard basis (identity matrix) to 1.
3. The determinant of the matrix $A = (a_{i,j})_{i,j=1}^n$ is a sum over permutations $\sigma \in S_n$,
$$\sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)},$$
where $\varepsilon : S_n \to \{\pm 1\}$ is the sign homomorphism, which takes value 1 if σ can be written as a product of an even number of transpositions, and value -1 otherwise. (We will see this homomorphism again in group theory.)

If you have time and inclination, you can prove that these are equivalent, but I'm not sure it's an enlightening exercise.

Definition 2 actually is a sort of proposition-definition, because it asserts (1) that there is a function satisfying those three criteria, and (2) that such a function is unique. This follows from the fact that the set of multilinear (condition (i)), alternating functions $(\mathbb{R}^n)^n \to \mathbb{R}$ forms a 1-dimensional vector space. (More generally, the set of multilinear, alternating functions $(\mathbb{R}^n)^k \to \mathbb{R}$ forms a vector space of dimension $\binom{n}{k}$. It's not too hard to prove this, though there is some technology which can make it easier.)

Exercise 13.

(a) Prove that similar matrices have the same determinant. (Which definition makes this proof one line?)

(b) Fix a matrix A. How does the determinant of A relate to the determinant of matrices obtained from A by doing one of the elementary column operations? (Which definition makes this proof one line?)

(c) Prove that $\det(A) = \det(A^T)$. (Which definition makes this proof one line?)

(d) Prove that a matrix with two identical columns has determinant 0. (Which definition...)

(e) (Important fact to know, but tricky proof. Skip this one and come back to it another time.) Prove that $\det(AB) = \det(A)\det(B)$. (Try thinking about the second definition.)

Exercise 14. Compute the determinant of

$$\begin{pmatrix} 1 & 3 & 4 \\ -1 & -1 & 2 \\ 2 & 6 & 2 \end{pmatrix}.$$

Characteristic polynomials

Saying that $Av = \lambda v$ is the same as saying that v lies in the kernel of $\lambda I - A$. So, to find all the eigenvalues of A, we can look at $\det(xI - A)$. This will be a polynomial of degree n called the **characteristic polynomial** of A. By the preceding, λ is an eigenvalue of A if and only if $\lambda I - A$ has nontrivial kernel, which happens if and only if $\det(\lambda I - A) = 0$, i.e. , if and only if λ is a root of the characteristic polynomial.

Exercise 15. Find the characteristic polynomial of

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

There's one caveat to the above discussion: a linear transformation may not have any eigenvalues/vectors in the field it is defined over, because the characteristic polynomial may fail to factor over that field. Exercise 16 gives an example where a matrix defined over $\mathbb{R}$ (in fact, over $\mathbb{Q}$) fails to have any real eigenvectors/values, while Exercise 17 asks you to show (among other things) that this perverse behavior cannot occur over algebraically closed fields like $\mathbb{C}$.

Exercise 16. Let

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(a) Compute the characteristic polynomial of A.

(b) Compute the eigenvalues of A.

(c) Compute the eigenvectors of A inside $\mathbb{C}^2$.

Exercise 17. (August 2014 Problem 3 (a)-(c))

(a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T\colon V \to V$ be a linear transformation. Show that T has an eigenvector.

(b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T\colon V \to V$ such that T does not have an eigenvector.

(c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

An important fact about the characteristic polynomial is the **Cayley-Hamilton Theorem:** a matrix always satisfies its characteristic polynomial. That means that if I take the characteristic polynomial $x^n + \cdots + c_1 x + c_0$, and I "plug in" A , I will get the 0 matrix: $A^n + \cdots + c_1 A + c_0 I = 0$.

Exercise 18. Try it with the matrices from Exercise 15 and Exercise 16.

Exercise 19. Let V be a finite-dimensional vector space over $\mathbb{C}$, and $A\colon V \to V$ a nonzero linear transformation.

(a) Give an example of a pair (V, A) such that the only eigenvalue of A is 0.

(b) Prove that the only eigenvalue of A is 0 if and only if A is nilpotent.

Trace

There are two definitions of the trace of a matrix (actually, I know a third, but it is only fit to be known by Masters of the Matrix like Josh Mundinger):

1. The trace of A is the sum of the diagonal entries of A.
2. The trace of A is the sum of the eigenvalues of A.

Again, time and energy permitting, you can choose to prove the equivalence of these two.

Exercise 20.

(a) Prove that the trace is a linear function $\mathbb{R}^{n\times n} \to \mathbb{R}$.

(b) Prove that the trace satisfies the cyclic relation $\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA)$.

One neat thing to note is that $\mathrm{Tr}(A)$ is (up to a sign) the coefficient of x^{n-1} in the characteristic polynomial, and $\det(A)$ is (again, up to a sign) the constant coefficient. Among other things, this provides a way to compute the characteristic polynomial of a 2×2 matrix pretty quickly:

$$\mathrm{char}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \lambda^2 - \mathrm{Tr}(A)\lambda + \det(A) = \lambda^2 - (a+d)\lambda + (ad - bc).$$

Linear Algebra 2

Even more eigen-stuff!

This section talks about what I think are the most important linear algebra constructions to know for the Qual: the minimal polynomial and Jordan Normal (or sometimes Canonical) Form.

Minimal polynomial

In the previous notes we discussed the Cayley-Hamilton Theorem, which says that A satisfies its own characteristic polynomial. Since A satisfies some polynomial, there must be a polynomial A satisfies of least degree. Such a polynomial is called the **minimal polynomial** of A.

Exercise 1. (Basic properties of the minimal polynomial) Let A be a linear operator on a finite-dimensional complex vector space V. Let $p(x)$ be the minimal polynomial of A.

(a) Prove that if f is a polynomial A satisfies, then $p \mid f$.

(b) Prove that only eigenvalues of A can be roots of p.

(c) Prove that if λ is an eigenvalue of A, then λ is in fact a zero of p. (Hint: let v be an eigenvector, and look at $p(A)v$.)

(d) Prove that two similar matrices have the same minimal polynomial.

(e) Prove that a matrix A is diagonalizable if and only if its minimal polynomial has no repeated roots. (Hint: think about $\dim\ker A - \lambda I$ as λ ranges over the roots of p.)

(f) Let $W \subset V$ be a subspace that is A-stable, that is, so that for every $w \in W$, $Aw \in W$. Then we can restrict the operator $A\colon V \to V$ to an operator $A|_W\colon W \to W$. Let $q(x)$ be the minimal polynomial of the restriction $A|_W$. What is the relation between $q(x)$ and $p(x)$?

How can the minimal polynomial differ from the characteristic polynomial? The next exercise offers one example.

Exercise 2. Take I_n the $n \times n$ identity matrix.

(a) What is its characteristic polynomial?

(b) What is its minimal polynomial?

(c) What is the characteristic/minimal polynomial of the shearing transformation

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}?$$

The above exercise suggests that the minimal polynomial has low degree when A has few eigenvalues and many eigenvectors. Indeed, in the next section we will see the exact sense in which this is true.

Generalized eigenspaces and Jordan Normal Form

As mentioned in the previous set of notes, v being an eigenvector for A is equivalent to v being a nontrivial vector in the kernel of $\lambda I - A$ for some λ. We saw two ways that a matrix could fail to be diagonalizable: if the field in question wasn't algebraically closed, the eigenvalues (and hence eigenvectors) might not be in the base field, or even over an algebraically closed field A could fail to have n linearly independent eigenvectors. Examples of the latter include

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

For both of these matrices, the obstacle preventing them from being diagonalizable is the same: they have only one eigenvalue, and the matrix $\lambda I - A$ is nilpotent, but not zero. We say a vector v is a **generalized eigenvector** of eigenvalue λ if v is in the kernel of $(\lambda I - A)^k$ for some natural number k. The smallest integer k such that $(\lambda I - A)^k v = 0$ I will call the **degree** of the generalized eigenvector v. (Wikipedia calls it the "rank", but I feel like that's a misleading term in this context because it is kinda the opposite of the rank of a matrix. I have chosen "degree" since it lines up with the terminology of "nilpotence degree".) Thus, an ordinary eigenvector is an eigenvector of degree 1.

The following exercise shows that the generalized eigenvectors are exactly the vectors we are "missing" if we want to find a good matrix similar to A.

Exercise 3. Let V be a finite-dimensional complex vector space, and $A\colon V \to V$ a linear transformation.

(a) For each eigenvalue λ of A, let $V_{\lambda,k} = \{v \in V : v \text{ a } \lambda\text{-generalized-eigenvector of degree } \leq k\}$. Prove that $V_{\lambda,k}$ is a subspace of V, and that $AV_{\lambda,k} \subset V_{\lambda,k}$. The subspace $V_{\lambda,k}$ is the **degree k λ-generalized-eigenspace** of A.

(b) Suppose v_k is a λ-generalized-eigenvector of degree k. Show that $v_{k-1} = (\lambda I - A)v_k$ is a λ-generalized-eigenvector of degree $k-1$. By induction, this shows that $(\lambda I - A)^\ell v_k$ is a generalized eigenvector of degree $k - \ell$. The sequence of vectors $v_k, v_{k-1}, \ldots, v_1$ is called a **Jordan chain**.

(c) Let $V_{\lambda,\max}$ be the maximal λ-generalized-eigenspace for A, that is, $V_{\lambda,\max} = V_{\lambda,k}$ for the maximal degree k of a λ-generalized-eigenvector of A. Let $n = \dim V_{\lambda,1}$. Show that $V_{\lambda,\max}$ has a basis consisting of n disjoint Jordan chains.

(d) Let $\mathcal{B}$ be a basis as in the previous part. What does the matrix of A with respect to $\mathcal{B}$ look like?

The form of matrix you found in part (d) above is called a **Jordan block**. The above exercise implies that for any matrix A, we can find a basis for V so that the matrix for A is almost diagonal: it is block-diagonal, with each block a Jordan block. This form of matrix is called the **Jordan normal form** of A. Moreover, it is easy to see that up to rearranging the blocks, this form is unique. This is worth stating as a theorem.

Theorem. Two matrices A, A' are similar if and only if they have the same Jordan normal form (up to rearranging the blocks).

Here's a more concrete exercise if you're having trouble connecting the discussion in Exercise 3 to the matrix form you expect to be seeing.

Exercise 4. (A concrete JNF computation) Consider the matrix

$$A = \begin{pmatrix} 4 & 1 & 0 \\ -11 & -3 & -1 \\ 5 & 2 & 2 \end{pmatrix}.$$

I will tell you right now that the only eigenvalue of A is 1.

(a) Given the fact that the only eigenvalue of A is 1, find a basis for the 1-eigenspace of A. (This is a kernel computation, which you know how to do.)

(b) Based on your computation from part (a), how many Jordan chains will you need to find in order to span $\mathbb{R}^3$?

(c) (Optional) Write down a basis for the degree 2 generalized eigenspace of A.

(d) Choose a generalized eigenvector for A of maximum degree. Write down its Jordan chain.

(e) Find a basis for $\mathbb{R}^3$ consisting of Jordan chains. Let B be the matrix with those vectors as its columns. Compute B^{-1}.

(f) Write down the matrix of A with respect to this basis by computing $B^{-1}AB$. Notice that this is the JNF.

(g) If you were writing an exercise for someone else to compute a JNF, how would you come up with a matrix that makes it so the problem isn't trivial, but also so that it isn't too hard? Asking for a friend...

Exercise 5.

(a) Given a matrix in Jordan normal form, how can you tell what its characteristic polynomial is?

(b) If the matrix A has only one Jordan block, what is its minimal polynomial?

(c) If the matrix A has two Jordan blocks (possibly for different eigenvalues) what is its minimal polynomial?

(d) In general, given a matrix in Jordan normal form, how can you tell what its minimal polynomial is?

Exercise 6.

(a) Give an example of two linear transformations with the same characteristic polynomial but different minimal polynomials.

(b) Give an example of two linear transformations on the same vector space with the same minimal polynomials which are not similar.

Commuting matrices

There's a standard trick involving commuting matrices which comes up often enough that I want to mention it.

Exercise 7. (August 2014 Problem 3 (d)) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U.

There are lots of variations on this theme, but the moral is that whenever you see two matrices which commute (or almost do so) you should be looking for a trick along these lines. One particular application of the above fact is given in the following exercise.

Exercise 8. Suppose that T and U are as in Exercise 7.

(a) Suppose T and U are diagonalizable. Then prove that in fact, there is a basis such that the matrices of T and U are *both* diagonal with respect to that basis. This is called **simultaneous diagonalization**, and it appears in various parts of math, but I learned of it because in number theory you simultaneously diagonalize Hecke operators to learn things about modular forms/elliptic curves.

(b) One might suspect that without the assumption of diagonalizability, we might be able to simultaneously *Jordanize* T and U (that is, put them into simultaneous Jordan form). Find a counterexample showing that this is false.

(c) Something weaker is true: if T and U commute, but are not necessarily diagonalizable, then there is a basis with respect to which T and U are both *upper triangular*. (Cheekily, this is called "simultaneously upper triangularization".) Here are some steps to proving this.

i. Prove that if T and U have a simultaneous eigenvector then they can be put into the simultaneous block form:
$$\begin{pmatrix} \lambda & R \\ 0 & T_2 \end{pmatrix}$$
where λ is the eigenvalue, R is some $1 \times (n-1)$ arbitrary row, 0 is a $(n-1) \times 1$ column of 0s, and T_2 is some $(n-1) \times (n-1)$ square submatrix.

ii. Suppose in this block form, the square block for T is T_2 and the square block for U is U_2. Show that T_2 and U_2 commute.

iii. Induct to obtain the result.

Inner products and bilinear forms

Inner products

Something that usually comes up in an introductory linear algebra class but which has so far not come up at all in these notes are inner products. The vector space $\mathbb{R}^n$ comes with an important function usually called the **dot product**, or sometimes the "standard inner product". It is defined by $u \cdot v = u_1v_1 + u_2v_2 + \cdots + u_nv_n$, where the u_i and v_i are the coordinates of the vectors u and v. This comes up all the time in physical applications.

The essential properties of the dot product are encoded in the definition of an **inner product**. An inner product on a real vector space V is a function $V^2 \to \mathbb{R}$, often denoted with angle brackets $(u, v) \mapsto \langle u, v\rangle$, with the following properties.

(i) (bilinearity) $\langle au_1 + bu_2, v\rangle = a\langle u_1, v\rangle + b\langle u_2, v\rangle$, and similarly on the other side.

(ii) (symmetry) $\langle u, v\rangle = \langle v, u\rangle$

(iii) (positive definite) We have $\langle v, v\rangle \geq 0$, with equality if and only if $v = 0$.

Inner products can be used to define the angle between two vectors in an arbitrary vector space. We won't need this fact, except to mention that two vectors are **orthogonal** if $\langle v, u\rangle = 0$. They also can be used to define the length of a vector via $\|v\| = \sqrt{\langle v, v\rangle}$. This does in fact induce a metric, which is a consequence of the Cauchy-Schwarz theorem.

Given a subspace U of a vector space V, the **orthogonal complement** $U^\perp$ is the set of all vectors which are orthogonal to everything in U: $U^\perp = \{v \in V : \forall u \in U,\ \langle v, u\rangle = 0\}$. The orthogonal complement to a subspace is "the best" complement to that subspace, because by design the vectors from the original subspace essentially do not interact at all with those in the orthogonal space.

Exercise 9.

(a) Give an example of an inner product on $\mathbb{R}^2$ that is not the dot product.

(b) Let $\mathcal{C}([0, 1])$ be the set of real-valued continuous functions on the interval $[0, 1]$. Define a pairing by
$$\langle f, g\rangle = \int_0^1 f(x)g(x)\ \mathrm{d}x.$$
Prove that this is an inner product.

Exercise 10. (August 2020 Problem 2) Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \to \mathbb{R}^n$; let A denote the transformation corresponding to M, and let B denote that corresponding to M^T.

(a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\mathrm{im}(B)$ with respect to the standard inner product (dot product).

(b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\mathrm{im}(B)$. That is to say, show that for any vector $u \in \mathbb{R}^n$, there exist unique vectors $v \in \ker(A)$ and $w \in \mathrm{im}(B)$ so that $u = v + w$.

Using inner products, one can create especially nice bases: orthonormal bases. In an **orthonormal basis**, each basis vector has length 1 and is orthogonal to all the other basis vectors. If $\{e_1, \ldots, e_n\}$ is an orthonormal basis, it is particularly easy to determine the coefficient of e_i in the expansion of an arbitrary vector v: it is just $\langle e_i, v\rangle$.

Exercise 11. Consider the functions $\{\sqrt{2}\sin(2n\pi x), \sqrt{2}\cos(2n\pi x)\} \subset \mathcal{C}([0,1])$. Prove that these functions are orthonormal with respect to the inner product from Exercise 9. (Hint: Look up your double angle formulas, and your product-to-sum formulas.)

(The functions $\{\sqrt{2}\sin(2n\pi x), \sqrt{2}\cos(2n\pi x)\}$ don't quite form a basis in the traditional sense, because this is a countable set and $\mathcal{C}([0,1])$ has dimension equal to the cardinality of the continuum. But, the span of these functions is dense, which in the context is a more natural condition to work with. Analysts call the normal definition of a basis an "algebraic basis" or "Hamel basis", and they will often call a system like this a "Schauder basis" or just a "basis" if there isn't an algebraist in earshot.)

Exercise 12. Let O be an $n \times n$ real matrix. Show that the following are equivalent.

(i) The columns of O form an orthonormal basis.

(ii) $O^{-1} = O^T$.

If O satisfies either of the conditions in the preceding problem, it is called an **orthogonal matrix**. The set of $n \times n$ orthogonal matrices is a group, called the **orthogonal group**, and denoted $\mathrm{O}(n)$. The subgroup of orthogonal matrices with determinant 1 is called the **special orthogonal group**, and is denoted $\mathrm{SO}(n)$.

Exercise 13. (Spectral theorem) Let V be a finite-dimensional inner product space. A linear transformation $A\colon V \to V$ is called **symmetric** if $\langle Au, v\rangle = \langle u, Av\rangle$.

(a) Let $\{e_1, \ldots, e_n\}$ be an orthonormal basis for V. Show that A is symmetric if and only if the matrix of A is symmetric with respect to the chosen basis.

(b) Let A be a symmetric linear operator. Show that there is some orthonormal basis such that the matrix of A is diagonal.

Other products

The term "inner product space" is also applied to complex vector spaces, although there it has a somewhat different definition. An inner product on a complex vector space V is a function $(V)^2 \to \mathbb{C}$ satisfying the following.

(i) (linearity in the second term) $\langle u, av_1 + bv_2\rangle = a\langle u, v_1\rangle + b\langle u, v_2\rangle$.

(ii) (conjugate-linearity in the first term) $\langle au_1 + bu_2, v\rangle = \bar{a}\langle u_1, v\rangle + \bar{b}\langle u_2, v\rangle$

(iii) (conjugate symmetry) $\langle u, v\rangle = \overline{\langle v, u\rangle}$

(iv) (positive definite) $\langle v, v\rangle \geq 0$, with equality if and only if $v = 0$. (Note that conjugate symmetry forces $\langle v, v\rangle \in \mathbb{R}$, so it makes sense to talk about whether or not it is greater than 0.)

Of course, it is a matter of convention whether or not you take an inner product to be conjugate linear in the first variable and linear in the second or visa versa. I think the convention above is the most common one, on account of the fact that it makes $u^T v$ into the standard inner product. A function satisfying only (i)-(iii) above is called a **Hermitian product**, and a function satisfying only (i)-(ii) is sometimes called a **sesquilinear product**. Over complex inner product spaces, the analog of an orthogonal matrix is called a **unitary matrix**.

You might wonder (as I have) why when we extend the definition of an inner product in a way that treats the two variables differently, rather than demanding bilinearity which seems more straightforward and natural. One reason is that without conjugate-linearity, we cannot guarantee that $\langle v, v\rangle \in \mathbb{R}$, which we want in order to talk about whether or not it is positive-definite. A related reason is that if an inner product were required to be symmetric and bilinear instead of Hermitian, you could get strange behavior extending an inner product from a real vector space contained inside a complex vector space. For example, take $\mathbb{R}^2$ with the dot product contained inside $\mathbb{C}^2$, and let $v_1, v_2 \in \mathbb{R}^2$ be orthonormal. If we extended the dot product to $\mathbb{C}^2$ bilinearly instead of sesquilinearly, we would get that $\langle v_1 + iv_2, v_1 + iv_2\rangle = \langle v_1, v_1\rangle + i^2\langle v_2, v_2\rangle = 1 - 1 = 0$, so $v_1 + iv_2$ would be a nonzero vector with "length 0".

So far our discussion has been limited to just vector spaces over $\mathbb{R}$ and $\mathbb{C}$, but it would be pretty sad if there were no analogs for vector spaces over other fields. Luckily, there are. For a vector space V over a field k, a bilinear function $(V)^2 \to k$ is called a **bilinear form**, or sometimes just a **pairing**. (Although sometimes you can have pairings between different vector spaces.) Often people think about symmetric bilinear forms in imitation of the inner product case, but the condition of positive-definiteness cannot be straightforwardly ported to an arbitrary field. In its place, the next best thing is to require a bilinear form to be **nondegenerate:** if $\langle u, v\rangle = 0$ for all $u \in V$, then $v = 0$. Symmetric bilinear forms are closely related to what are called "quadratic forms": polynomials in n variables each of whose terms has total degree 2.

Exercise 14. Show that the set of bilinear forms on V is a vector space.

There are other kinds of products considered in various parts of mathematics. For example, a **symplectic product** on a vector space is a nondegenerate bilinear form that is required to be *anti*-symmetric, rather than symmetric: $\langle u, v\rangle = -\langle v, u\rangle$.

The dual vector space

Discussing bilinear forms is a great time to mention a construction that is good to know: the dual vector space. Given a vector space V over a field k, the dual vector space is the set of all linear maps $V \to k$. This does in fact form a vector space, variously denoted $V^\vee$, V^*, V', $\overline{V}$, $\widetilde{V}$, and V^T. I will use $V^\vee$. The elements in $V^\vee$ are variously called **covectors** or **functionals** depending on context.

Exercise 15. Let V be a vector space over a field k.

(a) Suppose that $\dim V = n$ and that $\{v_1, \ldots, v_n\}$ is a basis for V. Define $\phi_i \colon V \to k$ to be the unique map such that $\phi_i(v_j) = 1$ if $i = j$ and 0 otherwise. Show that the ϕ_i form a basis for $V^\vee$ (and, in particular, that $\dim V = \dim V^\vee$). This basis is called the **dual basis** to $\{v_1, \ldots, v_n\}$.

(b) If you are a set theory lover, show that $\dim V$ need not equal $\dim V^\vee$ if $\dim V$ is not finite. (This is pretty similar to showing that the power set of $\mathbb{N}$ is uncountable.)

(c) Suppose $A \colon V \to V$ is an invertible linear transformation. Then $\{Av_1, \ldots, Av_n\}$ is also a basis for V. What is the relationship between the dual basis to $\{Av_1, \ldots, Av_n\}$ and the dual basis to $\{v_1, \ldots, v_n\}$?

Exercise 16. Let V, W be vector spaces over a field k. Denote by $\operatorname{Hom}_k(V, W)$ the set of all linear maps $V \to W$.

(a) Show that $\operatorname{Hom}_k(V, W)$ is a vector space.

(b) If $\dim V = m$ and $\dim W = n$, what is $\dim \operatorname{Hom}_k(V, W)$?

Exercise 17. I'm thinking of a linear isomorphism between the vector space of bilinear forms on V and $\operatorname{Hom}_k(V, V^\vee)$. This isomorphism can be written out explicitly, and doesn't require me to tell you anything about V or about k (in particular, it doesn't require me to tell you a basis for V). Can you find the isomorphism?

Given a linear map $A\colon V \to W$, there is an associated linear map $A^\vee\colon W^\vee \to V^\vee$ given by "pulling back functionals": $A^\vee(f)(v) = f(Av)$. **Exercise 18.** Check that if $V = \mathbb{R}^m$ and $W = \mathbb{R}^n$ with the standard bases, then the matrix of $A^\vee$ with respect to the dual bases is A^T.

"Generalized products" and tensor products

The great utility of inner products, and bilinear forms more generally, induced people to want more general kinds of products, which could be between two different vector spaces, and were required to satisfy nothing other than bilinearity. Given vector spaces V and W, we can introduce the symbol $v \otimes w$ to stand in for the value this "most general possible product" would take on the pair $(v, w) \in V \times W$. Then, the assumption of bilinearity forces upon us certain relations among these symbols:

- $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$, and conversely on the other side.
- $(av) \otimes w = a(v \otimes w) = v \otimes (aw)$.

This, in fact, has a name, given to it by Einstein in the 1910s: these symbols subject to these relations are called **tensors**. The vector space of all tensors from V and W is called the **tensor product** of V and W, denoted $V \otimes_k W$. A tensor can be a linear combination of symbols, like $v_1 \otimes w_1 + v_2 \otimes w_2 + \dots$. The tensors that can be written as $v \otimes w$ are called **simple tensors**.

Exercise 19. Suppose V, W are vector spaces over a field k, and that $\{v_1, \dots, v_m\}$ is a basis for V and $\{w_1, \dots, w_n\}$ is a basis for W. Show that the set $v_i \otimes w_j$ is a basis for $V \otimes_k W$.

Exercise 20. (January 2022 Problem 4) Let V and W be vector spaces over a field k, not necessarily finite dimensional. As usual, we denote by $\operatorname{Hom}_k(V, W)$ the space of linear maps from V to W, and by $V^\vee$ the dual of V, so that $V^\vee = \operatorname{Hom}_k(V, k)$.

(a) I am thinking of an injective linear map

$$\Phi\colon V^\vee \otimes_k W \to \operatorname{Hom}(V, W).$$

This map can be written down explicitly, and doesn't require me to tell you anything further about V, W or about k. Can you find the map?

(b) Prove that the operator Φ is an isomorphism if V and W are finite-dimensional.

(c) Prove a necessary and sufficient condition that a linear map $A \in \operatorname{Hom}(V, W)$ is in the image of Φ. (The assumption that V and W are finite-dimensional does not apply to this part.)

If you have a vector space V defined over a field K, and a field L which contains K, then (as we'll discuss when it comes to field theory) L is itself also a K-vector space. For example, $\mathbb{C}$ can be viewed as a two-dimensional real vector space. Then the tensor product $V \otimes_K L$ has a particularly nice interpretation: it is just the vector space V extended to be a vector space over L instead of over K. For example, $\mathbb{R}^3 \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}^3$ is just what we get when we take a basis for $\mathbb{R}^3$, and declare that we will form a complex vector space with exactly that basis. This process is called **extension of scalars** and it is very good to know about.

The next exercise shows how we can use extension of scalars to understand a little bit about the structure of orthogonal transformations. I have been told that people found this exercise very difficult and hard to get their brains around, which I think is fair. This is not part of what I would consider to be the "standard algebra curriculum", but something like this did appear on the qual in August 2023, so I have included it for completeness.

Exercise 21. (Hard) Let V be a finite-dimensional real inner product space. Let $O\colon V \to V$ be an orthogonal transformation. Let $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ be the **complexification** of V. The transformation O extends to a transformation $V_{\mathbb{C}} \to V_{\mathbb{C}}$, which by abuse of notation I will also call O.

(a) Show that all the eigenvalues of O in $\mathbb{C}$ lie on the unit circle.

(b) Show that O is diagonalizable. (Hint: $\langle v, w\rangle = \langle Ov, Ow\rangle$. Try looking at the orthogonal complement to an eigenvector.)

(c) If $v \in V_{\mathbb{C}}$ is an eigenvector of O of eigenvalue $\lambda \in \mathbb{C}\setminus\mathbb{R}$, then $\bar{v}$ is also an eigenvector of eigenvalue $\bar{\lambda}$. Consider the subspace W of the original V given by $V \cap \operatorname{span}(v, \bar{v})$. Show that $OW = W$.

(d) Let the argument (= angle in the complex plane) of λ be θ. Prove that O acts on W by rotation by $\pm\theta$. (To be honest, I didn't think about this one very long. Idk if it's easy or not.)

Rational matrices which are similar over $\mathbb{C}$

Discussing extension of scalars brings me to one more somewhat conceptually tricky topic. The above discussion of Jordan normal form was entirely over $\mathbb{C}$ for the purpose of making sure that a matrix always has an eigenvector. However, the theory works over any field, as long as the eigenvalues of A actually lie in that field.

I bring this up because of a trick that shows up occasionally. Sometimes, you want to show that two matrices A, B over $\mathbb{Q}$ or $\mathbb{R}$ are similar, but for some reason this is hard. However, after passing to a field where all the eigenvalues are defined (e.g. $\mathbb{C}$) it might become easy to show that the two matrices have the same JNF, and so are similar. A priori, the matrix C such that $A = CBC^{-1}$ could have complex entries (or entries in whatever field you passed to). Is it the case that we can choose C with entries in the field we started with?

The answer is yes, here's one way to think about it. Given A, B matrices over, say, $\mathbb{R}$, we can define a map $L\colon M_n(\mathbb{R}) \to M_n(\mathbb{R})$ via $C \mapsto AC - CB$. This is in fact a linear map on the real vector space $M_n(\mathbb{R})$, and a matrix C so that $A = CBC^{-1}$ would be the same thing as an element in $\ker L \cap \mathrm{GL}_n(\mathbb{R})$. If we know that A and B are similar over $\mathbb{C}$, that's the same as saying that once we extend scalars to consider L as a linear transformation $M_n(\mathbb{C}) \to M_n(\mathbb{C})$, we know that $\ker L \cap \mathrm{GL}_n(\mathbb{C})$ is nonempty. The next exercise asks you to show that at least this implies that $\ker L$ is nonempty over $\mathbb{R}$.

Exercise 22. Suppose V is a real vector space, and $L: V \to V$ is a linear transformation. Let $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ be the complexification of V, and extend L to a linear map $L_{\mathbb{C}}: V_{\mathbb{C}} \to V_{\mathbb{C}}$. Prove that if $\ker L_{\mathbb{C}}$ is nonempty, so too is $\ker L$. (Try to do so without invoking complex conjugation. This fact applies to any extension of fields $K \subset L$ whatsoever, so to prove it in maximum generality you should avoid using anything special about the relationship between $\mathbb{R}$ and $\mathbb{C}$.)

To finish the argument, we would need to show that this implies $\ker L \cap \mathrm{GL}_n(\mathbb{C})$ is nonempty, but that requires more than just linear algebra knowledge. You could do it by showing that $\mathrm{GL}_n(\mathbb{C})$ is dense in $M_n(\mathbb{C})$, but these aren't topology notes so I am not going to do that.

Exercise 23. Let $V, V_{\mathbb{C}}, L, L_{\mathbb{C}}$ be as in the previous problem. Suppose $L_{\mathbb{C}}$ has an eigenvector $v \in V_{\mathbb{C}}$ of eigenvalue $\lambda \in \mathbb{R}$. Prove that L also has an eigenvector in V of eigenvalue λ.

While not a consequence of the above, the next exercise has similarities to the reasoning.

Exercise 24. (January 2019 Problem 5) Let F be a field and n be a positive integer. Fix an $n \times n$ matrix S over F that is invertible and symmetric. Writing A^t for the transpose of a matrix, we let

$$V \coloneqq \{n \times n \text{ matrices } A \text{ over } F : A^t = SAS^{-1}\}.$$

Note that V is a vector space (you do not need to prove this). Find the dimension of V in terms of n.

Ring Theory 1

User's Guide

By my estimation, ring theory and module theory are both the most frequent qual questions (I estimate about they make up about 2/5 of every qual), as well as some of the toughest questions. For this reason, I have tried to make the notes on them thorough, at some cost to brevity, with the goal that the thoroughness will help you if you have less background in ring theory already, and will be a useful reference even if you already know more ring theory.

If you've already taken a course in ring theory, I would expect you to be familiar with most of the things discussed in these notes. For such a person, I think the most important things in these notes are:

- to know the basic properties of the three big examples
- the little bit of content discussed in the section on noncommutative ring theory, and
- the various examples and counterexamples that pop up throughout. (e.g. a UFD that is not a PID, a prime ideal that isn't maximal, ideals whose intersection is not equal to their product, etc.)

The next set of notes will discuss, among other things, **the lattice isomorphism theorem** and **localization**, which are both very important and appear on the qual quite frequently.

Conventions

In these notes, all rings will have unity, but may or may not be commutative, depending on the section. As a reminder, a homomorphism $R \to R'$ of unital rings is required to send 1_R (the multiplicative unit of R) to $1_{R'}$ (that of R'), and a subring $R'' \subset R$ is required to have the same unit as R does, so that the inclusion $R'' \hookrightarrow R$ is a unital ring homomorphism. These conventions differ from the conventions in Dummit and Foote, where they allow rings without unity, and they allow "subrings" of rings with unity to not have unity. In particular, they are happy to call ideals a special kind of "subring". I think these conventions were a little out of date at the time, and nowadays I think it'd be hard to find authors who allow rings without unity.

Exercise 1. Find a subset of $M_2(\mathbb{R})$ which has the structure of a ring with unity under the operations of matrix addition and multiplication, but which is not a subring of $M_2(\mathbb{R})$.

Examples

The three big ones

I think there are three main examples you should keep in mind for ring theory:

- the ring of integers $\mathbb{Z}$,
- the ring of complex polynomials in one variable $\mathbb{C}[x]$,
- the ring of complex 2×2 matrices $M_2(\mathbb{C})$.

Especially when we start talking about modules, it will be enlightening to be able to think about what modules look like for these three examples. (In the past I haven't emphasized non-commutative rings enough, but when preparing these notes I realized they show up on the qual somewhat frequently.)

Other good examples

Here are a few more rings you should have in your head for various examples/counterexamples in ring theory.

- The rings $\mathbb{R}[x]$, $\mathbb{Q}[x]$, and $\mathbb{Z}[x]$. The rings $\mathbb{R}[x]$ and $\mathbb{Q}[x]$ will have a lot in common with $\mathbb{C}[x]$, but if k is not algebraically closed you will get different irreducible elements. The ring $\mathbb{Z}[x]$ is closely connected to $\mathbb{Q}[x]$ (via Gauss' lemma, see the next set of notes), but also has additional interesting structure of its own.
- Rings of polynomials in more than one variable. For example, $\mathbb{C}[x, y]$ and $\mathbb{C}[x, y, z]$.
- Quotient rings of rings you already know. Some examples that come to mind are:
  - If $f \in \mathbb{Q}[x]$ is an irreducible polynomial, the ring $\mathbb{Q}[x]/(f)$ is a field extension of $\mathbb{Q}$. This kind of example is very important in field theory.
  - For k a field, the ring $k[x]/(x^2)$ is sometimes called "the ring of dual numbers". In this ring, the element x behaves like an infinitesimal in old-timey calculus, because x is not zero, but $x^2 = 0$.
  - There are lots of interesting quotients of polynomial rings in many variables. For example, $\mathbb{C}[x]/(x^2 + y^2 - 1)$ is a ring that is associated to the equation $x^2 + y^2 - 1 = 0$, or $x^2 + y^2 = 1$ as a sane person might write it. So, this ring captures some information about the unit circle. The ring $\mathbb{C}[x, y]/(xy)$ has come up a couple times on old quals. In a sense, the field of **algebraic geometry** is concerned with studying the rings that arise as quotients of multivariable polynomial rings.
- Other matrix rings such as $M_2(\mathbb{R})$ and $M_2(\mathbb{Z})$. Sometimes you might need to think about matrices larger than 2×2, but there's a meta-principle that "any behavior that can occur for matrices, can occur for 2×2 matrices". This, of course, is not literally true, but it's true often enough to be a useful guide.
- Given a group G, one can form the **group ring** $\mathbb{C}[G]$ (sometimes written $\mathbb{C}G$). The elements of the group ring are formal linear combinations $\sum c_i g_i$, where we treat the group elements as linearly independent vectors over $\mathbb{C}$. Addition is straightforward, and multiplication is defined by using the multiplication in the group. (The complex numbers are not important here, one can just as easily

form $\mathbb{R}[G]$, $\mathbb{Z}[G]$, or $R[G]$ for any ring R.) Group rings come up in the area of **representation theory**, which is no longer listed as a qual topic but can sometimes appear disguised as something else, such as ring theory questions about a group ring.

Basics about commutative rings

Ideals in commutative rings

For now, all our rings will be commutative. An **ideal** in a commutative ring R is a subset $I \subset R$ with the following two properties:

(i) I is closed under addition, and

(ii) I is *absorbing* under multiplication; that is, for any $r \in R$ and $i \in I$, $ri \in I$.

Later in this section I'll give some historical examples that motivated creating ideals. For now, I'll note that given some elements $g_1, \ldots, g_n \in R$, we can form an ideal *generated by* those elements, denoted either $\langle g_1, \ldots, g_n \rangle$ or just $(g_1, \ldots, g_n)$ when the parentheses will cause no confusion, which is simply the subset of R consisting of elements of the form $r_1 g_1 + \cdots + r_n g_n$. (This definition can be extended to ideals generated by infinite sets, with the condition that you are only ever allowed to take a finite sum of $r_i g_i$. Our rings don't have a notion of "convergent series" in them ... yet.) If an ideal can be written as $I = (g)$, then I is called **principal**.

A ring R is always an ideal of itself, often denoted (1) to distinguish when we're thinkging about R as a ring versus as an ideal of itself. This ideal is often called the *unit ideal.* There's one more ideal that every ring has: the ideal that consists just of the element 0, intuitively named the *zero ideal.*

Euclidean domains, PIDs, and UFDs

Turning to our examples, notice that both $\mathbb{Z}$ and $\mathbb{C}[x]$ have a notion of "division with remainder", which you hopefully learned about sometime during your K-12 education. They are examples of **Euclidean domains**: a ring R together with a function $f\colon R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ so that for any pair a, b, we can find q, r so that $a = bq + r$ and either $r = 0$ or $f(r) < f(b)$. In the case of $\mathbb{Z}$, the function in question is the usual absolute value, and in the case of $\mathbb{C}[x]$ the function takes a polynomial to its degree. It might seem strange to single out the element 0 to not be given a value by this function. The reason it is left out is that there are contexts in which you want the degree of 0 to be $-\infty$ to make some formula work, and there are instances in which you want the degree of 0 to be ∞ to make some formula work. So, all in all, it's best to not say anything about the degree of 0, and single it out for special treatment.

Let's start working on understanding $\mathbb{Z}$ and $\mathbb{C}[x]$. One extremely basic observation about these rings is that if the product of two elements is 0, one of the two elements is 0. A ring R is an **integral domain**, or simply a **domain**, if whenever $ab = 0$, either $a = 0$ or $b = 0$. An element z such that there exists a non-zero element b satisfying $zb = 0$ is called a **zero-divisor**. So, another equivalent thing to say is that in an integral domain, the only zero-divisor is 0. (Sometimes people define a zero-divisor to be non-zero.)

Another rephrasing is that integral domains are the rings in which *cancellation* is valid: if $ab = ac$, then either $a = 0$ or $b = c$. Notice that every Euclidean domain is also an integral domain.

Let's start working on understanding the ideals of $\mathbb{Z}$ and $\mathbb{C}[x]$. Define an element $d \in R$ to be a **greatest common divisor** (gcd) of two elements a and b if $d \mid a$, $d \mid b$, and for any other element n that divides a and b, we also have that $n \mid d$. Notice that I said *a* gcd, not *the* gcd, because such an element need not be unique: for example, if d is a gcd of a and b, then so is $-d$. More generally, if $u \in R$ is a **unit** (i.e. there exists $v \in R$ so that $uv = 1$) then ud is also a gcd of a and b. Two elements r, r' in a ring R that differ only by multiplication by a unit $r = ur'$ are called **associates**.

Exercise 2. (gcds) Let R be a Euclidean domain.

(a) Suppose $a = bq + r$, and suppose d is a gcd of a and b. Show that d is a gcd of b and r as well.

(b) Given two elements $a, b \in R$, we may perform division with remainder to write $a = bq_1 + r_1$ with either $r_1 = 0$ or $f(r_1) < f(b)$. Then, we may do it again to b and r_1: $b = q_2 r_1 + r_2$. Then to r_1 and r_2. And so on, and so on. Prove that eventually this process reaches 0.

(c) Let r_n be the last nonzero remainder found by the above process. Show that r_n is a gcd of a and b.

(d) The penultimate step of the process gives us an expression of the form $r_{n-2} = q_n r_{n-1} + r_n$, so rearranging we get $r_n = r_{n-2} - q_n r_{n-1}$. Do this all the way up the chain to show that there exist $x, y \in R$ so that $r_n = ax + by$.

(e) Perform the algorithm outlined above starting with the integers $a = 13623$ and $b = 9215$. Calculate their gcd d, then back-substitute to find integers x and y so that $d = 13623x + 9215y$. Yes, I'm serious.

Greatest common divisors are related to thinking about factorizations. An element $p \in R$ is called **irreducible** if whenever $a \mid p$, we have that a is either a unit or an associate of p. Whenever we can take gcds between elements, we can use gcds to factor a given element into irreducibles, like we can factor integers in $\mathbb{Z}$. An integral domain in which every element can be written as a product of irreducible elements, and that product is unique up to multiplying the whole thing by a unit, and up to replacing the irreducible elements in the product by associates, is called a **Unique Factorization Domain** (UFD). (For example, in $\mathbb{Z}$, we don't consider $6 = 2 \cdot 3 = (-2) \cdot (-3) = -(2 \cdot (-3))$ to really be different factorizations of 6.)

Exercise 3. (EDs are PIDs) Prove that if R is a Euclidean domain, then every ideal in R is principal. An integral domain in which every ideal is principal is called a **principal ideal domain** (PID).

Exercise 4. (PIDs are UFDs, Part 1) Let R be a PID. This exercise, along with Exercise 6, will prove that R is also a UFD.

(a) Prove that the ideal $(p) \subset R$ is maximal if and only if p is irreducible.

(b) Prove that every element $r \in R$ has an irreducible divisor. (Hint: every ideal is contained in a maximal ideal.)

The idea is basically to use the proof of unique factorization that is familiar in $\mathbb{Z}$, and adapt it to a general PID. For $\mathbb{Z}$, the outline of the proof is:

1. First, show that any positive integer has a positive prime factor.

2. Then, successively factor $n = p_1 n_1 = p_1 p_2 n_2 = \dots$. The sequence of integers $n, n_1, n_2, \dots$ is a decreasing sequence of positive integers, and since $\mathbb{Z}$ is well-ordered this sequence must end after finitely many steps. By design, it can only end when $n_\ell = 1$, at which point we will have achieved some prime factorization of n.

3. Finally, show that any two putatively distinct factorizations of n actually have the same primes to the same powers.

In the exercise above, we have already accomplished step 1 in a general PID. Step 3 is no harder for a general PID than it is for $\mathbb{Z}$. However, there is something difficult in step 2, because we cannot assume that our PID is well-ordered like the positive integers are. We need some other way to guarantee that this process cannot go on indefinitely, which is provided by the next exercise.

Exercise 5. Let R be a PID, and suppose we have an infinite chain of ideals

$$I_1 \subset I_2 \subset I_3 \subset \dots.$$

(a) Prove that $\bigcup_{j=1}^{\infty} I_j$ is an ideal of R.

(b) Prove that this sequence terminates, in the sense that there is some N so that $I_j = I_N$ for all $j \geq N$.

The property (b) above implies that a PID is **Noetherian**, which we will discuss further in the next set of notes. This tool fills the gap we had.

Exercise 6. (PIDs are UFDs, Part 2) This exercise continues Exercise 4. Let R be a PID.

(a) Prove that any $r \in R$ has a (finite) factorization into irreducible elements.

(b) Suppose $r = up_1^{e_1} \dots p_m^{e_m} = vq_1^{f_1} \dots q_n^{f_n}$, where u, v are units and p_i, q_j are irreducibles with the p_i distinct and the q_j distinct. Prove that every p_i is associate to some q_j, and that $e_i = f_j$. Conclude that also $n = m$. Thus, the factorization is unique in the manner desired, so R is a UFD.

(Actually, it may not be worth your time to think too much about part (b). It's a bit of a tedious argument, and the conclusion is what's important.)

This whole story about factorization is related to why ideals were originally created. People were looking at rings of the form $\mathbb{Z}[\alpha]$ where $\alpha \in \mathbb{C}$ was the root of a monic polynomial with integer coefficients. They hoped that arithmetic in these rings would work a lot like arithmetic in $\mathbb{Z}$. Tragically, it does not. For example, in the ring $\mathbb{Z}[\sqrt{-5}]$, I can factorize $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. All of the elements $2, 3, 1 \pm \sqrt{-5}$ are irreducible in this ring, but none of them are associate to one another (the only units in this ring are ± 1). Kummer observed that the problem here is that there is no number that can serve as $\gcd(2, 1 + \sqrt{-5})$ or $\gcd(3, 1 \pm \sqrt{-5})$. So, he introduced what he called "ideal numbers", which vaguely served to stand in for these missing gcds, and with them he achieved the factorization $6 = (2, 1 + \sqrt{-5})^2(3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5})$. Later, Dedekind solidified Kummer's idea by defining ideals as we do today, as subsets of a ring R.

So, one important takeaway is that an ideal $I = (g_1, \dots, g_n)$ can be thought of as a "generalized gcd" of the elements $g_1, \dots, g_n$.

Fun things to do with ideals

You can do several things to ideals to get new ideals:

- If I, J are ideals, then their **intersection** $I \cap J$ is an ideal. When R is a PID, $(a) \cap (b) = (\mathrm{lcm}(a,b))$, and in general you should think of $I \cap J$ as being like the least common multiple of I and J. **Exercise 7.** Prove that $I \cap J$ is an ideal.
- If I, J are ideals, then their **sum** $I + J = \{i + j : i \in I,\ j \in J\}$ is an ideal. When R is a PID, $(a) + (b) = (\gcd(a,b))$. The sum of two ideals I and J is the smallest ideal that contains both I and J. **Exercise 8.** Prove that $I + J$ is an ideal. Prove that for any ideal I' such that $I' \supset I$ and $I' \supset J$, $I' \supset I + J$.
- If I, J are ideals, then their **product** $IJ = \langle ij : i \in I,\ j \in J \rangle$ is an ideal. This ideal always sits inside $I \cap J$, and it can be strictly smaller. If R is a PID, $(a)(b) = (ab)$, and the fact that IJ can be strictly smaller than $I \cap J$ is similar to how the product of two numbers can be larger than their lcm if they have common factors. **Exercise 9.** Notice that in the definition of IJ I put angle brackets rather than curly brackets, because in general $\{ij : i \in I,\ j \in J\}$ is not an ideal. Let $R = \mathbb{C}[x,y]$, $I = J = (x,y)$. Show that there are elements in IJ that are not expressible as just a product of the form $ij,\ i \in I,\ j \in J$.

Exercise 10. Let $R = \mathbb{C}[x,y,z]$, and define the ideals $I = (x^3, x^2z)$ and $J = (x^4, x^2y^2)$.

(a) Find a generating set for $I + J$.

(b) Find a generating set for IJ.

(c) Find a generating set for $I \cap J$.

(d) Verify that $I \cap J \supsetneq IJ$ by demonstrating a polynomial in $I \cap J$ that is not in IJ.

Since ideals are like numbers, we have an analog of prime numbers. An ideal P is **prime** if it is proper (not all of R), and whenever $ab \in P$, either $a \in P$ or $b \in P$. This gives us one more way to rephrase what it means to be an integral domain: a ring is an integral domain if the ideal (0) is prime. One more kind of ideal that is very important, but which doesn't have a direct analog for numbers are maximal ideals. A proper ideal is **maximal** if it is not contained in any other proper ideal.

The following fact is good to know, but unenlightening to prove. (It is just an exercise in using Zorn's lemma.)

Fact. Given a ring R and an ideal $I \subset R$, there exists a maximal ideal $\mathfrak{m} \supset I$.

Given an ideal I, we can form the **quotient ring** R/I. The elements of the quotient ring are *additive cosets* $r + I = \{r + i : i \in I\}$, and the ring operations are more or less clear: $(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I$, $(r_1 + I)(r_2 + I) = r_1r_2 + I$. As usual, you can shortcut this in your mind as just setting all the elements of the ideal equal to zero, and working "modulo I". Notice that unlike with groups, we don't quotient rings by subrings, but rather by ideals, which have a somewhat different flavor.

Exercise 11.

(a) Prove that $\mathbb{C}[x]/(x-3) \cong \mathbb{C}$.

(b) Prove that $\mathbb{C}[x,y,z]/(x-3, y+1, z-2) \cong \mathbb{C}$.

(c) Prove that $\mathbb{C}[x]/(x^2+x+1)$ is a 2-dimensional $\mathbb{C}$-vector space.

Exercise 12. Let $\phi\colon R \to R'$ be a ring homomorphism. Show that $\ker\phi$ is an ideal of R.

Exercise 13.

(a) Prove that a maximal ideal is prime.

(b) Let I be an ideal of R. Prove that I is prime if and only if R/I is an integral domain.

(c) Let I be an ideal of R. Prove that I is maximal if and only if R/I is a field.

(d) Let $\phi\colon R \to R'$ be a ring homomorphism, and let $I \subset R'$ be an ideal. Prove that $\phi^{-1}(I)$ is an ideal of R. (How does this relate to Exercise 12 above?)

(e) Let $\phi\colon R \to R'$ be a ring homomorphism, and let $P \subset R'$ be a *prime* ideal. Prove that $\phi^{-1}(P)$ is a prime ideal of R.

Exercise 14. Suppose R is a ring with exactly two ideals. Prove that R is a field.

Exercise 15.

(a) Prove that the maximal ideals of $\mathbb{Z}$ are the ideals (p) where p is a prime number. What are the prime ideals of $\mathbb{Z}$? (The answer is different!)

(b) What are the prime and maximal ideals in $\mathbb{C}[x]$?

(c) Consider the ring $\mathbb{R}[x]$. Like $\mathbb{C}[x]$, this ring is also a Euclidean domain, hence a PID and a UFD as well. However, its prime ideals are different from $\mathbb{C}[x]$. Describe the prime ideals of $\mathbb{R}[x]$.

(d) The rings $\mathbb{Z}[x]$ and $\mathbb{C}[x,y]$ are both UFDs, but not PIDs. Choose one, and prove that it is not a PID by finding an ideal which is not principal, and proving that it is such an example.

The next exercise introduces two kinds of ideals which I won't discuss much further in these notes, but which are good to know in general: primary ideals and the radical of an ideal. Primary ideals don't tend to feature on the qual, and when they do they are defined. There has been a qual question asking you to know the definition of the radical before.

Exercise 16. (August 2019 Problem 2) This problem involves ideals in a commutative ring (with unit). A proper ideal I is *primary* if whenever $ab \in I$ and $a \notin I$, then $b^n \in I$ for some $n > 0$. The radical of an ideal I, denoted $\sqrt{I}$, is the set $\sqrt{I} = \{f : f^n \in I \text{ for some } n > 0\}$.

(a) Prove that if I is primary, then $\sqrt{I}$ is prime.

(b) Is the ideal $J = (x^2, xy) \subset \mathbb{Q}[x,y]$ primary? Be sure to carefully justify your answer.

Exercise 17. Let $I = (r)$ be an ideal in a PID R.

(a) Prove that $\sqrt{I}$ is generated by the largest squarefree factor of r.

(b) **(August 2022 Problem 1 (b))** What is the radical of the ideal $(x^3 - x^2)$ in $\mathbb{R}[x]$?

The Isomorphism Theorems

In this section I'll recap the three basic isomorphism theorems in the context of commutative rings. There will be a similar section in the notes on modules and groups, for the correct statements in those contexts. (The statements below hold for non-commutative rings with every instance of "ideal" replaced by "two-sided ideal".)

I won't dwell long here, but I want to draw some attention to the following **Motto: the Isomorphism Theorems for a given algebraic object are analogs of the Rank-Nullity Theorem for vector spaces.** Vector spaces are very simple objects, there is really only one invariant a vector space has, its dimension, and any two vector spaces of the same dimension are isomorphic. The reason nobody ever talks about the isomorphism theorems for vector spaces is that often distinctions that hold when speaking of e.g. groups, rings, or modules will collapse down to statements about dimensions of vector spaces adding to the correct number. The isomorphism theorems below capture facts that are true for vector spaces, and indeed follow from the Rank-Nullity Theorem, but require greater nuance in other contexts

This highlights a strategy, which is perhaps not always helpful on the qual, but is often helpful when doing math. A very general way of thinking about algebra when you are stuck is to see if you can imagine what an algebraic statement would say if everything was a vector space. This doesn't always work, but it can be enlightening, and sometimes the isomorphism theorems will let you directly apply your reasoning from one domain to the other. In the notes about modules, we will see that Nakayama's lemma is another theorem that allows one to transfer information about vector spaces back to rings/modules.

Theorem (first isomorphism theorem)**.** Let $\phi\colon R \to S$ be a ring homomorphism. Then $\ker\phi$ is an ideal of R (see Exercise 12), $\operatorname{im}\phi$ is a subring of S, and the induced homomorphism $R/\ker\phi \to \operatorname{im}\phi$ is an isomorphism. In particular, if ϕ is surjective, then $S \cong R/\ker\phi$.

Theorem (second isomorphism theorem)**.** Let R be a ring, $S \subset R$ a subring, and $I \subset R$ an ideal. Then $S+I = \{s+i : s \in S,\ i \in I\}$ is a subring of R, I is an ideal of $S+I$, $(S+I)/I$ is a subring of R/I, $S\cap I$ is an ideal of S, and the composition $S \hookrightarrow R \twoheadrightarrow R/I$ induces an isomorphism $S/(S\cap I) \xrightarrow{\sim} (S+I)/I$.

Theorem (third isomorphism theorem)**.** Let R be a ring and I an ideal. Then if A is a subring of R such that $I \subset A \subset R$, then A/I is a subring of R/I (we have already used this in the previous isomorphism theorem). Furthermore, if J is an ideal of R such that $I \subset J \subset R$, then J/I is an ideal of R/I, and the composition $R \twoheadrightarrow R/I \twoheadrightarrow (R/I)/(J/I)$ induces an isomorphism $R/J \xrightarrow{\sim} (R/I)/(J/I)$.

Some ideal theory in noncommutative rings

For noncommutative rings like $M_2(\mathbb{C})$, we can also define ideals, but now there's a little rub: an ideal can have the absorbing property under left multiplication, under right multiplication, or under both. These are called, respectively, left ideals, right ideals, and two-sided ideals. You can prove the analog of Exercise 12 by proving that the kernel of a homomorphism $R \to R'$ is a two-sided ideal of R.

The following exercise is here to give you some practice working with ideals in the most basic noncommutative rings.

Exercise 18. (ideals in matrix rings)

(a) **(August 2020 Problem 1 (c), August 2019 1 (a)** Let $R = M_2(\mathbb{C})$. Give an example of a left ideal that is not a right ideal. (In 2019, it was $R = M_2(\mathbb{Z})$, but same dif.)

(b) **(August 2022 Problem 1 (e))** How many two-sided ideals are there in $M_2(\mathbb{C})$? (The answer is going to be a small number.)

(c) **(January 2021 Problem 1 (b), August 2019 Problem 1 (b))** Let $R = M_2(\mathbb{Z})$. Give an example of a proper, non-zero, two-sided ideal.

Something that comes up for noncommutative rings that doesn't come up for commutative rings are simple rings. A **simple ring** is a ring with no proper, non-zero ideals. So, (spoilers) above you should have shown that $M_2(\mathbb{C})$ is simple. A commutative ring is simple if and only if it is a field, but for noncommutative rings there can be many examples of interesting simple rings.

Exercise 19. (January 2020 Problem 1 (a) and (b)) On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

(a) Write down a nonzero element of R which is a zero-divisor.

(b) Write down an element in the center of R which is not in $\mathbb{Z}$.

A few caveats about ideals in noncommutative rings

In noncommutative rings, the definition of a prime ideal is somewhat modified: a proper two-sided ideal P is prime if whenever the product $AB \subset P$ for *ideals* A, B, we in fact have either $A \subset P$ or $B \subset P$. Equivalently, P is prime if whenever the product of principle (two-sided) ideals $(a)(b) \subset P$, then either $a \in P$ or $b \in P$.

Exercise 20.

(a) Prove that in a commutative ring, the two conditions in the preceding paragraph are equivalent to the usual definition of a prime ideal.

(b) Prove that in $M_2(\mathbb{C})$, the zero ideal is prime using the above definition, but is not prime using the definition for commutative rings. (Ideals in a noncommutative ring which satisfy the commutative definition of "prime" are sometimes called **completely prime ideals**.)

Quotient rings can only be constructed when I is a two-sided ideal. If I is one-sided, you can still construct R/I as an abelian group, but you won't be able to make the multiplication work. However, you should keep these kinds of quotients in mind, because later they will be important ways to construct modules over noncommutative rings.

Ring Theory 2

Conventions

In these notes, all rings will be commutative with unity. Although some of the statements generalize (with more or less effort) to the noncommutative setting, we won't be concerned with stating/proving the most general version.

The universal property of quotient rings

Quotient rings (or groups, or modules) have a special property which not everybody sees/appreciates in their first algebra class. I'm going to spend a little space discussing htis property because of how fundamental it is.

Theorem. Let R, R' be rings, I an ideal of R, and $q\colon R \to R/I$ the quotient homomorphism. If $\phi\colon R \to R'$ is a homomorphism, and $\ker\phi \supset I$, then there is a unique map $\widetilde{\phi}\colon R/I \to R'$ such that $\phi = \widetilde{\phi} \circ q$.

Concretely, the map $\widetilde{\phi}$ sends the coset $r+I$ to $\phi(r)$. The content of the statement is mostly verifying that this is a well-defined function.

Of course, given any map $\psi\colon R/I \to R'$, we can readily obtain a map $R \to R'$ whose kernel contains I by taking the composition $\psi \circ q$. So, what this theorem is saying is that in fact, there is a bijection between the set of homomorphisms $R \to R'$ whose kernel contains I, and the set of all homomorphisms $R/I \to R'$. Here's another way of thinking about things: suppose somebody gave you a (admittedly very bizarre) puzzle or a brain teaser by giving you a specific $\phi\colon R \to R'$ satisfying $\ker\phi \supset I$, and saying "I'm thinking of a homomorphism $\psi\colon R/I \to R'$, and this homomorphism satisfies that $\psi \circ q = \phi$, can you figure out what ψ is?". What this theorem says is that this is a very well-designed puzzle, in the sense that there is exactly one answer, and the puzzle-maker has given you exactly the right amount of information to figure out what the answer is, no more no less.

On a practical level, I think about the importance of this theorem in the following way. As a set, the quotient ring R/I is pretty unwieldy. Manipulating cosets is not easy or intuitive, and there's an annoying indeterminacy about cosets where two seemingly different representatives might give the same coset. For example, $1 + 2\mathbb{Z} = 3 + 2\mathbb{Z}$. What the above theorem tells you is that if you want to define a homomorphism from a quotient ring R/I to some other ring R', then you can instead specify what that homomorphism should look like on elements of R, and all you have to do to make sure it descends to the quotient is verify that $\ker\widetilde{\phi} \supset I$, which is often an easy condition to check.

This theorem is perhaps best illustrated by drawing a picture containing all the homomorphisms involved. In diagram form, this theorem looks like this:

$$\begin{tikzcd} R \arrow[d, "q"'] \arrow[dr, "\phi"] & \\ R/I \arrow[r, dashed, "\widetilde{\phi}"'] & R' \end{tikzcd}$$

A restatement of the theorem is "for all homomorphisms ϕ such that $\ker \phi \supset I$, there exists a unique homomorphism $\widetilde{\phi}$ making the above diagram commute". (To say a diagram **commutes** just means that composing the homomorphisms along each directed path gives the same thing in the end, so in this diagram it says that $\phi = \widetilde{\phi} \circ q$.)

The lattice theorem: ideals in a quotient

The lattice isomorphism is often glossed over in a first course in ring theory. The third isomorphism theorem stated that we can find ideals in a quotient ring R/I by taking ideals $J \supset I$ and passing to the quotient J/I. The lattice isomorphism theorem asserts that this process in fact produces a bijection

$$\{\text{ideals of } R \text{ containing } I\} \xrightarrow{\sim} \{\text{ideals of } R/I\}.$$

It also states that this correspondence has a few further properties:

1. This correspondence preserves the subset relation: if J and J' are ideals of R containing I, then $J \subset J'$ iff then $J/I \subset J'/I$.
2. As a consequence, this correspondence preserves ideal sums and intersections: if J and J' are as above, then $(J + J')/I = (J/I) + (J'/I)$ and $(J \cap J')/I = (J/I) \cap (J'/I)$ as ideals of R/I.
3. This correspondence also preserves ideal products: if J and J' are as above, then $(J/I)(J'/I) = (JJ')/I$ as ideals of R/I.
4. As a further consequence of 1. and 3., this correspondence preserves prime and maximal ideals: if P is a prime/maximal ideal of R containing I, then P/I is a prime/maximal ideal of R/I, and all prime/maximal ideals of R/I arise in this way.

This is called the "lattice" isomorphism theorem because we can imagine the ideals of R in a lattice indicating their subset relationships. For example, (part of) the lattice of ideals in $\mathbb{Z}$ is shown in Figure 4.1.

The lattice isomorphism theorem says that the lattice of ideals for R/I is exactly the part of this lattice that lies above the ideal I. So, for example, the lattice for $\mathbb{Z}/6\mathbb{Z}$ is shown in Figure 4.2, deliberately illustrated to highlight the similarity in structure to the ideals of $\mathbb{Z}$ above the ideal (6). Furthermore, the lattice theorem says this correspondence also descends prime and maximal ideals, so for example the prime ideals of $\mathbb{Z}/6\mathbb{Z}$ are the ideals (2) and (3), because these descend from prime ideals of $\mathbb{Z}$.

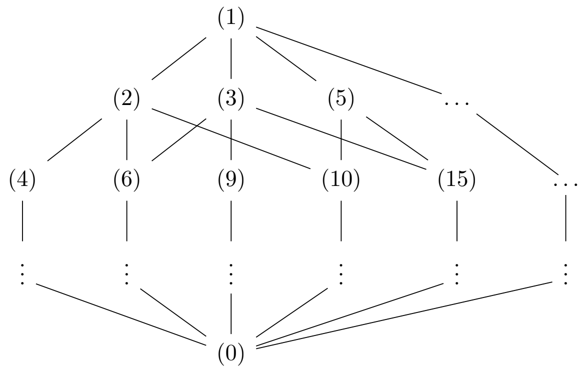


Figure 4.1: The lattice of ideals in $\mathbb{Z}$

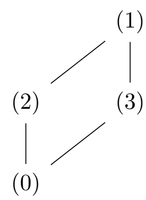


Figure 4.2: The lattice of ideals in $\mathbb{Z}/6\mathbb{Z}$

Exercise 1. (January 2020 Problem 3(b), August 2023 Problem 1 (d)) Give an example of a ring with finitely many elements that has exactly three prime ideals. (In 2023, the problem didn't specify "with finitely many elements". Can you find an example with infinitely many elements?)

The Chinese Remainder Theorem

Background on solving congruences

The Chinese Remainder Theorem (CRT) is one of my favorite theorems. The name comes because of computations appearing in an ancient Chinese mathematical treatise, the *Sunzi Suanjing*. Eric Bach here at UW often calls it "Sunzi's theorem" as a way of crediting the original author, rather than just having a name attributed to his nationality.

The original scope of the CRT is solving systems of congruence relations. For example, consider the system

$$\begin{aligned} x &\equiv 1 \pmod 2 \\ x &\equiv 2 \pmod 3 \\ x &\equiv 4 \pmod 7. \end{aligned}$$

Are there any integers x which satisfy these relations? The answer in this case is yes, for example,

$x = 11$ works, as does $x = 53$ and $x = 95$. However, consider the system

$$\begin{aligned} x &\equiv 1 \pmod 2 \\ x &\equiv 2 \pmod 3 \\ x &\equiv 4 \pmod 8. \end{aligned}$$

Now the answer is no, because if $x \equiv 1 \pmod 2$, then x must be odd, but if $x \equiv 4 \pmod 8$ then x must be even. What differs between the two examples is that the integers $\{2, 3, 7\}$ are "multiplicatively independent" of one another, by virtue of being pairwise coprime, whereas the integers $\{2, 3, 8\}$ are not.

Theorem (CRT, congruence form)**.** A system of congruences

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ &\vdots \quad \vdots \\ x &\equiv a_n \pmod{m_n} \end{aligned}$$

where the m_i are pairwise coprime always has an integer solution. Moreover, the set of solutions forms a residue class modulo $M = \prod_i m_i$.

More generally, if the m_i are not pairwise coprime, then a solution exists iff for all pairs i, j we have that $a_i \equiv a_j \pmod{\gcd(m_i, m_j)}$. In this case, the set of solutions forms a residue class modulo $M = \operatorname{lcm}\{m_i\}_i$

A more modern presentation

The statement of the CRT given above is actually a bit difficult to prove by purely elementary means, and also it would be a weird thing to include in these notes if there wasn't a way it applied to ring theory more generally. The modern way to interpret it is to consider the ring homomorphism $\mathbb{Z} \to \prod_{i=1}^n \mathbb{Z}/m_i\mathbb{Z}$ which sends an element $a \in \mathbb{Z}$ to the tuple of its reductions modulo each m_i: $(a \bmod m_1, a \bmod m_2, \ldots, a \bmod m_n)$. The first statement of the CRT above asserts that this map is surjective, and that its kernel is exactly the ideal $M\mathbb{Z}$, so by the first isomorphism theorem we get an isomorphism of rings $\mathbb{Z}/M\mathbb{Z} \cong \prod_i \mathbb{Z}/m_i\mathbb{Z}$.

In the more general setting where the m_i need not be coprime, for every pair i, j we get a *group* homomorphism $\mathbb{Z}/m_i\mathbb{Z} \times \mathbb{Z}/m_j\mathbb{Z} \to \mathbb{Z}/\gcd(m_i, m_j)\mathbb{Z}$ which takes the pair $(a \bmod m_i, b \bmod m_j)$ to $a - b \bmod \gcd(m_i, m_j)$. (This is not a ring homomorphism because the multiplicative identity in $\mathbb{Z}/m_i\mathbb{Z} \times \mathbb{Z}/m_j\mathbb{Z}$ is $(1, 1)$, which gets sent to 0.) The second statement above is equivalent to saying that a tuple $(a_1 \bmod m_1, \ldots, a_n \bmod m_n)$ is in the image of the map $\mathbb{Z}/M\mathbb{Z} \to \prod_i \mathbb{Z}/m_i\mathbb{Z}$ exactly when it is in the kernel of *all* of these $\binom{n}{2}$ difference-mod-gcd maps.

Rephrasing the statement in these terms both makes it easier to prove, and also makes it easier to generalize. In the first set of notes, we saw analogs for all of the above statements in terms of ideals in an arbitrary ring: gcd corresponds to ideal sum, lcm corresponds to ideal intersection. We are therefore able to state a version of the CRT valid for an arbitrary ring.

Theorem (CRT, general ring version)**.** Let R be a ring and $I_1, \dots, I_n$ be ideals. The ring homomorphism

$$\begin{aligned} R &\xrightarrow{\pi} \prod_{i=1}^{n} R/I_i \\ r &\mapsto (r \bmod I_1, \dots, r \bmod I_n) \end{aligned}$$

has kernel $\bigcap_{i=1}^{n} I_i$, so by first isomorphism theorem we get an injective ring homomorphism

$$\frac{R}{\bigcap_i I_i} \xrightarrow{\widetilde{\pi}} \prod_i R/I_i.$$

Furthermore, we have a group homomorphism (but not ring homomorphism, as discussed above)

$$\begin{aligned} \prod_i R/I_i &\xrightarrow{d} \prod_{i \neq j} R/(I_i + I_j) \\ (a_1, \dots, a_n) &\mapsto (a_i - a_j \bmod I_i + I_j)_{i \neq j}, \end{aligned}$$

and we have that $\ker d = \operatorname{im} \pi$.

In particular, if $I_i + I_j = (1)$ for every pair i, j, then $\widetilde{\pi}$ is an isomorphism.

Exercise 2. Prove that the group homomorphism $d \colon R/I \times R/J \to R/(I+J)$ defined in the statement above is a well-defined function.

The payout of the CRT is that we can sometimes decompose a ring into a direct product of simpler rings. In particular, the comment at the end of the statement of the general ring CRT indicates we can do this if we have the condition $I_i + I_j = (1)$. By analogy with $\mathbb{Z}$, if two ideals $I, J \subset R$ satisfy $I + J = (1)$, those ideals are called **coprime**, or **comaximal** (I slightly prefer the latter, but the former is more common).

Exercise 3. Prove that if I and J are coprime ideals, then $IJ = I \cap J$.

As a short example of a kind of commutative algebra calculation that gets simplified by CRT, consider the ring $\mathbb{C}[x]/(x^2-1)$. We can factor $x^2 - 1 = (x-1)(x+1)$, which also gives us a factorization on the level of ideals: $(x^2 - 1) = (x-1)(x+1)$. Also, the ideals $(x-1)$ and $(x+1)$ are coprime, since $x - 1$ and $x + 1$ are distinct irreducibles and $\mathbb{C}[x]$ is a PID. Thus, CRT tells us that

$$\frac{\mathbb{C}[x]}{(x^2-1)} = \frac{\mathbb{C}[x]}{(x-1)(x+1)} = \frac{\mathbb{C}[x]}{(x-1) \cap (x+1)} \cong_{\mathrm{CRT}} \frac{\mathbb{C}[x]}{(x-1)} \times \frac{\mathbb{C}[x]}{(x+1)}.$$

This is a simpler ring to understand, because $\mathbb{C}[x]/(x \pm 1) \cong \mathbb{C}$.

Exercise 4. Above, we verified that $(x-1)$ and $(x+1)$ are coprime ideals by appealing to the fact that they are generated by distinct irreducibles in a PID.

(a) Prove that if R is a PID, and f, g are distinct irreducible elements of R, then $(f, g) = (1)$.

(b) On the other hand, give an example of $f, g \in \mathbb{C}[x, y]$ such that $\gcd(f, g) = 1$ but $(f, g) \neq (1)$.

Comment. If you know a typed programming language, you will be dismayed that in my general statement of the CRT, I seem to be mixing functions of the type `ring homomorphism` with functions of the type `group homomorphism`. I'm gonna get a compilation error! To make sure my theorems compile correctly, I technically should say that everything is a homomorphism of R-modules, and some of those also happen to be homomorphisms of R-algebras. But this is a little bit of a piddling point.

Exact sequence formulation

When I was taking 742, Daniel Erman told me the following way to think about the CRT in the case of two ideals. I'm going to write it in terms of exact sequences, but I will not define exact sequences until the module theory notes. The CRT asserts that the following sequence of morphisms is exact

$$0 \to R/(I \cap J) \xrightarrow{\widetilde{\pi}} R/I \times R/J \xrightarrow{d} R/(I+J) \to 0.$$

We have an analogy that R/I and R/J are like two subsets of some set $\mathcal{S}$, $R/(I \cap J)$ is like the intersection of the two sets, and $R/(I+J)$ is like their union. In this analogy, $R/I \times R/J$ is like the disjoint union of the two sets, which is what you would get if you took two disjoint copies of $\mathcal{S}$, and cut out R/I from one of them and R/J from the other one. What the CRT is saying is that if we take these two disjoint copies, and glue them together along the intersection $R/(I \cap J)$, we would exactly reconstruct the union $R/(I+J)$. (Algebraic geometers will object that I should have told you that $R/(I \cap J)$ is like a union and $R/(I+J)$ is like an intersection.)

You see ideas similar to this in Topology, where the Mayer-Vietoris sequence looks very similar to the above sequence, and encodes information about intersections, disjoint unions, and unions. The van Kampen theorem does something similar, although it doesn't have a nice formulation in terms of a sequence.

Noetherian Rings

All of the rings discussed so far have had a nice property called the **ascending chain condition** (ACC), which was first "isolated" by Emmy Noether in her work on invariant theory. In her honor, rings with this property are called **Noetherian rings** (sometimes not capitalized, the same way we sometimes don't capitalize abelian even though it is named after Abel). The condition goes like this: a ring satisfies the ascending chain condition if any (possibly infinite) chain of ideals

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

eventually stabilizes. That is, there exists an integer N such that for all $n > N$, $I_n = I_N$.

You should think of this in the context of $\mathbb{Z}$ or $\mathbb{C}[x]$. Since those are PIDs, we can instead look at the generators of the ideals, and $(r_n) \subset (r_{n+1})$ means that $r_{n+1} \mid r_n$. So, in these contexts, the ACC says that we can't have an infinitely long chain of divisibilities $\dots r_n \mid r_{n-1} \mid \cdots \mid r_1$. This makes sense in $\mathbb{Z}$ because in that case the divisors have to get progressively smaller, and in $\mathbb{C}[x]$ they have to have lower and lower degree. So, a ring being Noetherian is a kind of "finiteness" condition, and it plays a similar role for general rings that well-orderedness/induction plays for $\mathbb{Z}$.

In general, what kind of thing does the ACC rule out? The standard example of a non-Noetherian ring is the polynomial ring in infinitely many variables $\mathbb{C}[x_1, x_2, \ldots]$. This ring has an infinitely ascending chain of ideals $(x_1) \subset (x_1, x_2) \subset (x_1, x_2, x_3) \subset \ldots$. The ascending chain condition doesn't allow this kind of perverse behavior.

Exercise 5.

(a) Prove that in any Noetherian ring, every ideal can be generated by finitely many elements.

(b) In the first set of notes, we proved that any PID is Noetherian. Suppose more generally that a ring R has the property that every ideal can be finitely generated. Prove that R is Noetherian.

(c) **(January 2023 Problem 1 (e), modified)** True or False: any ideal $I \subset \mathbb{C}[x, y]$ can be generated by (at most) 2 elements.

Exercise 6.

(a) Suppose R is a commutative Noetherian ring, and $I \subset R$ is an ideal. Prove that R/I is Noetherian.

(b) If R is Noetherian, and R' is a subring of R, it is *not* necessarily true that R' is also Noetherian! Prove this by finding a subring of $\mathbb{C}[x, y]$ that is not Noetherian. (Hint: use the non-Noetherian example above as inspiration. Try to find infinitely many elements so that no element can be obtained from the rest via sums and products. Think about the degrees with respect to x and to y.)

Exercise 7. (Hilbert's basis theorem) The goal of this exercise is to prove that if R is Noetherian, so too is $R[x]$. We will do this by proving that any ideal of $R[x]$ is finitely generated, which is sufficient by Exercise 5 (c).

(a) Let $I \subset R[x]$ be any ideal. Prove that the set of leading coefficients of polynomials in I is an ideal $J \subset R$.

(b) Suppose that I is not finitely generated, and let $f_1, f_2, \ldots$ be elements such that $f_n \in I$, $f_n \notin$ \($f_1, \ldots, f_{n-1}$), and f_n has minimal degree among polynomials with those two properties. Define the ideals $I_n = (f_1, \ldots, f_n) \subset R[x]$, and define J_n to be the ideal of R generated by the leading coefficients of $f_1, \ldots, f_n$. Prove that $J_N = J$ for some N.

(c) Let N be an integer such that $J_N = J$. Prove that $f_{N+1} \in I_N$. Since this is a contradiction with the construction of the fs, conclude that I is finitely generated.

There is a complementary condition to the ACC called the **descending chain condition** (DCC), and rings satisfying the DCC are called **Artinian rings** (same thing about the capitalization). Despite the formal similarity, a ring being Artinian is a *much* more restrictive class of rings. In particular, every Artinian ring is in fact Noetherian, although proving this is not straightforward.

As an example, $\mathbb{Z}$ is Noetherian but not Artinian, while for any $n > 1$, $\mathbb{Z}/n\mathbb{Z}$ is both Noetherian and Artinian.

Comment. Artinian rings were first discussed in the context of central simple algebras, the classification of which is the subject of the Wedderburn-Artin theorem. It is possible that in the notes about modules I will include some further comments on this.

Exercise 8. Suppose R is a commutative Artinian ring. Prove that any prime ideal $P \subset R$ is maximal. (Hint: prove that R/P is a field.)

Fields of fractions and Gauss' Lemma

Given an integral domain R, there's an important field attached to R called its **field of fractions**, denoted $\operatorname{Frac} R$, which consists of all fractions r/s where $r, s \in R$ and $s \neq 0$. Addition and multiplication work just the way you think they should for fractions. For example, the field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$. An example you might not already be familiar with is the field of fractions of $\mathbb{C}[x]$ is $\mathbb{C}(x)$, the field of rational functions. Next section we will discuss localization, of which the field of fractions is just one example.

The field of fractions is useful because often things are easier to prove for fields, and sometimes we can transfer properties from the field of fractions back to the original ring R. For example, we have already proven that $k[x]$ is a Euclidean domain whenever k is a field. The next exercise is to prove **Gauss' lemma**, a super important lemma which, among other things, shows that if R is a UFD, then so is $R[x]$. Hence, so is $R[x_1, \dots, x_n]$ for all n. In particular, this proves the assertions above that $\mathbb{Z}[x]$ and $\mathbb{C}[x, y]$ are UFDs.

(I feel sometimes that whenever there's a hard proof in ring theory, the proof always reduces to either Gauss' lemma or the Cayley-Hamilton theorem, or both!)

Exercise 9. (Gauss' lemma)

(a) Let R be a UFD. Given a polynomial $f \in R[x]$, the *content* of f, $\operatorname{cont}(f)$, is the gcd of all the coefficients in f. Prove that $\operatorname{cont}(fg) = \operatorname{cont}(f)\operatorname{cont}(g)$.

(b) A polynomial such that $\operatorname{cont}(f) = 1$ is called *primitive*. Show that fg is primitive if and only if f and g are primitive.

(c) Let $K = \operatorname{Frac} R$. Prove that a primitive polynomial $f \in R[x]$ is irreducible if and only if it is irreducible when considered as an element of $K[x]$.

(d) Prove that $R[x]$ is a UFD.

Comment. There are actually many related results that get called "Gauss' lemma", some of which can look significantly different from the formulation above.

Localization

This final section is the longest, and most important in these notes. Localization refers to the process of adding fractions to a ring. Although this is a very simple premise, this turns out to enjoy many useful formal properties that makes it very helpful, which we will begin exploring here, and will continue to discuss in the notes on modules. If you're curious about why this process is called "localization", there are two optional exercises in the last section attempting to illustrate the connection between fractions and "local" information, in a geometric or topological sense.

We already know some examples of localization, because, as previously mentioned, taking an integral domain to its field of fractions is a localization. The procedure we will describe will generalize this in two directions:

1. We will be able to talk about fractions in an arbitrary ring, no longer restricted to just integral domains.
2. We will have the option to add only some denominators and not others, and this flexibility will be valuable.

The next exercise is an opportunity for you to reflect on what the algebraic implications are of adding in denominators.

Exercise 10.

(a) Consider the ring $\mathbb{Z}[\frac{1}{2}]$, which consists of the integers, the fraction $\frac{1}{2}$, and the minimum of other numbers which must be in the set in order for it to be a ring. Describe the set of all possible denominators a fraction in this ring can have.

(b) Consider the ring $R = \mathbb{Z}/12\mathbb{Z}$. As in the previous example, I want to be able to make sense of fractions with 2 in the denominator, but now it's more complicated because 2 is a zero-divisor. Denote the ring where I can take denominators of 2 by $R[\frac{1}{2}]$.

(i) Use your knowledge of how to manipulate fractions to convince yourself that $\frac{3}{2} = 0$ in $R[\frac{1}{2}]$. (Hint: you can multiply the top and bottom of a fraction by the same number to get an equivalent fraction.)

(ii) On the other hand, convince yourself that if $n \in R$ is not in the ideal (3), then $\frac{n}{2} \neq 0$ in $R[\frac{1}{2}]$.

(iii) The ring $R[\frac{1}{2}]$ is finite. Use what you deduced above to figure out how many inequivalent elements it has.

(c) Still thinking about $R = \mathbb{Z}/12\mathbb{Z}$, what ring would we get if we decided to take 5 in the denominator? What would we get if we decided to take 6 in the denominator?

In general, what do we need from a set of possible denominators $S \subset R$ in order for it to make sense to add in all those denominators to our ring? We always want to be able to have a denominator of 1, so we will demand that $1 \in S$ always. In order for the familiar property $\left(\frac{r_1}{s_1}\right)\left(\frac{r_2}{s_2}\right) = \frac{r_1 r_2}{s_1 s_2}$ to hold, we need $s_1 s_2 \in S$ for any $s_1, s_2 \in S$. Any set S that satisfying these two properties is called **multiplicatively closed**. Given a multiplicatively closed subset $S \subset R$, we define a new ring $S^{-1}R$ in imitation of the definition of the rational numbers, with one twist:

- The elements of $S^{-1}R$ are symbols of the form $\frac{r}{s}$ where $r \in R$ and $s \in S$.
- You might expect two symbols $\frac{r_1}{s_1}, \frac{r_2}{s_2}$ to be equal if their cross difference vanishes, $r_1 s_2 - r_2 s_1 = 0$. However, this is the twist: we actually say they are equal if *there is some* $s \in S$ *such that* $s(r_1 s_2 - r_2 s_1) = 0$.
- Multiplication is defined as you would expect. Addition is defined by taking common denominators:

$$\frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}.$$

I think about the modification in the second bullet point like this. We all agree how addition and subtraction of fractions should work, and we all agree that $\frac{a}{b} = \frac{sa}{sb}$ for any $s \in S$. The change in the second bullet point over what you would naively expect is there to account for situations where you can chain these two rules to get unexpected results. In the example computed in Exercise 10 (b), we could have

$$4 - 1 = 3 = \frac{4 \cdot 3}{4} = \frac{0}{4} = 0,$$

so in fact we find that $4 = 1$ once we allow in $\frac{1}{2}$.

Comment. The notation S is standard for a multiplicatively closed subset, and also as the name of a second ring after R is taken. I've avoided using the latter so that I can have S always be a multiplicatively closed subset. Some authors, such as Atiyah-Macdonald and also me in other writings, adopt the French convention in which rings are A (for "anneau"), and further rings are B, C, which has the added benefit of leaving S available to always be a multiplicatively closed subset.

Example 1. Here are two very commonly used kinds of localization.

- Given a ring R and an element $f \in R$, the set $S = \{1, f, f^2, \ldots\}$ consisting of all the powers of f is multiplicatively closed. In this case, the localization $S^{-1}R$ is commonly denoted by either R_f or $R[\frac{1}{f}]$. The former is both more traditional and faster to write, while the latter is much more unambiguous. When written R_f, it is commonly said aloud as "R localized at (the element) x", and when written $R[\frac{1}{f}]$ it is said as "R adjoin $\frac{1}{f}$".
- Given a ring R and a prime ideal P, the set $S = R \setminus P$, all the elements of R that are *not* in P, is multiplicatively closed. In this case, the localization $S^{-1}R$ is denoted by R_P, and it is said aloud as "R localized at (the ideal) P". Sometimes I have heard people say "R localized *away from* P", to emphasize that S is the complement of P. **Exercise 11.** Prove that $S = R \setminus P$ is multiplicatively closed. Prove that in R_P, the ideal generated by the elements of P, $\langle \frac{p}{1} : p \in P \rangle$, is a maximal ideal.

As one example, if R is an integral domain, we have already seen that (0) is a prime ideal of R. Then the localization $R_{(0)}$ is just the same thing as throwing in $1/r$ for every nonzero $r \in R$, which is the same thing as taking the fraction field. So, $\mathrm{Frac}(R) = R_{(0)}$.

Comment. Considering the element $x \in \mathbb{C}[x]$, note that the ideal (x) is prime. So, I can either form the localization $\mathbb{C}[x]_x$ or $\mathbb{C}[x]_{(x)}$, and these are two *different* rings! This is why in many cases I would prefer to write $\mathbb{C}[x]_x$ as $\mathbb{C}[x, x^{-1}]$. What is worse, when $R = \mathbb{Z}$, group theorists and topologists will write $\mathbb{Z}_n$ to mean the cyclic group $\mathbb{Z}/n\mathbb{Z}$, and number theorists will write $\mathbb{Z}_p$ to mean the p-adic integers (which are closely related to localization, but distinct). So, in the context of $\mathbb{Z}$, you should always write $\mathbb{Z}[\frac{1}{n}]$, and if you want to localize at (away from) the prime ideal (p), make sure you include the parentheses: $\mathbb{Z}_{(p)}$.

The localization homomorphism

Whenever you learn a way to get a new object from old objects, you should ask "how does this interact with homomorphisms?". In many respects, localization behaves a lot like a quotient ring. There is a homomorphism $R \to S^{-1}R$ that takes $r \to \frac{r}{1}$. The quotient homomorphism $R \to R/I$ is characterized by taking every element of I to 0 in the quotient. The localization homomorphism is characterized by taking every element $s \in S$ to a unit in the localization, which makes sense because, after all, we're explicitly adding in elements of the form $\frac{1}{s}$. The localization homomorphism also has a property similar to the one discussed above in the section **The universal property of quotient rings**:

Theorem. Let R, R' be rings, let S be a multiplicatively closed subset of R, and let $\ell\colon R \to S^{-1}R$ be the localization homomorphism. If $\phi\colon R \to R'$ is a homomorphism such that for every $s \in S$, $\phi(s)$ is a unit in R', then there is a unique map $\widetilde{\phi}\colon S^{-1}R \to R'$ such that $\phi = \widetilde{\phi} \circ \ell$.

Concretely, the map $\widetilde{\phi}$ sends the fraction $\frac{r}{s} \mapsto \phi(r)\phi(s)^{-1}$. As before, the content of the statement is mostly verifying that this is a well-defined function.

The discussion of how to think about/interpret the similar property for quotient rings applies equally well to this theorem. Similar to quotient rings, if we tried to write down functions out of a localization on the level of the elements of the localization, it would be difficult because (1) the localization can have many new elements that weren't in the original ring, and (2) the elements of the localization might have difficult-to-anticipate relationships between them, as we saw above when we showed that $4 = 1$ in $(\mathbb{Z}/12\mathbb{Z})[\frac{1}{2}]$. This theorem gives us an alternative way of defining these maps, which can be much easier to check in general.

Here is a diagram for the universal property of the localization homomorphism:

$$\begin{tikzcd} R \arrow[d, "q"'] \arrow[dr, "\phi"] & \\ S^{-1}R \arrow[r, dashed, "\widetilde{\phi}"'] & R' \end{tikzcd}$$

A restatement of the theorem is "for all homomorphisms ϕ such that $\phi(s)$ is a unit for all $s \in S$, there exists a unique homomorphism $\widetilde{\phi}$ making the above diagram commute".

Exercise 12.

(a) Let $f, g \in R$. Prove that $R[\frac{1}{fg}] \cong R[\frac{1}{f}, \frac{1}{g}]$. (Try doing this once using the definition of localization, and once using the theorem above to construct homomorphisms in both directions, then showing they are inverses. Reflect on how these two arguments feel different to think about.)

(b) Prove that the localization homomorphism $\ell\colon R \to S^{-1}R$ is injective if and only if S contains no zero-divisors.

(c) Prove that the localization $S^{-1}R$ is the zero ring if and only if $0 \in S$.

Exercise 13. Give an example of a ring homomorphism $\phi\colon R \to S$, and a maximal ideal $\mathfrak{m} \subset S$ such that $\phi^{-1}(\mathfrak{m})$ is *not* a maximal ideal of R. (Hint: this problem is in the section on localization.)

Ideals in a localization

There is an analog to the lattice isomorphism theorem for ideals in a localization.

Theorem. Let R be a ring, S a multiplicatively closed subset, and I an ideal of R. Then the ideal $S^{-1}I = \langle \frac{i}{1} : i \in I \rangle$ is an ideal of $S^{-1}R$. Further, given an ideal $J \subset S^{-1}R$, we can always take its inverse image under the localization homomorphism $\ell^{-1}(J)$, and this will be an ideal of R. Pictorially:

$$\begin{tikzcd}
\{\text{ideals } I \subset R\} \arrow[r, bend left, "I \mapsto S^{-1}I"] & \{\text{ideals } J \subset S^{-1}R\} \arrow[l, bend left, "J \mapsto \ell^{-1}(J)"]
\end{tikzcd}$$

This correspondence has the properties:

- Starting with an ideal $J \subset S^{-1}R$ and first taking its inverse image then taking its localization gets us back the original ideal: $S^{-1}\ell^{-1}(J) = J$. In particular this implies that the map $J \mapsto \ell^{-1}(J)$ is injective, and the map $I \mapsto S^{-1}I$ is surjective.
- Both maps preserve subsets and intersections. So, for example, if $I \subset J$, then $S^{-1}I \subset S^{-1}J$. (It can fail to preserve strict subsets, as seen below in Figure 4.3.)
- **(important)** The above maps induce a bijection between the set of prime ideals $P \subset R$ that does not meet S (meaning $P \cap S = \emptyset$) and the set of prime ideals of $S^{-1}R$.

When the localization is of the form R_P, where P is a prime ideal, then often instead of writing $S^{-1}I$ people will write I_P. Similarly, when the localization is of the form R_f, where $f \in R$ is an element, often people will write I_f. This is the main situation in which the notation R_f seems better than the notation $R[\frac{1}{f}]$, because nobody writes (or wants to write) $I[\frac{1}{f}]$, and writing something more apt like $IR[\frac{1}{f}]$ is a bit cumbersome. (But people do write it, e.g. number theorists will write things like $3\mathbb{Z}[\frac{1}{2}]$ quite happily.)

The moral here is that if we add in a fraction $\frac{1}{f}$, then we effectively eliminate all the ideals that contain f, because now any ideal that contains f also contains $f \cdot \frac{1}{f} = 1$, and so is the unit ideal. So, for example, referencing our lattice of the ideals in $\mathbb{Z}$ from earlier, the lattice of ideals in $\mathbb{Z}[\frac{1}{6}]$ is illustrated in Figure 4.3, where I have slashed out the ideals that have become the unit ideal after localizing. Notice that any ideal divisible by only (2) and/or (3) has now become the unit ideal, and several other ideals have merged, like (5) = (10) = (15) = (30).

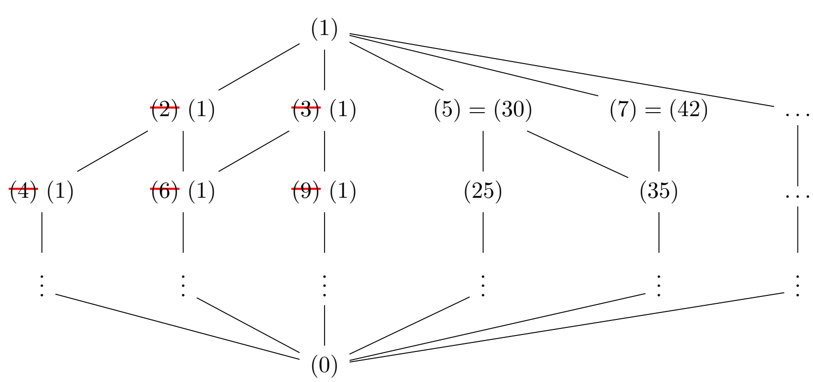


Figure 4.3: The lattice of ideals in $\mathbb{Z}[\frac{1}{6}]$

Exercise 14. (January 2020 Problem 3 (a), (c)) Let R be a commutative ring, I an ideal in R, and S a multiplicative subset of R. There are two standard correspondences involving prime ideals:

(i) Prime ideals in R/I are in bijection with prime ideals in R which contain I.

(ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S.

(a) Prove (i) and (ii). (The original problem only asked for one or the other, but you should be able to prove both.)

(c) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

Exercise 15. (August 2021 Problem 1 (a), (b)) On this problem, only the answer will be graded.

(a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?

(b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)

Exercise 16. The bijection between primes of $S^{-1}R$ and primes of R that do not meet S might induce you to believe that there is a bijection between the set of *all* ideals of $S^{-1}R$ and the set of ideals of R that do not meet S. (In particular, it induced me to say that in an earlier version of these notes.) Give a counterexample to this proposal. (Hint: you can find one from your answer to Exercise 15.)

Local rings

A ring is called a **local ring** if it has exactly one maximal ideal. All fields are local rings (because (0) is maximal in a field).

Exercise 17.

(a) Suppose R is a local ring with unique maximal ideal $\mathfrak{m}$. Prove that any $x \in R \setminus \mathfrak{m}$ is a unit.

(b) Conversely, suppose that R is a ring, and I is an ideal of R with the property that every $x \in R \setminus I$ is a unit. Prove that R is local and I is its unique maximal ideal.

(c) Find an example of a quotient of $\mathbb{Z}$ or $\mathbb{C}[x]$ that is a local ring that is not a field.

(d) Let R be a ring, P a prime ideal. Prove that R_P is a local ring with maximal ideal P_P (using the notation from above).

If fields are easiest rings to work with, local rings are for most things the second easiest rings to work with, and if you do anything with commutative algebra, algebraic geometry, or the algebraic side of number theory, you will encounter local rings of the form R_P very frequently. A proof strategy that is used very commonly goes like this:

1. First, prove a statement is true for fields. This is usually very easy.
2. Next, prove that the statement for fields implies the statement for local rings. If R is a local ring with maximal ideal $\mathfrak{m}$, then $R/\mathfrak{m}$ is called the **residue field** of R, and many properties of the residue field can be lifted to R, for example by using Nakayama's lemma (which will be discussed more in the notes on modules).

3. Finally, prove that if R is an arbitrary ring, then the statement holds for R if and only if it holds for all the rings R_P, where P ranges over all prime ideals of R. This is usually the hardest part of the process, and is frequently not true without pretty strong hypotheses on the ring R.

Exercise 18. (January 2018 Problem 1) For this problem, your answer will be graded on correctness alone, and no justification is necessary.

(a) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization $R_f = 0$.

(b) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization map $R \to R_f$ is neither injective nor surjective.

(c) Give an example of a *local* ring R and an element $f \in R$ where $R_f \neq 0$ but R_f is no longer a local ring.

What makes local rings "local"?

The following two exercises are optional. Their purpose is to give a hopefully enlightening example showing in what sense "local rings" and "localization" can capture "local" information, in a topological or geometric sense.

Exercise 19. (localization in geometry) Take the plane $\mathbb{R}^2$ with the usual topology via open disks.

(a) For any open set $U \subset \mathbb{R}^2$, show that $\mathcal{C}(U)$, the set of continuous functions $U \to \mathbb{R}$, is a commutative ring under the operations of pointwise addition and multiplication.

(b) For any $f\colon U \to \mathbb{R}$, show that $f^{-1}(0)$ is a closed subset of U.

(c) Show conversely that for any closed subset $K \subset U$, there exists a function $f\colon U \to \mathbb{R}$ such that $f^{-1}(0) = K$. (Don't spend too long on this, it's not worth it to do it in great detail.)

(d) If U, V are open sets with $V \subset U$, then given any $f \in \mathcal{C}(U)$, we can restrict f to V to get a function $f|_V \in \mathcal{C}(V)$. Show that this defines an injective ring homomorphism $\mathcal{C}(U) \to \mathcal{C}(V)$. (Notice that the map goes the opposite direction from the set inclusion!)

(e) If V is open, $V \subset U$, then $U \setminus V$ is a closed subset of U. Call this closed subset K. Show that if $f \in \mathcal{C}(U)$ satisfies $f^{-1}(0) \subset K$, then $f|_V$ is a unit in $\mathcal{C}(V)$. Thus, by restricting to a smaller open set, we have added $1/f$ to the ring for all such f.

Exercise 20. (continuation of Ex. 19) Take again the plane $\mathbb{R}^2$.

(a) Let $U \subset \mathbb{R}^2$ be any open set containing the origin. Show that the set $\mathfrak{m}_{0,U} := \{f \in \mathcal{C}(U) : f(0,0) = 0\}$, the set of functions which vanish at the origin, is a maximal ideal of $\mathcal{C}(U)$. (Hint: show that $\mathcal{C}(U) \to \mathbb{R}$ given by $f(x,y) \mapsto f(0,0)$ is a ring hom.)

(b) Let U and V be any two open sets containing the origin, and declare a function $f \in \mathcal{C}(U)$ to be equivalent to a function $g \in \mathcal{C}(V)$ if there is some open set $W \subset U \cap V$, $0 \in W$, so that $f|_W = g|_W$. That is, two functions are equivalent if they look the same once we zoom in close enough to $(0,0)$.

Check that this is an equivalence relation on the set of pairs (U, f), where U is an open set containing the origin, and $f \in \mathcal{C}(U)$.

(c) Define a ring $\mathcal{C}_0$ as follows: the elements of $\mathcal{C}_0$ are pairs (U, f), where U is an open set containing the origin, and $f \in \mathcal{C}(U)$, subject to the equivalence relation above. The equivalence classes under this relation are called **germs** of functions. To add or multiply two germs (U, f) and (V, g), we pass to some smaller open set $W \subset U \cap V$, and then add or multiply $f|_W$ with $g|_W$.

Prove that $\mathcal{C}_0$ is a local ring, whose unique maximal ideal consists of germs of the form (U, f), where $f \in \mathfrak{m}_{0,U}$. (Hint: prove that if a germ is not in $\mathfrak{m}_{0,U}$, then it is a unit.)

(d) In fact, prove that if U is an open set containing the origin, then $\mathcal{C}_0 \cong \mathcal{C}(U)_{\mathfrak{m}_{0,U}}$.

You should think of the germ of a function f as essentially remembering its Taylor expansion at the origin (disregarding the fact that I stated the above for continuous rather than differentiable or analytic functions). It remembers the value $f(0,0)$, and all the infinitesimal information about f's behavior near 0, but it doesn't remember the value of $f(x)$ for any $x \neq 0$. One could say that the germ of f remembers all, and only, the local information about f near the point $x = 0$.

Notice that in the two proceeding exercises, it was never particularly important that the space we were considering was $\mathbb{R}^2$, and that the functions were landing in $\mathbb{R}$. In some sense, all we used in the constructions was:

- the space $\mathbb{R}^2$ is a topological space,
- the ring $\mathbb{R}$ is a ring,
- for any two points $x, y \in \mathbb{R}^2$, there is a function $f \colon \mathbb{R}^2 \to \mathbb{R}$ so that $f(x) = 0$ and $f(y) \neq 0$.

Because of how lax these requirements are, similar constructions can be (and are) used to study many different kinds of geometric objects from an algebraic perspective: manifolds, complex analytic spaces, varieties, and more.

Module Theory 1

First examples and basic properties

A module is what you get if you take the definition of a vector space and replace every instance of the word “field” with the word “ring”. A homomorphism of R-modules is an R-linear map: $f(x+y) = f(x)+f(y)$, $f(rx) = rf(x)$. If the ring is not commutative, one has to specify whether the scalar multiplication is on the right or the left. I’m going to focus most of my attention on commutative rings, and write multiplication on the left, but most of what I say will either apply verbatim to noncommutative rings or require only minor adjustments. Just keep in mind that over noncommutative rings, the difference between a left and a right module can be substantive.

Examples

We will begin with modules over our three big examples.

Example 1.

- A $\mathbb{Z}$-module is just an abelian group. Think of your favorite abelian group, that’s a $\mathbb{Z}$-module baby.
- How can we describe a $\mathbb{C}[x]$-module? The elements of $\mathbb{C}[x]$ can be built out of constants from $\mathbb{C}$ and the element x, along with addition and multiplication. So, to define a $\mathbb{C}[x]$-module M, we need to say how to multiply by constants from $\mathbb{C}$, and also how to multiply by the element x. Saying we need to multiply by constants from $\mathbb{C}$ is the same as saying that M must be a $\mathbb{C}$-vector space (we will generalize this below to any situation involving a subring). **Exercise 1.** Check from the module axioms that the “multiplication by x” map $x\cdot\colon M \to M$ must be a linear operator on M as a complex vector space.

  On the other hand, given any complex vector space V and any linear operator $T\colon V \to V$, we can make V into a $\mathbb{C}[x]$-module by declaring that for every vector $v \in V$, $x \cdot v = T(v)$. Thus, the data of a $\mathbb{C}[x]$-module is exactly the same as the data of a pair (V,T) of a complex vector space and a linear operator. (Note that one and the same V can have non-isomorphic realizations as a $\mathbb{C}[x]$-module if we choose different linear operators T.)
- What do the (left) modules of $M_2(\mathbb{C})$ look like? There is one $M_2(\mathbb{C})$-module that is hopefully easy to think of: the vector space $\mathbb{C}^2$ where $M_2(\mathbb{C})$ acts by left multiplication. Here’s another one: take the vector space $\mathbb{C}^4$, but write each element as a pair of vectors $(v_1,v_2) \in \mathbb{C}^2 \times \mathbb{C}^2$. This is an $M_2(\mathbb{C})$-module where $M(v_1,v_2) = (Mv_1,Mv_2)$. In a similar fashion we could make $(\mathbb{C}^2)^n$ into an

$M_2(\mathbb{C})$-module, but all of these modules seem ... idk a little plain? Are there any spicier $M_2(\mathbb{C})$-modules? Actually, the answer is no, these are the only (finitely generated) $M_2(\mathbb{C})$-modules. We will prove this later, this is related to the fact that $M_2(\mathbb{C})$ is a simple ring, as discussed in the Ring Theory 1 notes.

Below, in the section Finitely-generated modules over a PID, we will give another way to look at $\mathbb{Z}$-modules and $\mathbb{C}[x]$-modules. In particular, viewing a $\mathbb{C}[x]$-module as a pair (V, T), we will obtain some nice applications of module theory to linear algebra.

Exercise 2.

(a) How can you describe a $\mathbb{C}[x]$-submodule of a given $\mathbb{C}[x]$-module (V, T)?

(b) How can you describe a $\mathbb{C}[x]$-module homomorphism from (U, S) to (V, T)?

Example 2. Let us turn now to modules over an arbitrary ring R. Here are some ways to find some R-modules.

- Given any ring R, R is a module over itself. Moreover, the Cartesian product R^n is a module over R, e.g. $\mathbb{Z}^2$ is a $\mathbb{Z}$-module, and $\mathbb{C}[x]^2$ is a $\mathbb{C}[x]$-module.
- Given a ring R, an ideal $I \subset R$ is an R-module. In fact, the definition of an ideal is the same as "R-submodule of R". For example, $2\mathbb{Z}$ is a $\mathbb{Z}$-module, and the ideal $(x, y) \subset \mathbb{C}[x, y]$ is a $\mathbb{C}[x, y]$-module.
- Given a ring R and an ideal $I \subset R$, the quotient R/I is an R-module, e.g. $\mathbb{Z}/6\mathbb{Z}$ or $\mathbb{C}[x]/(x^2+x+1)$.
- Given a commutative ring R, any ring of polynomials in one or more variables $R[x], R[x, y], \ldots$ is an R-module.
- Given a commutative ring R, one can also form the ring of **noncommutative polynomials** in n variables over R, denotes $R\langle x_1, \ldots, x_n\rangle$. These are polynomials where the variables don't commute with one another. In one variable, this makes no difference, but in 2 or more variables you can have expressions like $xy - yx$, which is nonzero in $R\langle x, y\rangle$. I mention this because every now and then a ring of this flavor shows up on the qual.

Exercise 3. Let f be a polynomial of degree d. The $\mathbb{C}[x]$-module $\mathbb{C}[x]/(f)$ is also a complex vector space. What is its dimension?

Example 3. We have ways to make new modules out of old.

- Given any two R-modules M, N, we can form their **direct sum** $M \oplus N$. The underlying set of $M \oplus N$ is the Cartesian product $M \times N$, and scalar multiplication works as you might expect: $r(m, n) = (rm, rn)$. So, above, $R^n = \underbrace{R \oplus \cdots \oplus R}_{n \text{ times}}$. This gives us other examples, such as $\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z}$.
- Given an R-module M and a submodule $N \subset M$, the quotient group M/N is an R-module. This generalizes our observation above that R/I is an R-module.

Exercise 4. Given R-modules M, N, let $\mathrm{Hom}_R(M, N)$ be the set of all R-module homomorphisms $f\colon M \to N$.

(a) Prove that $\mathrm{Hom}_R(M, N)$ is itself an R-module.

(b) What are the isomorphism classes of the following $\mathbb{Z}$-modules?

  (i) $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z})$

  (ii) $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$

  (iii) $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/6\mathbb{Z}, \mathbb{Z})$

  (iv) $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/5\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}, \mathbb{Z}/25\mathbb{Z})$

(c) For any ring R and any R-module M, there is an isomorphism $\mathrm{Hom}_R(R, M) \cong M$ that doesn't require me to tell you anything else about the ring R or the module M. Can you find it?

Exercise 5.

(a) Consider $\mathrm{Hom}_R(M, M)$. Show that under pointwise addition and function composition, $\mathrm{Hom}_R(M, M)$ is a ring. This ring, which is often not commutative, is called the **ring of endomorphisms** of M, $\mathrm{End}_R(M)$.

(b) Prove that for $r \in R$, the "multiplication by r" map $\phi_r\colon M \xrightarrow{r\cdot} M$ is an endomorphism of M.

(c) Prove that the function $R \to \mathrm{End}_R(M)$ sending $r \in R$ to $\phi_r \in \mathrm{End}(M)$ is a ring homomorphism.

(d) In the case V is a finite-dimensional k-vector space, what is $\mathrm{End}_k(V)$? What is the image of the ring homomorphism $k \to \mathrm{End}_k(V)$?

Example 4. We can make modules for some rings out of modules for others.

- If $R' \subset R$ is a subring, and M is an R-module, then M is moreover an R'-module. For example, we said above that any $\mathbb{C}[x]$-module is moreover a $\mathbb{C}$-vector space. Another example is we can view any $\mathbb{C}$-vector space as an $\mathbb{R}$-vector space of twice the dimension, or view any $\mathbb{Q}$-vector space as a $\mathbb{Z}$-module (abelian group). Often, this way of forming modules is referred to as **restriction of scalars**. (At the end of the linear algebra notes I mentioned extension of scalars. As the example of $\mathbb{C}$- versus $\mathbb{R}$-vector spaces illustrates, restriction of scalars does not quite undo extension of scalars, but it does something pretty close, which I may or may not mention later.)
- More generally, if $f\colon R' \to R$ is any ring homomorphism, and M is an R-module, then we can view M as an R'-module via $r' \cdot m := f(r')m$. In particular, R itself becomes an R'-module in this way.

Generating sets, bases, and torsion

Let's compare some behavior of modules to their analogs in vector spaces. The analog of a finite-dimensional vector space would be a **finitely-generated** module. A module M over a ring R is finitely-generated (abbreviated to f.g.) if there is some finite set of elements $m_1, \ldots, m_n \in M$ so that for any other $m \in M$ we can find ring elements $r_1, \ldots, r_n$ such that $m = r_1 m_1 + \cdots + r_n m_n$. The set $\{m_1, \ldots, m_n\}$ is said to **span** or **generate** M.

Exercise 6. In these notes we will primarily occupy ourselves with finitely-generated modules, but don't go thinking you can avoid infinitely-generated modules! Prove that the polynomial ring $R[x]$ is infinitely-generated as an R-module.

We can define a basis for a f.g. module M over a ring R in much the same way we do for vector spaces. A collection $m_1, \dots, m_n \in M$ is said to be **linearly independent** if whenever $r_1 m_1 + \cdots + r_n m_n = 0$, we must have that the r_i are all 0. A **basis** for the module M is a linearly independent spanning set.

(There are similar definitions for spanning/generating sets and linear independence in infinitely generated modules, I just don't care about writing them down.)

A basis for a module is a much more special thing than for a vector space. Two special properties that vector spaces have is that if V is a (say d-dimensional) vector space, any maximal set of linearly independent vectors is a basis, and any minimal set of spanning vectors is a basis. Neither of these is true for modules. Consider $\mathbb{Z}$ as a module over itself. This module *does have* a basis, namely the set $\{1\}$ (or the set $\{-1\}$). However, the set $\{2\}$ is a set of linearly independent elements of $\mathbb{Z}$, and no element of $\mathbb{Z}$ can be added while maintaining linear independence, but nevertheless this set is not a basis. Similarly, the set $\{2, 3\}$ spans $\mathbb{Z}$, and removing either element makes it not span $\mathbb{Z}$, but it is not a basis. (Note that 2 and 3 are not linearly independent of each other because $3 \cdot 2 - 2 \cdot 3 = 0$. In a module, saying two elements are linearly dependent is *not* the same as saying one is a multiple of the other, because we may not be able to divide in the ring R.)

But modules are not just vector spaces where you have to be careful not to divide, there is genuinely new behavior possible in modules that is not possible in vector spaces. We observed above that $\mathbb{Z}/6\mathbb{Z}$ is a $\mathbb{Z}$-module, and in $\mathbb{Z}/6\mathbb{Z}$ the (class of the) element $[2]$ is not the 0 element of the module, but after scaling by the nonzero element $3 \in \mathbb{Z}$ it becomes 0. Nothing of the sort can happen in a vector space.

The two behaviors we have seen already are important enough to the study of modules that they get their own names:

- A module with a basis is called a **free module**. For any ring R, R^n is a free module over R, with basis $e_i = (0, \dots, 0, 1, 0, \dots, 0)$, where the 1 is in the ith place. For any free module F, choosing a basis gives an isomorphism $F \cong R^n$ by sending the chosen basis to the e_i.

- For a module M over an integral domain R, an element $m \in M$ is **torsion** if there exists some nonzero $r \in R$ so that $rm = 0$. (There is much vocabulary related to the idea of rm being a way of "twisting" the element m.) A module M is called **torsion** if all of its elements are torsion. Any module M has a submodule $T(M)$ consisting of all of its torsion elements, reasonably enough called the "torsion submodule". A module is **torsion-free** if its only torsion element is 0. (Like with zero-divisors, sometimes people will exclude $0 \in M$ from being torsion, mostly depending on whether it makes a given statement simpler or not.)

Comment. Some authors don't define torsion over rings with zero-divisors, but you can do so if you want. All that changes is you replace "exists some nonzero $r \in R$" with "exists some non-zero-divisor $r \in R$". This is a somewhat subtle thing, for example, inside $\mathbb{Z}/6\mathbb{Z}$ we have the ideal $3\mathbb{Z}/6\mathbb{Z}$. You might think this is a torsion $\mathbb{Z}/6\mathbb{Z}$-module, because $3 \in 3\mathbb{Z}/6\mathbb{Z}$ satisfies $2 \cdot 3 = 0$. However, the above definition tells us this module is *torsion-free*, because $2 \in \mathbb{Z}/6\mathbb{Z}$ is a zero-divisor!

This subtlety will not come up in the rest of the notes, and I haven't seen it come up on the qual, because they are careful to ask about torsion only over integral domains.

Exercise 7. Let M be a module over an integral domain R, and suppose M has a nontrivial torsion submodule, $T(M) \neq \{0\}$. Prove that M has no basis.

Exercise 8. Let R be an integral domain. Prove that an ideal $I \subset R$ is a free module iff it is principal.

Exercise 9.

(a) Suppose (V, T) is a $\mathbb{C}[x]$-module, and V is finite-dimensional. Prove that V is finitely-generated as a $\mathbb{C}[x]$-module.

(b) Under the same assumptions, prove that (V, T) is torsion.

Exercise 10.

(a) Let M be a f.g. torsion module over an integral domain R. Prove that there is some nonzero $r \in R$ so that $rM = 0$.

(b) Give an example of a module M over $\mathbb{Z}$ or over $\mathbb{C}[x]$ which is torsion, but for which no nonzero r satisfies $rM = 0$.

Exercise 11. Let M be a torsion-free module. Prove that any submodule of M is also torsion-free.

A very important perspective on modules is that we can view any f.g. module as a quotient of a free module. (Again, this is also true for infinitely-generated modules.) If $m_1, \ldots, m_n$ generate M, then there is an R-linear surjection $R^n \xrightarrow{g} M$ which sends e_i to m_i. (Notice the similarity to the map you found in Exercise 4 (c).) Then the first isomorphism theorem tells us that $M \cong R^n / \ker g$. Because of this, knowing properties of M is tied up with knowing properties of $\ker g$.

Finitely-generated modules over a PID

The nicest case of modules are f.g. modules over PIDs. The main attraction here is the following classification theorem.

Theorem. Let R be a PID, and let M be a f.g. R-module. Then

$$M \cong R^m \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_n)$$

for some integers m, n and non-unit ring elements $a_1, \ldots, a_n$ satisfying the divisibility relations $a_1 \mid a_2 \mid \cdots \mid a_n$. The elements a_i are called the **invariant factors** of M, and they are unique up to units.

Alternatively, using the Chinese Remainder Theorem, we can rewrite

$$M \cong R^m \oplus R/(p_1^{e_1}) \oplus R/(p_2^{e_2}) \oplus \cdots \oplus R/(p_n^{e_n})$$

Where the p_i are all (not necessarily distinct) irreducible elements of R. The set of elements $p_i^{e_i}$ that appears are called the **elementary divisors** of M, and again they are unique up to units.

Either one of the above forms can be called the **canonical form** of M, or sometimes informally will just be called its **isomorphism class**. (Maybe somebody might call it the "normal form" of M? I've never seen it, but it might happen.)

We will eventually prove parts of this theorem, but probably not all of it unless I let these notes really get away from me. What I want to focus on is a few immediate consequences, and then some applications to linear algebra.

Facts.

- Taking $R = \mathbb{Z}$, the classification theorem specializes to the classification of finitely-generated abelian groups.
- If M is a f.g. torsion-free module over a PID, then in fact M must be a free module. We will see this is not true for modules over non-PIDs.
- The power m appearing in the R^m term in the canonical form of M is often called the **rank** of M, even though M is not itself free. (There is a slightly more general context in which people will talk about the rank of a non-free module, which I will mention in future notes.) For example, in number theory there are things called "elliptic curves", which have the structure of a finitely-generated abelian group. Thus, the group is isomorphic to some finite torsion module plus some number of copies of $\mathbb{Z}$. The torsion is understood relatively well, but people still work very hard to understand the ranks of elliptic curves.

Applications to linear algebra

Recovering Jordan Normal Form: As we said earlier, a $\mathbb{C}[x]$-module is the same thing as a complex vector space V along with a chosen linear operator $T\colon V \to V$. In linear algebra, we often find ourselves in the position of starting with a vector space and some linear operator, and trying to deduce properties of the operator, so it seems natural that we could use the canonical form to get the information we're after.

So, suppose V is finite-dimensional. In Exercise 9 you showed that V is a f.g. torsion module, so the classification theorem tells us that we can break V up in terms of its elementary divisors:

$$(V, T) \cong \mathbb{C}[x]/(f_1^{e_1}) \oplus \mathbb{C}[x]/(f_2^{e_2}) \oplus \cdots \oplus \mathbb{C}[x]/(f_m^{e_m}),$$

with each f_i an irreducible polynomial. Here the linear transformation on the right is implicitly "multiplication by x". For now, let's consider the case when $n = 1$, so that $V \cong \mathbb{C}[x]/(f^e)$ for some irreducible polynomial f. Luckily, over $\mathbb{C}$, we know that $f = (x - \lambda)$ for some (suggestively named) real number $\lambda \in \mathbb{C}$.

Exercise 12.

(a) Prove that $\mathcal{B} = \{(x-\lambda)^{e-1}, (x-\lambda)^{e-2}, \ldots, 1\}$ is a vector space basis for $\mathbb{C}[x]/(f^e)$. (This space does not have a basis as a $\mathbb{C}[x]$-module, because no torsion module is free by Exercise 7.)

(b) As mentioned, "multiplication by x" is a $\mathbb{C}$-linear operator $x\cdot\colon \mathbb{C}[x]/(f^e) \to \mathbb{C}[x]/(f^e)$. In the case $e = 3$, write $x \cdot (x-\lambda)^i$ in terms of the basis $\mathcal{B}$ for $i = 2, 1, 0$.

(c) In the case $e = 3$, what is the matrix of $x\cdot$ with respect to the basis $\mathcal{B}$?

The above exercise extends to the case with more than one factor $\mathbb{C}[x]/(f_i^{e_i})$, and completely recovers the Jordan Normal Form!

Rational Canonical Form: We also get information by looking at the invariant factors of the decomposition. If we write

$$(V, T) \cong \mathbb{C}[x]/(g_1) \oplus \cdots \oplus \mathbb{C}[x]/(g_n)$$

where $g_1 \mid g_2 \mid \cdots \mid g_n$, then we can see that g_n is the minimal polynomial of T. As we saw in the ring theory notes, there is not much special about $\mathbb{C}[x]$ as far as polynomial rings go, any polynomial ring over a field k is a PID. For the same reasons that a $\mathbb{C}[x]$-module is a $\mathbb{C}$-vector space with a linear operator, a $k[x]$-module is a k-vector space with a linear operator, and we can get an analogous decomposition of a $k[x]$-module (V, T)

$$(V, T) \cong k[x]/(g_1) \oplus k[x]/(g_2) \oplus \cdots \oplus k[x]/(g_n).$$

Similar to the discussion above of how choosing a certain basis can give the Jordan Normal Form of the operator T, choosing a certain basis here gives us a new matrix form.

Exercise 13. Let g be a polynomial of degree d, and let $\mathcal{B} = \{1, x, x^2, \ldots, x^{d-1}\}$ be chosen as a k-vector space basis for $k[x]/(g)$. Denote the "multiplication by x" map as usual by $x\cdot\colon k[x]/(g) \to k[x]/(g)$.

(a) Suppose $g = x^3 + ax^2 + bx + c$, where $a, b, c \in k$. Write $x \cdot x^i$ in terms of the basis $\mathcal{B}$ for $i = 0, 1, 2$.

(b) In the same case as part (a), what is the matrix of $x\cdot$ with respect to the basis $\mathcal{B}$?

(c) In general, if $g = x^d + \sum_{i=0}^{d-1} a_i x^i$ is monic, what will the matrix of $x\cdot$ look like with respect to the basis $\mathcal{B}$?

The matrix form obtained above is called a **companion matrix** to the monic polynomial g. Using the decomposition coming from the classification theorem, we obtain that any linear operator T has some basis under which:

1. the matrix of T is a block diagonal matrix where
2. the blocks are the companion matrices associated to the invariant factors g_i.

This matrix form is called the **rational canonical form** (RCF) or **Frobenius normal form**. (I have never heard someone say the latter, but it's what Wikipedia calls it.)

The RCF is not my favorite matrix normal form, but it is beloved by Matrix Master Josh Mundinger. Here are some reasons you might like the RCF.

1. The RCF is defined over any field, not just over algebraically closed fields.
2. The RCF doesn't change if you change your field of definition from a smaller field to a larger one. So, for example the RCF of some $n \times n$ matrix with entries in $\mathbb{R}$ doesn't depend on whether you view that matrix as a linear operator $\mathbb{R}^n \to \mathbb{R}^n$ or as an operator $\mathbb{C}^n \to \mathbb{C}^n$. This is because the invariant factors will not change as you change your field, that's what makes them *invariant*.
3. As a corollary to the above, the existence of RCF implies that two matrices are similar over $\mathbb{R}$ iff

they are similar over $\mathbb{C}$ (and also the statement for any other field extension). At the very least, this is evidence of the power of module theory, because as we saw, this statement is not particularly easy to prove with linear algebra techniques alone.

I think Dummit and Foote give some completely contorted algorithm that ostensibly computes the RCF of any given matrix, but it seems useless as a matter of practice. My impression is that the RCF is sort of like Cramer's rule, where it's good to know about because it provides quick proofs of certain statements, but you would never employ it for any practical computation.

Cyclic modules and cyclic subspaces: Our construction of the RCF involved the fact that $k[x]/(g)$ is generated as a $k[x]$-module by the element $1 \in k[x]/(g)$, which we used in the form that $1, x \cdot 1, x^2 \cdot 1, \ldots, x^{d-1} \cdot 1$ form a k-basis for $k[x]/(g)$. In the case that $(V,T) \cong k[x]/(g)$, there must be some vector $v \in V$ that corresponds to the element $1 \in k[x]/(g)$, and this vector v has the two equivalent properties:

- v generates (V,T) as a $k[x]$-module.
- The set $\{v, Tv, T^2v, \ldots, T^{d-1}v\}$ forms a basis for V.

A module (over any ring R) is called **cyclic** if it is generated by a single element as an R-module. So, a cyclic group is just the same thing as a cyclic $\mathbb{Z}$-module. A cyclic $k[x]$-module is what we described above: a pair (V,T) so that there exists some vector v with the property that $\{v, Tv, \ldots\}$ span V. Given a k-vector space V and a linear operator T, a vector with this property is called a **cyclic vector** for T. A subspace $W \subset V$ is called a **cyclic subspace** for T if $(W, T|_W)$ is a cyclic submodule of (V,T). That is to say, (1) W is T-stable, in the sense that for all $w \in W$, $Tw \in W$, and (2) restricting T to W turns W into a cyclic $k[x]$-module.

Exercise 14. Let $V = \mathbb{C}^2$.

(a) Give an example of a linear operator T so that the $\mathbb{C}[x]$-module (V,T) is cyclic.

(b) Give an example of a T so that (V,T) is not cyclic.

Exercise 15. Let (V,T) be a $\mathbb{C}[x]$-module, and suppose $v \in V$ is a cyclic vector.

(a) Prove that if $f(T)v = 0$ for some polynomial $f \in \mathbb{C}[x]$, then $f(T)u = 0$ for all $u \in V$. (This can be done quickly with the classification theorem, but also can be done with pure linear algebra.)

(b) Conclude that if $f(T)v = 0$, then f is a multiple of the minimal polynomial of T.

In this new vocabulary, we can rephrase what the classification theorem/RCF are telling us in purely linear algebra terms. The classification theorem says that given any linear operator $T \colon V \to V$, we can write V as the direct sum of its T-cyclic subspaces. RCF says that on each cyclic subspace, choosing a cyclic vector makes the matrix of T into a companion matrix.

Here is one of the hardest qual problems in recent memory:

August 2022 Problem 5, edited Let V be a vector space over $\mathbb{C}$ of dimension $n \geq 2$. Let

$$A \colon V \to V$$

denote a $\mathbb{C}$-linear map with n mutually distinct eigenvalues. Prove that V contains one-dimensional subspaces $V_1, \dots, V_n$ such that

(i) we have

$$V = \sum_{i=1}^{n} V_i$$

and if $i \neq j$, $V_i \cap V_j = \{0\}$ (**NB** these two conditions are equivalent in this case);

(ii) $AV_i \subseteq V_i + V_{i+1}$ for $1 \leq i \leq n-1$ and $AV_n \subseteq V_n + V_1$; and

(iii) V_i is not an eigenspace of A for $1 \leq i \leq n$.

Exercise 16 goes through some basic rephrasing of the problem, and Exercise 17 walks you through a clever proof that Josh Mundinger told me. (This is not the only way to solve the problem, nor indeed the solution I came up with.)

Exercise 16.

(a) Convince yourself that the conditions in the problem are equivalent to saying: V has a basis $v_1, \dots, v_n$ such that

(i) $Av_i = a_{ii}v_i + a_{i+1,i}v_{i+1}$ for $1 \leq i \leq n-1$ and $Av_n = a_{nn}v_n + a_{1n}v_1$, for some set of $2n$ scalars a_{ij}.

(ii) The scalars $a_{i+1,i} \neq 0$ for $1 \leq i \leq n-1$, and $a_{1n} \neq 0$.

(b) Convince yourself that if you had such a basis, you could easily get a basis so that $a_{i+1,i} = 1$ for $1 \leq i \leq n-1$. (So, really, there are only $n+1$ scalars that we need to worry about.)

(c) Suppose $V = \mathbb{C}^4$, and suppose you already had found a basis $v_1, \dots, v_4$ as above. What would the matrix of A look like with respect to this basis.

Exercise 17.

(a) Suppose you had a basis of the form given in Exercise 16(b). Prove that v_1 would be a cyclic vector for A.

(b) Prove that the assumptions on A imply that the minimal polynomial of A equals the characteristic polynomial of A.

(c) Prove that A has a cyclic vector.

(d) Suppose for now that we have fixed the scalars $a_{11}, \dots, a_{nn}$ arbitrarily. Starting from a cyclic vector v_1, find vectors $v_2, \dots, v_n$ so that $Av_i = a_{ii}v_i + v_{i+1}$ for $1 \leq i \leq n-1$.

(e) For the vectors you constructed above, write Av_n in terms only of A, v_1, and the scalars $a_{11}, \dots, a_{nn}$.

(f) How should the scalars $a_{11}, \dots, a_{nn}$ be chosen so in order to ensure that $Av_n = a_{nn}v_n + a_{1n}v_1$ for some scalar a_{1n}?

(g) What makes this problem difficult? And conversely, what clues could have led you to thinking of a solution like this? (Those clues must exist, because Josh followed them.)

A theorem on $M_2(\mathbb{C})$-modules

Earlier, I claimed that every f.g. left $M_2(\mathbb{C})$-module was of the form $(\mathbb{C}^2)^n$ for some n, where the action of a matrix M was "coordinate-wise" on the n different 2×1 vectors. The exercise below walks you through how to prove this fact.

Exercise 18. Write $V = \mathbb{C}^2$, considered as a left $M_2(\mathbb{C})$-module via matrix-vector multiplication. Recall that previously you proved that V was a simple module, and that V is the only simple left $M_2(\mathbb{C})$-module up to isomorphism.

(a) Let M be any left $M_2(\mathbb{C})$-module. Prove that the image of any homomorphism $V \to M$ must either be 0 or a submodule of M isomorphic to V.

(b) Prove that if $N_1, N_2 \subset M$ are any two distinct submodules of M isomorphic to V, then either $N_1 = N_2$ or $N_1 \cap N_2 = \{0\}$.

(c) Prove that $M_2(\mathbb{C}) \cong V \oplus V$ as a left module over itself.

(d) Conclude that any free left module $M_2(\mathbb{C})^{\oplus k}$ is isomorphic to $V^{\oplus 2k}$. (Here, $A^{\oplus n}$ means the direct sum of n copies of A. I apologize that the notation is weird and somewhat awkward, it is standard in order to distinguish between taking direct sums and tensor powers.)

(e) Let M be any finitely-generated $M_2(\mathbb{C})$-module. Prove that M is the internal direct sum of finitely many submodules isomorphic to V.

Module Theory 2

Properties of general modules

In the previous set of notes, we focused on the very nice properties of modules over PIDs. In these notes, we will talk about more general modules, and how much their behavior can differ from that special case.

Free modules

We might hope that for a given free module F, the number of elements in any basis for F is the same. It turns out that for some noncommutitive rings this can actually fail, but luckily for commutative rings it is true. I am including the proof of this statement because it is one example of proving a theorem for modules over a ring by somehow reducing that theorem to a statement about vector spaces. This might seem like a tricky trick the first time you see it, but it's an important method in algebra generally.

Proposition. If R is a commutative ring, then any basis for the free module R^m has m elements.

Proof. We can prove this by reducing it to the situation of a vector space over a field. Suppose we have an isomorphism $R^m \xrightarrow{\sim} R^n$, and take any maximal ideal $\mathfrak{m} \in R$. (On the Ring Theory 1 sheet we asserted that every ring has a maximal ideal, and hinted at an argument involving Zorn's Lemma.) Then the isomorphism must take the submodule $\mathfrak{m}^m \subset R^m$ to $\mathfrak{m}^n \subset R^n$, so it passes to an isomorphism on the quotients $(R/\mathfrak{m})^m \xrightarrow{\sim} (R/\mathfrak{m})^n$. But now this is an isomorphism of vector spaces over the field $R/\mathfrak{m}$, so we know $m = n$.

This property is sometimes called the Invariant Basis Number (IBN) property. For rings with IBN, the number of elements in a basis for a free module is called the **rank** of the free module. For integral domains, the rank has another nice description: given a free module F, we can extend scalars to $K = \operatorname{Frac} R$ to get a vector space F_K. A basis for F remains a basis for F_K, so the rank of F is equal to the dimension of F_K.

Warning. In this context, the word "dimension" is reserved exclusively to refer to the dimension of a vector space over a field. The number of generators needed to generate a module unfortunately does not have a name of its own (at least, not one I'm aware of), but you should conceptually separate in your mind the generators of a module from any notion of dimension. Even in the case of free modules over integral domains, where the rank is equal to the dimension of some vector space, nevertheless it should be referred to as the "rank" and not the "dimension".

One reason for this is that a module might be both an R-module and *also* a vector space, for example if $R = \mathbb{C}[x]$, and the number of generators required as an R-module could be different from the dimension of the vector space. For example, $\mathbb{C}[x]/(x^4)$ is a 4-dimensional complex vector space, but it is spanned by the single element 1 as a $\mathbb{C}[x]$-module.

One might guess, based on our prior experience with vector spaces and modules over PIDs, that a submodule of a free module $G \subset R^n$ should be another free module, and that $\operatorname{rank} G \leq n$. Unfortunately, this is not necessarily the case.

Exercise 1. Let $R = \mathbb{C}[x, y]$, and consider R as a free module over itself of rank 1. Find a submodule of R that is not free.

The good news is that our proof above that any basis for a free module over a commutative ring has the same number of elements can be adapted to show that if $G \subset R^n$ happens to be a free module, then $\operatorname{rank} G \leq n$.

Any free module is torsion-free, and hence so is any submodule of a free module. However, your example in Exercise 1 shows that over a non-PID there can be torsion-free modules that are not free, even over a UFD, which seems like it would be the next best thing. The next exercise shows that, at the very least, for any finitely-generated torsion-free module, that module can be realized as a submodule of a free module.

Exercise 2. Let M be a finitely-generated torsion-free module over an integral domain R. We will prove that M can be injected into a free module of finite rank.

(a) Prove that M contains a finite-rank free module.

(b) Let $F \subset M$ be a maximal free submodule. Prove that M/F is torsion.

(c) Prove that there is some $r \in R$ so that $rM \subset F$. (Hint: look at Module Theory 1, Exercise 10.)

(d) Prove that the "multiplication by r" map above is injective, so this realizes the injection we wanted.

Exercise 2 gives us the power to prove that any f.g. torsion-free module over a PID is free. (I stated this as a consequence of the classification theorem, but I don't know of a proof of the classification theorem that doesn't use this fact.) All we need to do is show that a submodule of a f.g. free module is free. This is the goal of the next exercise. (In fact, this is true without the f.g. assumption, and if you're of a logical bent you are welcome to try to prove it.)

Exercise 3. Let R be a PID, and let R^n be a free module with basis $e_1, \ldots, e_n$, and $M \subset R^n$ a nonzero submodule. We will prove this by induction on n.

(a) Show the base case $n = 1$.

(b) Now suppose $n > 1$. Show that there is an i so that $M \cap \langle e_1, \ldots, e_i \rangle \neq \{0\}$.

(c) Take the smallest such i, and consider $p_i \colon M \cap \langle e_1, \ldots, e_i \rangle \to R$ which takes $r_1 e_1 + \cdots + r_i e_i \mapsto r_i$ (i.e. the projection onto the last coordinate). Prove that this map is injective. Conclude that $M \cap \langle e_1, \ldots, e_i \rangle$ is generated by a single element m_1.

(d) Prove that $M/\langle m_1 \rangle$ is isomorphic to a submodule of $R^n/\langle e_1, \ldots, e_i \rangle \cong R^{n-i}$. Induct to conclude that $M/\langle m_1 \rangle$ is free.

(e) Using a basis for $M/\langle m_1\rangle$, construct a basis for M.

Other important properties a module can have

A module is **simple**, or sometimes **irreducible**, if it has no nonzero proper submodules. Note that a ring is simple iff it is a simple module over itself. For a module M, the **annihilator** of M, $\operatorname{Ann}(M)$, is the set of all $r \in R$ that kill every $m \in M$. In symbols: $\operatorname{Ann}(M) = \{r \in R : \forall m \in M, rm = 0\}$.

Exercise 4. Prove that $\operatorname{Ann}(M)$ is an ideal of R.

Exercise 5.

(a) Prove that a module M is simple if and only if it is cyclic and any nonzero $m \in M$ is a generator.

(b) Prove that if M is simple, then $\operatorname{Ann}(M)$ is a maximal ideal.

(c) Prove that if M is simple, then $M \cong R/\operatorname{Ann}(M)$.

(d) Give an example of two non-isomorphic simple $\mathbb{Z}$-modules.

(e) Give an example of two non-isomorphic simple $\mathbb{R}[x]$-modules.

(f) **(Frequent Problem 1 example)** How many non-isomorphic simple (left) $M_2(\mathbb{C})$-modules can you think of? Prove that you found them all.

A module is **Noetherian** if its submodules satisfy the Ascending Chain Condition. As with "simple", a ring is Noetherian iff it is a simple module over itself.

Example 1.

- Similar to rings, a module is Noetherian iff all of its submodules are finitely generated.
- Any finitely-generated module over a Noetherian ring is also Noetherian.
- There can be Noetherian modules over non-Noetherian rings. The zero module is always Noetherian, as is any simple module, and we showed above that every ring has a simple module since every ring has a maximal ideal.
- There can be non-Noetherian modules over Noetherian rings. **Exercise 6.** Prove that $\mathbb{Q}$ is not a Noetherian $\mathbb{Z}$-module by demonstrating an infinite ascending sequence of submodules. (Incidentally, this proves that $\mathbb{Q}$ is not a finitely-generated $\mathbb{Z}$-module.)

The isomorphism theorems

Below, all modules are modules over a single ring R.

Theorem (first isomorphism theorem)**.** Let $\phi\colon M \to N$ be a homomorphism of R-modules. Then $\ker\phi$ is a submodule of M, $\operatorname{im}\phi$ is a submodule of N, and the induced homomorphism $M/\ker\phi \to \operatorname{im}\phi$ is an isomorphism. In particular, if ϕ is surjective, then $N \cong M/\ker\phi$.

Theorem (second isomorphism theorem)**.** Let M be an R-module, $N_1, N_2 \subset M$ submodules. Then $N_1 + N_2 = \{n_1 + n_2 : n_1 \in N_1,\ n_2 \in N_2\}$ is a submodule of M, N_2 is a submodule of $N_1 + N_2$,

$(N_1 + N_2)/N_2$ is a submodule of M/N_2, $N_1 \cap N_2$ is a submodule of N_1, and the composition $N_1 \hookrightarrow M \twoheadrightarrow M/N_2$ induces an isomorphism $N_1/(N_1 \cap N_2) \xrightarrow{\sim} (N_1 + N_2)/N_2$.

Theorem (third isomorphism theorem)**.** Let M be an R-module, N a submodule. Then if P is an submodule of M such that $N \subset P \subset M$, then P/N is a submodule of M/N, and the composition $M \twoheadrightarrow M/N \twoheadrightarrow (M/N)/(P/N)$ induces an isomorphism $M/P \xrightarrow{\sim} (M/N)/(P/N)$.

Theorem (lattice theorem)**.** The correspondence that takes a submodule $P \subset M$ containing N to the quotient P/N gives a bijection

$$\{\text{submodules of } M \text{ containing } N\} \xrightarrow{\sim} \{\text{submodules of } M/N.\}$$

Furthermore, this bijection preserves inclusions, sums, and intersections.

Exact sequences

It feels like about time to talk about exact sequences. If A, B, C are R-modules, a sequence of homomorphisms $A \xrightarrow{f} B \xrightarrow{g} C$ is called **exact** if $\operatorname{im} f = \ker g$. A longer sequence of homomorphisms

$$A_1 \to A_2 \to A_3 \to A_4 \to \dots$$

is called exact if every three-module sequence (two-arrow sequence) is exact.

Example 2.

- For every module A, there is only one homomorphism from the zero module to A. A sequence of the form $0 \to A \xrightarrow{f} B$ is exact if and only if $\ker f = 0$, that is, iff f is injective.
- For every module C, there is only homomorphism from C to the zero module. A sequence of the form $B \xrightarrow{g} C \to 0$ is exact if and only if $\operatorname{im} g = C$, that is, iff g is surjective.
- Combining the above, saying that $\phi\colon A \to A'$ is an isomorphism is the same as saying that the sequence $0 \to A \xrightarrow{\phi} A' \to 0$ is exact.
- A situation common enough to get its own name is the situation where you have an exact sequence that looks like
$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0.$$

  Such a sequence is called a **short exact sequence** (often abbreviated SES), not because it's the shortest possible exact sequence, but because in practice it's the shortest exact sequence that really matters. By the above, saying that this is a short exact sequence is the same as saying that

  - f is injective,
  - g is surjective, and
  - $\operatorname{im} f = \ker g$.

  You should think about a short exact sequence as encapsulating all the information of the first isomorphism theorem: the first isomorphism theorem can be rephrased as saying that for any short exact sequence as above, we have that $C \cong B/\operatorname{im} f$.

- We said earlier that a module is finitely generated iff it admits a surjective map from a finite-rank free module. We could recast this as saying a module is finitely generated iff there exists an exact sequence

  $$R^n \xrightarrow{f} M \to 0$$

  If R is a PID, then the kernel of this map is a submodule of R^n, and hence is free and of rank $k \leq n$, so we get a short exact sequence

  $$0 \to \ker f \cong R^k \to R^n \xrightarrow{f} M \to 0.$$

  One way to prove the classification theorem for modules over a PID is to prove that any map $R^k \xrightarrow{g} R^n$ can be written in a special form that reveals information about the quotient $R^n / \operatorname{im} g$. This is the theory of **Smith Normal Form**, which I love but will not discuss further in these notes.

- If R is not a PID, then the kernel of the surjection $R^n \to M \to 0$ need not be another free module, but it will be a finitely-generated module, so we can get another surjection $R^{n_2} \to \ker f \to 0$. Then we can look at the kernel of f_2, and get a surjection from a free module onto that, and so on. In the end, we get what is called a "free resolution" of the module M, which is an exact sequence:

  $$0 \leftarrow M \xleftarrow{f_1} R^{n_1} \xleftarrow{f_2} R^{n_2} \xleftarrow{f_3} R^{n_3} \leftarrow \ldots.$$

  (I sometimes write long sequences like this "backwards", which is a habit I learned from Daniel Erman, and which many of his students do as well. Among the benefits of writing it this way is that the indexing on the page goes in the correct order.)

The concept of exact sequences is more general than modules, you can also talk about exact sequences of groups, or pretty much any other algebraic thing. For example, there is a short exact sequence of groups

$$1 \to \{\pm 1, \pm i\} \to Q \to \mathbb{Z}/2\mathbb{Z} \to 1,$$

where here I'm writing 1 to mean the trivial group, and Q is the quaternion group $\{\pm 1, \pm i, \pm j, \pm k\}$.

You can find a nice visual metaphor for exact sequences in Ravi Vakil's lovely Puzzling Through Exact Sequences. It might be helpful to read, but don't feel bad if at some point you get lost because it does get quite advanced (eventually discussing spectral sequences).

Splitting short exact sequences

A short exact sequence

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is said to **split on the right** if there is a map $s \colon C \to B$ so that $g \circ s = \mathrm{id}_C$ (that is, for every $c \in C$, $g(s(c)) = c$). The map s is called a **(right) splitting homomorphism/map**, or often just a **splitting**. You will often see splittings notated in the following way:

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "f"] & B \arrow[r, "g"] & C \arrow[l, bend left, "s"] \arrow[r] & 0. \end{tikzcd}$$

Similarly, a SES is said to **split on the left** if there is a map $\sigma\colon B \to A$ so that $\sigma \circ g = \mathrm{id}_A$.

Exercise 7. Let

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

be a short exact sequence of R-modules.

(a) Show that the sequence splits on the right if and only if there is a submodule $C' \subset B$ so that $g\colon C' \to C$ is surjective and B is the internal direct sum of $f(A)$ and C'. (This is why the map s is called a "splitting". Its existence "splits" B up into the direct sum of two smaller things.)

(b) Show that the sequence splits on the left if and only if it splits on the right.

Thus, for modules, actually people will usually just say a sequence **splits** (or sometimes "is split" or sometimes "is split exact") if it splits on either side, since they are the same.. You might be confused about why I told you about splitting on the left versus right at all if I was just going to turn around and tell you they're the same. The reason is that they happen to be the same for modules, but they are not the same for groups (to be discussed in the notes on groups), and they may not be the same in some other algebraic context you encounter later in life. In general, splitting on the right is the less restrictive notion, and it's usually what people will default to when they talk about a SES "splitting".

Exercise 8.

(a) (Low ball) Give an example of an exact sequence of abelian groups ($\mathbb{Z}$-modules) that splits.

(b) Give an example of an exact sequence of abelian groups that does *not* split.

(c) Give an example of an exact sequence of $\mathbb{C}[x]$-modules that does not split.

Exact sequences that don't split are frequently asked-for examples in Problem 1s on the qual. See August 2021 Problem 1 (e) and August 2018 Problem 1 (a).

Exercise 9. If you don't remember much group theory, come back to this question later.

(a) Prove that the exact sequence of groups

$$1 \to \{\pm 1, \pm i\} \to Q \to \mathbb{Z}/2\mathbb{Z} \to 1$$

does not split on either the right or the left.

(b) Prove that the exact sequence of groups

$$1 \to \langle r \rangle \to D_5 \to \mathbb{Z}/2\mathbb{Z} \to 1$$

splits on the right but not on the left. (Here, D_5 is the dihedral group of order 10, presented as $\langle r,\, s \mid r^5 = s^2 = srsr = e\rangle$.)

Exercise 10. Let A, B be R-modules, and suppose there is a short exact sequence

$$0 \to A \xrightarrow{f} B \xrightarrow{g} R^n \to 0.$$

(a) Prove that this sequence splits. (Hint: show it splits on the right.)

(b) Suppose $R = k$ is a field, so that A, B are k-vector spaces. Show that part (a) is the same thing as the rank-nullity theorem. (In fact, your argument is also the argument used to prove the rank-nullity theorem, but saying "module" everywhere instead of "vector space".)

Exercise 11. Suppose you had a function λ that took in a module and output some integer. We say λ is **additive** if for every SES

$$0 \to A \to B \to C \to 0$$

we have $\lambda(B) = \lambda(A) + \lambda(C)$.

(a) Show that the function $\dim V$ is additive on finite-dimensional k-vector spaces.

(b) Let λ be an additive function, and let

$$0 \to M_0 \to M_1 \to \cdots \to M_n \to 0$$

be an exact sequence of modules. Show that we can do a sort of "inclusion-exclusion" on the sequence to get

$$\sum_{i=0}^{n} (-1)^i \lambda(M_i) = 0.$$

Why are they called "exact" sequences?

The terminology of exactness originally came from algebraic topology. In topology, you come across the situation in which you have a sequence of homomorphisms of abelian groups (a.k.a. $\mathbb{Z}$-modules)

$$0 \leftarrow C_0 \xleftarrow{\partial_1} C_1 \xleftarrow{\partial_2} C_2 \xleftarrow{\partial_3} \ldots$$

and for geometric reasons these maps satisfy $\partial_i \circ \partial_{i+1} = 0$. Another way of saying that is that $\operatorname{im} \partial_{i+1} \subset \ker \partial_i$. One thing that matters in this situation is when it is the case that $\operatorname{im} \partial_{i+1} = \ker \partial_i$, that is, you care about when the image of one is *exactly* the kernel of the next, not just some smaller thing inside the kernel. That is to say, you want to know if the above sequence is exact at i, and if it isn't you want to quantify in some way how not-exact it is.

Module Theory 3

Localizing modules

Just as we can localize rings, we can also localize modules. Starting with a ring R, an R-module M, and a multiplicatively closed subset $S \subset R$, we can form $S^{-1}M$, which will be an $S^{-1}R$-module. The process of doing so is pretty much the same as for forming the ring $S^{-1}R$ in the first place:

- The elements of $S^{-1}M$ are symbols of the form $\frac{m}{s}$, where $m \in M$ and $s \in S$.
- Two symbols $\frac{m_1}{s_1}, \frac{m_2}{s_2}$ are equal if their cross difference is S-torsion, that is, if there exists $s \in S$ so that $s(s_2 m_1 - s_1 m_2) = 0$.
- The addition in $S^{-1}M$ is defined by taking common denominators

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{s_2 m_1 + s_1 m_2}{s_1 s_2}.$$

- The $S^{-1}R$-module multiplication is probably what you expect: $\frac{r}{s_1} \cdot \frac{m}{s_2} = \frac{rm}{s_1 s_2}$.

Just like with rings, if S is the compliment of a prime ideal $\mathfrak{p}$, then the $R_\mathfrak{p}$-module $S^{-1}M$ is usually written as $M_\mathfrak{p}$. I am not sure that I've ever seen special notation for a module over a localization of the form $R_f = R[\frac{1}{f}]$.

Exercise 1. Consider the $\mathbb{Z}$-module $\mathbb{Z}/12\mathbb{Z}$.

(a) Let S be the set of powers of 2, so $S^{-1}\mathbb{Z} = \mathbb{Z}[\frac{1}{2}]$. What is the isomorphism class of the $\mathbb{Z}[\frac{1}{2}]$-module $S^{-1}\mathbb{Z}/12\mathbb{Z}$? (Hint: look back at Ring Theory 2, Exercise 10.)

(b) Now let S be the set of powers of 3. What is the isomorphism class of the $\mathbb{Z}[\frac{1}{3}]$-module $S^{-1}\mathbb{Z}/12\mathbb{Z}$?

(c) Show that if S contains no powers of 2 or 3, then $S^{-1}\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/12\mathbb{Z}$.

The module $S^{-1}M$ is *also* an R-module, because we have the localization homomorphism $R \to S^{-1}R$ (see Module Theory 1, Example 4). However, being an $S^{-1}R$ module is a more special property.

Exercise 2. Prove that an $S^{-1}R$-module is the same thing as an R-module M such that for every $s \in S$, the "multiplication by s" map $s\cdot\colon M \to M$ is an isomorphism.

The localization homomorphism and universal property

As with localizing rings, we get a homomorphism $\ell\colon M \to S^{-1}M$, also called the "localization homomorphism", by sending $m \mapsto \frac{m}{1}$. The homomorphism ℓ is a homomorphism *of R-modules*, so

$\ell(rm) = r\ell(m)$ for all $r \in R$.

Exercise 3. This exercise generalizes the statement that the localization homomorphism $R \to S^{-1}R$ is injective iff S contains no zero-divisors.

(a) Let M be an R-module. Take some $m \in M$, and let $\mathfrak{a} = \operatorname{Ann}(m)$. Prove that $\ell(m) = 0$ in $S^{-1}M$ if and only if $S \cap \mathfrak{a} \neq \emptyset$.

(b) Prove that ℓ is injective if and only if M has no s-torsion for any $s \in S$.

(c) Prove that $S^{-1}M$ has no s-torsion for any $s \in S$.

(d) Explain your answers to Exercise 1.

It is common to hear people paraphrase this by saying that localization *kills torsion.*

The localization homomorphism has a pretty similar universal property to the one for rings, but it's worth stating on its own.

Theorem. Let R be a ring, M an R-module, S a multiplicatively closed subset of R, and $\ell\colon M \to S^{-1}M$ be the localization homomorphism. Let N be an $S^{-1}R$-module, and recall that N is also an R-module. If $\phi\colon M \to N$ is an R-module homomorphism, then there exists a unique function $\widetilde{\phi}\colon S^{-1}M \to N$ such that $\phi = \widetilde{\phi} \circ \ell$. The map $\widetilde{\phi}$ will be a homomorphism of $S^{-1}R$-modules.

Concretely, the map $\widetilde{\phi}$ sends the fraction $\frac{m}{s} \mapsto \frac{\phi(m)}{s}$.

I mention this explicitly because, in addition to carrying all the interpretation of previous universal properties that we have seen, this universal property sort of connects two different worlds: the world of R-modules and the world of $S^{-1}R$-modules. This is illustrated in the diagrammatic interpretation below, where I have circled and labelled the $S^{-1}R$-modules.

$$\begin{tikzcd} M \arrow[d, "q"'] \arrow[dr, "\phi"] & \\ S^{-1}M \arrow[r, dashed, "\widetilde{\phi}"'] & N \end{tikzcd} \qquad S^{-1}R\text{-modules}$$

A restatement of the theorem is "for any R-module homomorphism ϕ from M to an $S^{-1}R$-module N, there exists a unique $S^{-1}R$-module homomorphism $\widetilde{\phi}$ making the above diagram commute". You will sometimes hear people say that a universal property like this is a kind of "approximation theorem". We already said that being an $S^{-1}R$-module is more restrictive than being an R-module, but we can think of $S^{-1}M$ as being the $S^{-1}R$-module that "best approximates" M.

Here's a comparison: consider the ceiling function $\lceil * \rceil\colon \mathbb{R} \to \mathbb{Z}$. It has the property that whenever n is an integer and $n \geq x$, then $n \geq \lceil x \rceil$. This shows that among {integers greater than x}, $\lceil x \rceil$ is the best approximation to the real number x. I could similarly make a diagram:

$$\begin{array}{ccccc} & & x & & \\ & \rotatebox{45}{$\leq$} & & \rotatebox{-45}{$\geq$} & \\ \lceil x \rceil & & \leq & & n \end{array}$$

Localization and exact sequences

I'm going to tell you the payoff of this section up front, to help motivate why the rest of the section exists. Suppose I want to understand the $\mathbb{C}[x,y]$-module $(\mathbb{C}[x,y]/(xy))_x$. To spell it out, first I take the ring $\mathbb{C}[x,y]$ and mod out by the ideal (xy), and then I take that ring and I localize by inverting the element x. What I'd really like to say is that I can switch the order of the localization and the quotient, because if I could I could compute

$$\begin{aligned}\left(\frac{\mathbb{C}[x,y]}{(xy)}\right)_x &\cong \frac{\mathbb{C}[x,y]_x}{(xy)_x}\\ &\cong \frac{\mathbb{C}[x,y]_x}{(y)_x}\\ &\cong \left(\frac{\mathbb{C}[x,y]}{(y)}\right)_x\\ &\cong \mathbb{C}[x]_x\\ &= \mathbb{C}[x,x^{-1}].\end{aligned}$$

Here, the second line follows from the first because in $\mathbb{C}[x,y]_x$, x is a unit, so the ideal generated by y is the same as the ideal generated by xy. We have swapped the order of localization and quotients twice, once in the first line, and once going from the second line to the third.

Hopefully you see why being able to do these manipulations is appealing. In this albeit contrived circumstance, it allows us to think about the ring $\mathbb{C}[x,x^{-1}]$, which seems much easier to understand than $(\mathbb{C}[x,y]/(xy))_x$. And I promise that in number theory, commutative algebra, and algebraic geometry, computations like this come up not that infrequently.

What exactly would it take to prove something like this? If you read the title of this section, you might have an inkling about what my answer is, but I do genuinely want you to pause to think about how you might justify the manipulations I was doing above. I think that I maybe could do it straight from the definitions, but it would be an absolute slog. Luckily, there is a better way, thought of by very clever people. Here is the first place that being conversant in exact sequences, and being willing to draw lots of diagrams, will really streamline our ability to do algebra.

Whenever we have a homomorphism of R-modules $f\colon A \to B$, we can compose f with the localization map $\ell_B\colon B \to S^{-1}B$. Concretely, the composition map $\ell_B \circ f\colon A \to S^{-1}B$ sends $a \mapsto \frac{f(a)}{1}$. Now, the composition will be a map from A to an $S^{-1}R$-module, so by the universal property there will exist an $S^{-1}R$-module homomorphism so that the following diagram commutes.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\ell_A"] & B \arrow[d, "\ell_B"] \\ S^{-1}A \arrow[r, dashed, "\widetilde{f}"] & S^{-1}B \end{tikzcd}$$

I said it way back in Ring Theory 2, so I'll remind you that a diagram **commutes** if composing the homomorphisms along each directed path gives the same thing in the end. In this diagram, that means that $\widetilde{f} \circ \ell_A = \ell_B \circ f$.

Now, if instead of just one homomorphism we had a sequence of homomorphisms

$$A \xrightarrow{f} B \xrightarrow{g} C$$

we would get an induced sequence of homomorphisms

$$S^{-1}A \xrightarrow{\tilde{f}} S^{-1}B \xrightarrow{\tilde{g}} S^{-1}C.$$

The important question here turns out to be "if the original sequence of homomorphisms was exact, is the induced sequence exact as well?".

Exercise 4. (localization preserves exact sequences) Prove that if the original sequence of homomorphisms was exact, the induced sequence is exact as well. (Hint: you need to prove two sets are equal. So, first prove one containment, then the reverse containment.)

This immediately gives us the justification we were looking for in the first place. If

$$0 \to A \to B \to C \to 0$$

is a SES of R-modules, then by Exercise 4

$$0 \to S^{-1}A \to S^{-1}B \to S^{-1}C \to 0$$

is a SES of $S^{-1}R$-modules. The first SES tells us that $C = B/A$, so $S^{-1}C = S^{-1}(B/A)$ is the order "quotient first, then localize". On the other hand, the second SES tells us that $S^{-1}C = S^{-1}B/S^{-1}A$, which is the order "localize first, then quotient".

Comment. I am not going to use any words pertaining to the theory developed by Eilenberg and Mac Lane in these notes, but I will note that people usually will say "localization is exact" rather than saying "localization preserves exact sequences".

Checking properties locally

One final important feature of localizing modules is that it gives us a way to check properties "locally", the way we might in topology. Here's a couple examples to get us started.

Exercise 5. (being the zero module is a local property) Prove that the following are equivalent:

(i) $M = 0$.

(ii) For every multiplicative subset $S \subset R$, $S^{-1}M = 0$.

(iii) For every prime ideal $\mathfrak{p} \subset R$, $M_{\mathfrak{p}} = 0$.

(iv) For every maximal ideal $\mathfrak{m} \subset R$, $M_{\mathfrak{m}} = 0$.

Exercise 6. Suppose $f\colon M \to N$ is an R-module homomorphism.

(a) Prove that f is surjective if and only if for all maximal ideals $\mathfrak{m} \subset R$, $\widetilde{f}\colon M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is surjective.

(b) Prove the same statement with the word "surjective" replaced by "injective".

Exercise 7. Choose either "injective" or "surjective". Give an example of a homomorphism between finitely generated $\mathbb{Z}$-modules $f\colon M \to N$ so that

(i) f is not whichever adjective you chose.

(ii) For all but one maximal ideal $\mathfrak{m} \subset \mathbb{Z}$, $\widetilde{f}\colon M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ *is* that adjective.

I know the first time I learned these facts, they bothered me because it seemed like they "reduced" an easy problem like checking if a homomorphism is injective/surjective to the hard problem of checking infinitely many homomorphisms of much more opaquely-defined modules to see if they are injective/surjective. The utility of statements like these is that what will usually happen is that most of the maps $\widetilde{f}$ will automatically have whatever property, so you will only need to check like one or two actual maximal ideals $\mathfrak{m}$. Furthermore, although I don't have enough space to convince you here, modules over local rings have lots of nice properties that modules over general rings do not, so it actually can be an improvement to trade one global computation for several local computations.

Tensor products and flat modules

We come now to the big doozy. I will give it my best shot, but I'm not sure I can do better than Keith Conrad.

Tensor products over commutative rings

I'm going to start with the commutative case, because it is easier to explain and more common. Given two R-modules M, N, we can form their **tensor product**, notated $M \otimes_R N$, which will again be an R-module.

- As a set, $M \otimes_R N$ consists of symbols of the form $m \otimes n$, where $m \in M$ and $n \in N$, as well as all finite R-linear combinations of these elements. These symbols are subject to the following two relationships.
- For all $m_1, m_2 \in M$, $n \in N$, we have $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$, and similarly on the right.
- For all $r \in R$, $(rm) \otimes n = m \otimes (rn) = r(m \otimes n)$.

The symbols $m \otimes n$ are called "simple tensors", and by definition they span the tensor product $M \otimes_R N$. As briefly mentioned in the Linear Algebra notes, the simple tensors can be thought of as representing some sort of "general product" between the elements $m \in M$ and $n \in N$, constrained only to behave like an R-bilinear map would.

Facts.

- Every simple tensor of the form $0 \otimes n$ or $m \otimes 0$ is equal to the zero element of the the tensor product $M \otimes_R N$.

- There is an isomorphism $M \otimes_R N \cong N \otimes_R M$ that just swaps all the simple tensors, so it doesn't really matter what order you take the tensor product in. (But probably don't swap on the fly in the middle of a paper or something.)

Example 1. Take the k-vector spaces $V = k^m$, $W = k^n$. Give V the basis e_i, and W the basis f_j. Then $V \otimes_k W \cong k^{mn}$, with a basis given by the simple tensors $e_i \otimes f_j$.

Example 2. Consider the tensor product $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z}$. We have already said that $0 \otimes n = 0 \otimes 0$ for all $n \in \mathbb{Z}/3\mathbb{Z}$. On the other hand, since $1 \equiv 3 \pmod 2$, we have $1 \otimes n = 3 \otimes n = 1 \otimes 3n = 1 \otimes 0 = 0 \otimes 0$ for all $n \in \mathbb{Z}/3\mathbb{Z}$. So, all the simple tensors of this tensor product are equal to 0, and hence $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$ as a module. This shows that the relation $(rm) \otimes n = m \otimes (rn)$ above can *really* collapse a tensor product down.

Comment. The subscript R on a tensor product can be important. This is because often one and the same abelian group can be considered as a module over different rings, and you need to declare which one you're thinking of so the tensor product knows which simple tensors to identify when it comes to the relation $(rm) \otimes n = m \otimes (rn)$. The smaller the ring, the fewer things will be identified, so the tensor product can be much larger. For example, we have said that every $\mathbb{C}[x]$-module is also a $\mathbb{C}$-vector space, but $\dim_{\mathbb{C}}\big(\mathbb{C}[x]/(x^2 + x + 1) \otimes_{\mathbb{C}[x]} \mathbb{C}[x]\big) = 2$, while $\dim_{\mathbb{C}}\big(\mathbb{C}[x]/(x^2 + x + 1) \otimes_{\mathbb{C}} \mathbb{C}[x]\big) = \infty$.

On the other hand, lots of times the ring in question is clear from context, so people will suppress the subscript.

Comment. There is a lot of confusion in the world over the correct scope of the word "tensor", because it gets used to mean "an element of any tensor product", or to mean "an element of a really specific kind of tensor product arising in physics". The physicists do not seem to know that's what they mean.

The universal property

Just like everything else, the tensor product also has a universal property, but it's a little bit of a different flavor from the other ones. As I've mentioned, tensor products were invented originally as a kind of "generalized bilinear product". This is true in the sense that given any homomorphism $f\colon M \otimes_R N \to C$, we can define an R-bilinear map $M \times N \to C$ by just sending (m, n) to $f(m \otimes n)$. Bilinearity comes for free because it's built into the definition. What the universal property says is that this relationship goes the other way as well.

Theorem. Let M, N, C be R-modules. If $\phi\colon M \times N \to C$ is an R-bilinear map, then there exists a unique R-module homomorphism $\widetilde{\phi}\colon M \otimes_R N \to C$ such that $\widetilde{\phi}(m \otimes n) = \phi(m, n)$.

Like all of our other universal properties, this gives us a way to define maps out of a tensor product, which otherwise might seem like a somewhat daunting task given that there can be so much weird collapsing that can happen inside a tensor product.

The next few exercises showcase some fundamental things you need to know about tensor products. If you are pretty unfamiliar with tensor products, you should take the results of Exercise 8 to Exercise 14 as axioms, and use them to attempt some problems like Exercise 15, or some qual problems. If you're more comfortable with tensor products, you should deepen your understanding by trying to prove each of these statements using the universal property of tensor products.

Exercise 8. Prove that $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$.

Exercise 9.

(a) Prove that for any R-module M, $M \otimes_R R = M$. If you think of the tensor product as like a multiplication on modules, the module R is like the "unit module" of this multiplication.

(b) Prove that for any multiplicative subset $S \subset R$, $S^{-1}R \otimes_R M \cong S^{-1}M$.

Exercise 10. Prove that $M \otimes_R (N_1 \oplus N_2) \cong (M \otimes_R N_1) \oplus (M \otimes_R N_2)$. This makes sense because multiplication distributes over addition.

Exercise 11.

(a) Suppose R is a subring of a larger ring R', so that R' is an R-module. Prove that $R' \otimes_R R[x] \cong R'[x]$.

(b) Consider carefully: what is $R[x] \otimes_R R[x]$?

Fact. A very important fact we will prove in a second, but which is somewhat hard to prove just from what we have seen so far, is that for any R-module M and ideal $I \subset R$, $M \otimes_R (R/I) \cong M/IM$.

Exercise 12. Using the above fact, prove that for any ring R, $R/I \otimes_R R/J \cong R/(I+J)$. Notice that this generalizes Exercise 8.

Exercise 13.

(a) You now have all the information you need to take the tensor product of any two finitely-generated $\mathbb{Z}$-modules. Pick some f.g. $\mathbb{Z}$-modules and tensor those puppies together!

(b) Pick an f.g. $\mathbb{Z}$-module A and tell me:

  (i) What is $A \otimes \mathbb{Z}[\frac{1}{2}]$?

  (ii) What is $A \otimes \mathbb{Z}_{(2)}$?

  (iii) What is $A \otimes \mathbb{Q}$?

Exercise 14. (from Vakil) Suppose R_1, R_2, R_3 are rings, and suppose there are ring homomorphisms $R_1 \to R_2$ and $R_1 \to R_3$, so that both R_2, R_3 are R_1-modules.

(a) Let M be an R-module. Prove that $R_2 \otimes_{R_1} M$ is an R_2-modules (in addition to an R_1-module).

(b) Prove that the tensor product $R_2 \otimes_{R_1} R_3$ is a ring.

Exercise 15. A classic example you should be familiar with is computing $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$. Since $\mathbb{C}$ is 2-dimensional as a real vector space, we know from the Example 1 above that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is a 4-dimensional real vector space. From Exercise 14, we know also that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is a ring. I only know of three rings that are also 4-dimensional real vector spaces:

- $M_2(\mathbb{R})$,
- $\mathbb{C} \oplus \mathbb{C}$,
- $\mathbb{H}$, Hamilton's algebra of quaternions.

Of these, only $\mathbb{C} \oplus \mathbb{C}$ is commutative, so it would be my guess that this is what we want.

(a) Show or recall that $\mathbb{C} \cong \mathbb{R}[x]/(x^2+1)$.

(b) Combine several of the above properties to prove that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$.

Comment. Perhaps you encountered this already with the above exercises, but often one of the most difficult parts of a problem involving tensor products is verifying that a given tensor is not 0. Although it is often pretty easy to spot when a tensor *is* 0, it is usually nigh impossible to argue directly from the definition of the tensor product that there is *no* sequence of valid manipulations that can convert a given tensor into the zero tensor. The only piece of technology at your disposal in such situations is the universal property: to prove that your tensor is not 0, use the universal property to construct a map where the image of that tensor is not 0. The first time you do this successfully you will feel more powerful than God.

Tensor products, homomorphisms, and flat modules

For now, fix some R-module N. If we have a homomorphism $f\colon M_1 \to M_2$, then we get an R-bilinear map $M_1 \times N \to M_2 \otimes_R N$ that sends $(m_1, n) \mapsto f(m_1) \otimes n$. Thus, the universal property tells us that we get a homomorphism $\widetilde{f}\colon M_1 \otimes_R N \to M_2 \otimes_R N$. Just like we talked about with localization, if we have a sequence of homomorphisms

$$A \to B \to C,$$

we get a sequence of homomorphisms

$$A \otimes_R N \to B \otimes_R N \to C \otimes_R N,$$

and if our original sequence is exact, we can ask if the sequence we obtain after tensoring is also exact. However, unlike for localization, this need not be the case, as the following classic counterexample demonstrates:

Example 3. Let $R = \mathbb{Z}$, $N = \mathbb{Z}/2\mathbb{Z}$, and consider the SES

$$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

Tensoring with N, we get the sequence

$$0 \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to 0.$$

Using our tensor product properties above we can rewrite the terms of this more nicely:

$$0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

However, if we examine the map on the left, we started with the muliplication by 2 map, so when we tensor with $\mathbb{Z}/2\mathbb{Z}$ it becomes the multiplication by 0 map! And that is definitely not injective, so the resulting sequence fails to be exact on the left side. (But notice, remains exact in the middle and on the right.)

You can cook up a similar example with $R = \mathbb{C}[x]$, $N = \mathbb{C}[x]/(x)$, and the multiplication by x map. Keep these sequences in mind, they are the best counterexamples to a lot of statements, and frequently appear in one form or another on the qual.

Thus, tensor products can fail to preserve exactness on the left term of a SES, which is the same as saying that if $A \to B$ is injective, it is not necessarily the case that $A \otimes_R N \to B \otimes_R N$ is also injective. On the other hand, it is *always* true that they preserve exactness in the middle and on the right. That is, for any homomorphism $f\colon A \to B$, we have that $(B/\operatorname{im} f) \otimes_R N \cong (B \otimes_R N)/(\operatorname{im} \widetilde{f})$, and in particular, if f is surjective then $\widetilde{f}$ is as well. To capture this, people say that **tensor product is right exact**.

Comment. Somehow this has failed to come up anywhere except in the preceding paragraph. If $f\colon A \to B$ is a homomorphism, the module $B/\operatorname{im} f$ is called the **cokernel** of f, $\operatorname{coker} f$. It is analogous to the kernel, in the sense that the kernel measures how far a map is from being injective, and the cokernel measures how far it is from being surjective. For any homomorphism, we thus get the following four-term exact sequence (the 0s don't count as terms):

$$0 \to \ker f \to A \xrightarrow{f} B \to \operatorname{coker} f \to 0.$$

I could rephrase what I said in the previous paragraph by saying "tensor product preserves cokernels", and people will actually say that! It's not even to be highfalutin, at some point in your life you realize that is the correct way to say it.

This gives us a quick proof of the fact that I mentioned earlier: $M \otimes_R R/I \cong M/IM$. (This fact is really invaluable!) We start with the SES

$$0 \to I \to R \to R/I \to 0,$$

and we tensor the sequence by M to get a right-exact sequence

$$M \otimes I \to M \otimes R \cong M \to M \otimes R/I \to 0.$$

Now, the map $M \otimes I \to M$ may not be injective, but just by looking at the definition of the map we get from the universal property of the tensor product, we can see that this map sends $m \otimes i \mapsto im$, so we know its image indeed is IM. Thus, exactness on the right tells us that

$$M \otimes R/I \cong M/(\operatorname{im} M \otimes I \to M) = M/IM.$$

Comment. It might seem like $M \otimes_R I$ should literally just be isomorphic to the submodule IM, and the map on the right just be the inclusion, which is injective. This seems supported by our proof that $M \otimes_R R \cong M$, because what we did there was rewrote every simple tensor $m \otimes r$ as $rm \otimes 1$, using the R-bilinearity. So, why can't we just rewrite $m \otimes i$ as $im \otimes 1$? The reason is that the simple tensors in $M \otimes_R I$ *absolutely must* have an element of I on the right side, and except in the trivial situation that $I = (1)$, 1 is not going to be an element of I, so $im \otimes 1$ is just an illegal expression to write down.

We will see in a moment that in a sense, the failure of $M \otimes_R I$ to be isomorphic to IM is the fundamental thing causing the tensor product with M to not be exact on the left.

The next exercise I took from a Michigan qual because I thought it was neat. It demonstrates the behavior I was talking about in the above comment, and ties together several other things we have discussed so far. I have broken it up into a lot of small parts, because there is a subtlety of the shape I mentioned before: you need to prove that the tensor $x \otimes y - y \otimes x$ is not 0, and doing so is not straightforward imo.

Exercise 16. (reworked from a Michigan qual) Let $R = \mathbb{C}[x, y]$, and let $\mathfrak{m} = (x, y)$. Note that $\mathfrak{m}$ is a maximal ideal of R, and that $R/\mathfrak{m} \cong \mathbb{C}$. Since $\mathfrak{m}$ is an ideal of R, it is a torsion-free R-module. The main goal of this problem will be to show that $\mathfrak{m} \otimes_R \mathfrak{m}$ is *not* torsion-free. (!)

(a) Verify that $x(x \otimes y) = x(y \otimes x)$, so that $x \otimes y - y \otimes x$ is x-torsion. (It is also y-torsion, for the same reasons.) So, if we can show that $x \otimes y - y \otimes x \neq 0$, we will have found a torsion element of $\mathfrak{m} \otimes \mathfrak{m}$.

(b) Consider the vector space $\mathbb{C}$ as an R-module where x and y both act by multiplication by 0. Show that the map $\mathfrak{m} \times \mathfrak{m} \to \mathbb{C}$ given by $(f, g) \mapsto f_x(0, 0)g_x(0, 0)$ is R-bilinear, and so gives a map $\mathfrak{m} \otimes \mathfrak{m} \to \mathbb{C}$. Here, f_x means the partial derivative of f with respect to x.

(c) Mentally verify that if I change f_x to f_y and/or g_x to g_y, I get another R-bilinear map.

(d) Prove that in $\mathfrak{m} \otimes_R \mathfrak{m}$, the elements $x \otimes y$ and $y \otimes x$ are both nonzero and not equal to each other. Thus, $x \otimes y - y \otimes x \neq 0$.

(e) There is a short exact sequence of R-modules

$$0 \to \mathfrak{m} \to R \to R/\mathfrak{m} \to 0.$$

Tensoring with $\mathfrak{m}$ gives the exact sequence

$$\mathfrak{m} \otimes_R \mathfrak{m} \to R \otimes_R \mathfrak{m} \cong \mathfrak{m} \to (R/\mathfrak{m}) \otimes_R \mathfrak{m} \to 0.$$

Prove that the map on the left is not injective.

The failure of tensor products to be exact on the left makes it a special occasion when you find a module N so that tensoring with that module *is* exact. A module N is called **flat** if whenever

$$0 \to A \to B \to C \to 0$$

is a SES, the tensor product sequence

$$0 \to A \otimes N \to B \otimes N \to C \otimes N \to 0$$

is also exact. As noted above, we already know that the tensor product sequence will be exact in the middle and on the right, so really the content is that N is flat if tensoring with N *preserves injections.*

Comment. It is also the case that if C is flat, and $0 \to A \to B \to C \to 0$ is an SES, then for any module M,

$$0 \to A \otimes_R M \to B \otimes_R M \to C \otimes_R M \to 0$$

is still exact. This was pretty surprising to me when I learned it, it seems like somehow the flatness of C is able to have a mysterious long-range effect on the injectivity of $A \otimes_R M \to B \otimes_R M$. Proving this fact involves some amount of homological algebra, and I don't want to get into that in these notes.

Example 4. We have already met some examples of flat modules.

- R is always a flat module over itself, because if we tensor with R we just get our original modules and our original homomorphism, so of course it preserves injections.
- As a consequence, any free module is flat, since R is flat and tensor products distribute over direct sums.
- For any multiplicative subset $S \subset R$, you showed that $S^{-1}R \otimes_R M \cong S^{-1}M$. Thus, since we already showed that localization is exact, tensoring with $S^{-1}R$ is exact, and so $S^{-1}R$ is a flat R-module.
- As a particular case of the previous point, $\mathbb{Q}$ is a flat $\mathbb{Z}$-module, since $\mathbb{Q} = \mathbb{Z}_{(0)}$. Keep this example in mind! We have already shown that $\mathbb{Q}$ is not a finitely-generated $\mathbb{Z}$-module in a previous set of notes, and having a non-finitely-generated flat module on hand is helpful for counterexamples, e.g. coming soon in the section on projective modules.

The definition of flatness seems like it would be impossible to check, because in general it seems like we would have to somehow predict every possible injective homomorphism of modules and verify that all of them stay injective. It would be a pretty sorry state if that was the case, and the following fact tells us that really we only need to check certain homomorphisms that we are already well-acquainted with.

Fact. A module M is flat if and only if for every ideal $I \subset R$, tensoring the injection $I \to R$ by M remains an injection. Equivalently, M is flat if and only if for every ideal I, the map $M \otimes_R I \to IM$ which sends $m \otimes i \mapsto im$ is an isomorphism. (It is automatically a surjection, so it is enough to check that it is injective.)

Exercise 17. Let R be a ring, and M an R-module.

(a) Prove that if M is flat, then M is torsion-free.

(b) Suppose R is a PID. Use the Fact above to prove that if M is torsion-free, then M is flat.

(c) What are the finitely-generated flat $\mathbb{Z}$-modules?

The above exercise, and others that have appeared in this section, suggest that the failure of a module to be flat in general has something to do with torsion. In the case of PIDs, you have shown that the torsion determines whether or not the module is flat, but as we saw in Exercise 16, over a non-PID you can have torsion-free modules that fail to be flat. There is a story to be told about trying to measure "how much a module fails to be flat", which is closely related to torsion. I will say no more about it other than to say that this is the reason why Tor is called Tor, if that means something to you.

Flat modules don't appear very often on the qual, which I personally think is a pity, but they did appear in January of this year, in the following problem. I have not yet defined what an R-algebra is, but we've already met them in Exercise 14: an R-algebra is an R-module that also happens to be a commutative ring, so that the ring multiplication commutes with scalar multiplication.

Exercise 18. (January 2024 Problem 4, notation altered) Let R be a commutative ring, and let A and B be commutative R-algebras.

(a) Prove that if B is flat over R, and f is a non zero divisor in A, then $f \otimes 1$ is a non zero divisor in $A \otimes_R B$.

(b) Prove that the flatness condition above is necessary by giving an example where part (a) fails if B is not flat over R.

Comment. I could alternatively have said, a commutative R-algebra A is a commutative ring A with a ring homomorphism $R \to A$, so that A can also be thought of as an R-module. This definition needs to be slightly modified if you want to allow A to be noncommutative, which you should because $M_2(\mathbb{C})$ should be a $\mathbb{C}$-algebra.

Why are they called "flat" modules?

Great question, I'd like to know as well. So would Serre, and he coined the term! (Unless it was maybe Eilenberg or Cartan, but he was the first one to use it in a publication.) If I had to guess, I'd hazard that the motivation might have had to do with the fact that flat modules are torsion-free, so perhaps the phrase "flat" was supposed to capture some idea of being "non-twisted". That's pure speculation, but this much is clear: Serre isolated flatness in a purely algebraic context as the right condition to preserve certain properties he was interested in. According to Brian Conrad on that mathoverflow post, Serre credits Grothendieck with the insight that, in the context of algebraic geometry, flatness is the right condition to ensure that a family of geometric objects varies "continuously" in the right way.

Daniel Erman's answer to that same mathoverflow post discusses this a little more. In that answer, he gives the example (which I'm rephrasing so I don't have to explain what Spec means) of the ring map $k[t] \to k[x, y, t]/(xy - t)$ which sends $t \mapsto t$. Basically, you can understand this ring map geometrically by graphing $xy = t$ on Desmos and watching what happens when you hit play on the t parameter. You can see that the shape of the graph varies nicely continuously, with only a little weirdness at $t = 0$, where the two branches of the hyperbola meet to become the union of the two coordinate axes. But visually, that seems like what they should do. On the other hand, his second example involves the non-flat map $k[t] \to k[x, y, t](t(xy - 1))$. If you were to graph $txy = t$ on Desmos, whenever $t \neq 0$ you would just get the hyperbola $xy = 1$, but when $t = 0$ you would suddenly get the vacuous equation $0 = 0$, whose solutions are all the points of the plane. This kind of bizarre behavior is the sort of thing flatness prevents.

If you're interested, this old course webpage has links to the original papers in which Serre uses the term "flat" (French "plat") in both the original French and English translations.

Tensor products over noncommutative rings

I have avoided talking about tensor products over noncommutative rings, because I didn't want to clog up the exposition fussing about left versus right modules, but it is important to know that you can take tensor products over noncommutative rings as well. Here are the key things that can be different for noncommutative rings:

1. In order to form the tensor product $M \otimes_R N$, M must be a right R-module and N must be a left R-module.
2. The relation $(rm) \otimes n = m \otimes (rn) = r(m \otimes n)$ for commutative rings is replaced by the relation $mr \otimes n = m \otimes rn$.
3. Because we can no longer "pull the r out", the resulting tensor product will not necessarily be an R-module, only an abelian group. It will only be an R-module if one of M or N also has an additional R-module structure on the other side. A module over a noncommutative ring that has a structure as both a left and a right R-module is called an **R-bimodule**.
4. The universal property will no longer reference R-bilinear maps, but rather will reference what are called R-balanced maps. An **R-balanced map** $\phi\colon M \times N \to P$ is a function that is
   - additive in each component, i.e. $\phi(m_1 + m_2, n) = \phi(m_1, n) + \phi(m_2, n)$, and similarly on the right; and,
   - allows elements of R to "pass across the arguments": $\phi(mr, n) = \phi(m, rn)$.

The most important tensor product properties are still true for tensor products over noncommutative rings. In particular:

- $M \otimes_R (N_1 \oplus N_2) \cong (M \otimes_R N_1) \oplus (M \otimes_R N_2)$, and similarly if the direct sum is on the other side.
- If I is a left ideal of R, then we can form the quotient R/I as an abelian group (but not as a ring), and it will be a left R-module. In this case, we have that $M \otimes_R R/I \cong M/MI$. There is a similar property for right ideals.

Exercise 19. (January 2021 Problem 2, modified) Let $R = M_2(\mathbb{C})$, the non-commutative ring of 2×2 matrices over $\mathbb{C}$.

(a) Give examples of a simple left R-module M and a simple right R-module N.

(b) Can you find another simple left R-module M' that is not isomorphic to M? (Either find one or explain why there is none such.)

(c) Compute $\dim_{\mathbb{C}}(N \otimes_R M)$. (**NB** You can do this by working directly with the tensors.)

Projective modules, and maybe a hair about injective modules

We return to a bit of possibly more familiar territory, to discuss a few more properties of the modules $\mathrm{Hom}_R(M, N)$.

Our previous experience with localizations and tensor products suggests that it would be interesting to find out how the modules $\mathrm{Hom}_R(M, N)$ interact with sequences. For now, we fix a module M. If we have a homomorphism $\phi\colon X \to Y$, we get a map $\phi_*\colon \mathrm{Hom}(M, X) \to \mathrm{Hom}(M, Y)$ by sending $f \mapsto \phi \circ f$. (You may hear this being called a "pushforward".) I think it's helpful to have a picture in your head of this:

$$\begin{tikzcd} & M \arrow[dl, "f"'] \arrow[dr, dashed, "\phi_*(f):=\phi\circ f"] & \\ X \arrow[rr, red, "\phi"] & & Y \end{tikzcd}$$

What I'm saying above is that given any red arrow ϕ, we can transform any $f\colon M \to X$ into a function $\phi_*(f)\colon M \to Y$ by "sliding the tip of f along ϕ". It sort of resembles the definition of vector addition. Now, given a SES

$$0 \to A \to B \to C \to 0$$

we get a sequence

$$0 \to \mathrm{Hom}_R(M, A) \to \mathrm{Hom}_R(M, B) \to \mathrm{Hom}_R(M, C) \to 0.$$

Is this sequence exact?

No, and we can turn to our old counterexample

$$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

If we apply $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$ to each term of the sequence we get the sequence

$$0 \to 0 \to 0 \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

This new sequence is not exact because the it's not exact at the $\mathbb{Z}/2\mathbb{Z}$ term on the right, since the map $0 \to \mathbb{Z}/2\mathbb{Z}$ is not surjective.

Exercise 20. We can modify this sequence a bit to make it a little less trivial-looking. Consider the sequence

$$0 \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

(a) Apply $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$ to each of the terms of this sequence, and write down the sequence you get at the end.

(b) Confirm that this is again an example where the resulting sequence is not exact at the right term, but is exact in the middle and on the left.

This behavior holds in general: it is always true that taking $\mathrm{Hom}_R(M, -)$ preserves exactness in the middle and on the left, but it may not preserve exactness on the right, i.e. may not preserve surjections. We say that **Hom is left exact**. Saying that the sequence stays exact on the middle and left is the same as saying that for any homomorphism $\phi\colon A \to B$, we have that $\mathrm{Hom}_R(M, \ker\phi) = \ker\phi_*$.

Just like with tensor products, the fact that taking Hom can fail to be exact makes it interesting whenever we find a module M so that taking Hom *is* exact. A module M is called **projective** if whenever $0 \to A \to B \to C \to 0$ is exact, so too is

$$0 \to \mathrm{Hom}_R(M, A) \to \mathrm{Hom}_R(M, B) \to \mathrm{Hom}_R(M, C) \to 0.$$

Exercise 21. Prove that a free module is projective. That is, let F be free, and let $\phi\colon B \to C$ be surjective, and prove that $\phi_*\colon \mathrm{Hom}_R(F, B) \to \mathrm{Hom}_R(F, C)$ is also surjective. (You may assume that F is finite rank, but it is not necessary.)

It turns out we can be a lot more specific with characterizing projective modules than we can with flat modules.

Exercise 22. Prove that the following are equivalent.

(i) The module P is projective.

(ii) P has the following **lifting property**: whenever $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ is a SES, and $\phi\colon P \to C$ is any map, there exists a map $\widetilde{\phi}\colon P \to B$ so that $\phi = g \circ \widetilde{\phi}$. In a diagram:

$$\begin{tikzcd} & & & P \arrow[dl, dashed, "\widetilde{\phi}"'] \arrow[d, "\phi"] & \\ 0 \arrow[r] & A \arrow[r, "f"] & B \arrow[r, "g"] & C \arrow[r] & 0 \end{tikzcd}$$

(iii) Any exact sequence $0 \to A \to B \to P \to 0$ splits.

(iv) The module P is a direct summand of a free module.

Hints:

(i) $\Rightarrow$ (ii) Apply $\operatorname{Hom}_R(P, -)$ to the sequence. Notice that we know that $\mathrm{id}_P \in \operatorname{Hom}_R(P, P)$.

(iii) $\Rightarrow$ (iv) You may use that any module is a quotient of a free module. (We already discussed this in the case of f.g. modules.)

(iv) $\Rightarrow$ (i) If F is a free module and $F = P \oplus Q$, show that any homomorphism $P \to C$ can be extended to a homomorphism $F \to C$. Then, use that free modules are projective.

Example 5.

- If R is a PID, then any submodule of a free module is free (we proved this in the f.g. case in Module Theory 2, but it is true even for infinitely-generated free modules). As a consequence, a module over a PID is projective iff it is free.
- Consider $\mathbb{Z}/2\mathbb{Z}$ as a $\mathbb{Z}/6\mathbb{Z}$-module. By CRT, we have that $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, so $\mathbb{Z}/2\mathbb{Z}$ is a direct summand of a free module, hence is projective. However, any free f.g. $\mathbb{Z}/6\mathbb{Z}$-module has order a power of 6, so $\mathbb{Z}/2\mathbb{Z}$ cannot be a free module.
- For a noncommutative example, consider the ring $M_2(\mathbb{C})$ and the simple left module $V = \mathbb{C}^2$. We showed at the beginning of this document (Modules 3) that $M_2(\mathbb{C}) \cong V \oplus V$, so V is a projective module, but similar to the previous example, V cannot be free because any free module has dimension divisible by 4 as a complex vector space.
- A long time ago we talked about the ring $R = \mathbb{Z}[\sqrt{-5}]$, which had the curious property that 6 has two factorizations into irreducibles: $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. It turns out that the ideal $I = (2, 1 + \sqrt{-5}) \subset R$ is not a principal ideal, and therefore it is not a free R-module. However, it *is* a projective R-module. In fact,

$$v_1 = \begin{pmatrix} 2 \\ 1 - \sqrt{-5} \end{pmatrix}, \; v_2 = \begin{pmatrix} 1 + \sqrt{-5} \\ 2 \end{pmatrix}$$

  forms a basis for $I \oplus I$, so $I \oplus I \cong R^{\oplus 2}$.

Exercise 23. (January 2022 Problem 3 (b)) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$-module.

Exercise 24. (January 2021 Problem 3) Two questions about projective modules.

(a) Show that a projective module over an integral domain is torsion free.

(b) Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:

(i) If M_1 and M_2 are projective, then so is M_3.

(ii) If M_2 and M_3 are projective, then so is M_1.

Exercise 25. (Projective modules are flat)

(a) Prove that a projective module is flat.

(b) Prove that $\mathbb{Q}$ is not a projective $\mathbb{Z}$-module, so we need not have an implication in the other direction.

A fun fact that I didn't learn until recently is that for f.g. modules over *any* ring, M is flat iff M is projective. The way you would prove this is by showing that both conditions are equivalent to M being **locally free**, which means that for every maximal ideal $\mathfrak{m} \subset R$, $M_{\mathfrak{m}}$ is a free $R_{\mathfrak{m}}$-module.

Homs the other way

We have talked a lot about taking Homs like $\mathrm{Hom}_R(M, -)$, but you could equally well talk about taking Homs like $\mathrm{Hom}_R(-, M)$. This presents an interesting new behavior: given a homomorphism $\phi\colon X \to Y$, we get a homomorphism going *the other direction* $\phi^*\colon \mathrm{Hom}_R(Y, M) \to \mathrm{Hom}_R(X, M)$ (also called a "pullback"). In diagram form:

$$\begin{tikzcd} X \arrow[rr, "\phi"] \arrow[dr, dashed, "\phi^*(f) := f \circ \phi"'] & & Y \arrow[dl, "f"] \\ & M & \end{tikzcd}$$

This time, we slide the tail of $f\colon Y \to M$ backwards along the red arrow ϕ to get a function $\phi^*(f)\colon X \to M$. Notice again the similarity to vector addition. To distinguish between the two kinds of Hom, people will call $\mathrm{Hom}_R(M, -)$ **covariant Hom** (because the homomorphisms go the same way), and $\mathrm{Hom}_R(-, M)$ **contravariant Hom** (because the homomorphisms swap).

Like before, one can ask what happens to exact sequences

$$0 \to A \to B \to C \to 0.$$

They flip around, and it turns out they don't remain exact, but

$$\mathrm{Hom}_R(C, M) \to \mathrm{Hom}_R(B, M) \to \mathrm{Hom}_R(A, M) \to 0$$

will still be exact. It's a little bit of a puzzle to decide what sidedness of exactness to call this, but the common consensus is that we should call contravariant Hom **left exact**, just like covariant Hom. (It's the left side of the original diagram that it preserves, it just mirror images it along the way.)

Just like with covariant Hom, there's a special name for the modules that make this exact. They are called **injective modules**. They're a lot less nice that projective modules, but there are good reasons to learn about them that I won't get into here. I will simply note that the $\mathbb{Z}$-modules $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are both injective, and in general injective modules have a feeling of having "highly dividable elements".

Algebras over a ring

I just wanted to mention briefly the definition of an "algebra" over a commutative ring R. We've already discussed that an R**-algebra** is an R-module A that is also a ring (not necessarily commutative), with the property that scalar multiplication by elements of R commutes with the ring multiplication in A. We can also define an R-algebra as a (not necessarily commutative) ring A together with a ring homomorphism $\phi\colon R \to A$, sometimes called the **structure homomorphism**. In order to make sure that scalar multiplication commutes with ring multiplication, i.e. $a_1(\phi(r)a_2) = \phi(r)a_1a_2$, we need for the image of ϕ to consist entirely of elements that commute with every $a \in A$. The subset of A consisting of elements that commute with every other element is called the **center** of A, and is denoted $Z(A)$.

An R**-algebra homomorphism** between two R-algebras A, B is a ring homomorphism $A \to B$ that is also an R-module homomorphism. Equivalently, if A, B have structure homomorphisms ϕ_A, ϕ_B, an R-algebra homomorphism between them is a ring map $f\colon A \to B$ so that $\phi_B = f \circ \phi_A$. That is, so that the following diagram commutes.

$$\begin{tikzcd} A \arrow[r, "f"] & B \\ R \arrow[u, "\phi_A"] \arrow[ur, "\phi_B"'] & \end{tikzcd}$$

You already know several examples of algebras.

Example 6.

- Given a commutative ring R, any quotient ring is an R-algebra.
- Any polynomial ring $R[x_1, \ldots, x_n]$ is an R-algebra, as is any quotient of a polynomial ring.
- The ring of $n \times n$ matrices with entries in R, $M_n(R)$, is an R-algebra.
- More generally, for any R-module M, $\mathrm{End}_R(M)$ is an R-algebra.
- Given a field K, any field extension L/K is a K-algebra.
- The field of real numbers $\mathbb{R}$ has a very interesting 4-dimensional algebra, Hamilton's algebra of quaternions $\mathbb{H}$. There is a more general notion of something called a "quaternion algebra" over any field K, inspired by Hamilton's quaternions.

Comment. A minor warning, some people in the world will use the word "algebra" to mean something more general than what it means here. For example, some people want to let the octonions $\mathbb{O}$ be an $\mathbb{R}$-algebra, and would prefer to call the algebras we've defined "associative algebras".

Group Theory 1

Examples

Here are some groups that you should be familiar with. If you want more examples than you could ever know what to do with, check out this page.

1. Finite abelian groups.
2. The dihedral group D_n, which is the symmetries of a regular n-gon.
3. The symmetric group S_n, which is the group of permutations of a set with n elements.
4. The alternating group A_n, which is the group of even permutations.
5. The quaternion group $Q = \{\pm 1, \pm i, \pm j, \pm k\}$ subject to the usual laws of quaternion multiplication $i^2 = j^2 = k^2 = ijk = -1$ and $ab = -ba$.
6. The linear groups $\mathrm{GL}_n(R)$, invertible $n \times n$ matrices with entries in R, and $\mathrm{SL}_n(R)$, determinant 1 matrices.
7. The free group F_n on n generators or "letters", which is the group of all "words" (i.e. arbitrary strings) in the generators and their inverses, g_i, g_i^{-1}, subject only to the obvious relation $g_i g_i^{-1} = g_i^{-1} g_i = e$.
8. Given any group G, we can form another group consisting of all the isomorphisms $G \xrightarrow{\sim} G$. This is called the **automorphism group of** G, denoted $\mathrm{Aut}(G)$. For example, $\mathrm{Aut}(\mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})^\times$, and for a free module R^n over a commutative ring R, $\mathrm{Aut}(R^n) = \mathrm{GL}_n(R)$.

Comment.

- It is common to write the group operation on an arbitrary abelian group as addition, but the group operation on any other arbitrary group as multiplication. There are two reasons why abelian groups get to be different. First, most abelian groups actually do come to us as additive groups of rings. Second, the automorphisms of a group like $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{Z}^2$ are naturally written as multiplication, so if we wrote these groups as multiplication we would be left trying to write their automorphisms as exponentiation, which is somewhat awkward. However, there are times when an abelian group is genuinely better written multiplicatively, e.g. the group of nth roots of unity in $\mathbb{C}$ naturally has group operation multiplication.

  For these reasons, I will sometimes distinguish between $\mathbb{Z}/n\mathbb{Z}$ and C_n, the latter being the cyclic group of order n but with group operation written multiplicatively. So, for example, I'd prefer to say "Q admits a surjective homomorphism to C_2" rather than "to $\mathbb{Z}/2\mathbb{Z}$".

- It is standard to denote the identity element of an arbitrary multiplicatively-written group as either e or 1, or e_G or 1_G if the group needs to be specified. In the context of writing exact sequences involving groups, I use 1 to mean the trivial group.
- It is standard to denote a subgroup of a group by using the $<$ symbol rather than $\subset$. For example, "let $H < G$ be a subgroup".
- Some people denote the dihedral groups as D_{2n}, so they write e.g. D_8 to mean what I will call D_4. In this stackexchange post Keith Conrad claims that this is the preferred convention among group theorists (and is the convention in Dummit and Foote).
- The linear groups GL and SL are usually defined over a field, on past quals usually $\mathbb{C}, \mathbb{R}$, or $\mathbb{F}_p$, but you can define these for any commutative ring R, and there has been a question about $\mathrm{SL}_n(\mathbb{Z})$ before (albeit not a very hard one). Note that if R is not a field, then $\mathrm{GL}_n(R)$ consists of invertible $n \times n$ matrices with coefficients in R whose inverse *also* has coefficients in R.
- Topologists and group theorists will sometimes write $\mathbb{F}_n$ for the free group on n letters.

Reminder about normal subgroups

For groups you have to be careful because you can't take a quotient by any subgroup, but rather only by **normal subgroups**. A subgroup $N < G$ is normal if for any $g \in G$, $gNg^{-1} = N$, or equivalently $gN = Ng$, or equivalently for all $n \in N$ there exists $n^g \in N$ so that $gng^{-1} = n^g$. (This condition is required so that coset multiplication works the way you want it to: $(g_1N)(g_2N) = g_1g_2N$.) Normal subgroups are denoted with a triangle: $N \triangleleft G$.

Exercise 1. Let $\phi\colon G \to H$ be a homomorphism. Prove that $\ker\phi$ is a normal subgroup of G.

Exercise 2. (January 2024 Problem 1 (c)) Give an example of a group G and a subgroup H, and a subgroup K of H, such that K is normal in H, and H is normal in G, but K is not normal in G.

The next problem is the first problem I could actually solve when I was studying for the qual.

Exercise 3. (August 2018 Problem 2) For a finite group G, denote by $s(G)$ the number of subgroups of G.

(a) Show that $s(G)$ is finite.

(b) Show that if H is a nontrivial normal subgroup of G, then $s(G/H) < s(G)$.

(c) Show that $s(G) = 2$ if and only if G is cyclic of prime order.

(d) Show that $s(G) = 3$ if and only if G is cyclic of order p^2 for a prime p.

Conjugation and the center

Given an element $h \in G$, an element of the form ghg^{-1} is said to be **conjugate** to h. The set of all elements conjugate to h is called the **conjugacy class** of h. In these notes, I am going to denote the conjugacy class of h by $[h]$. Both Dummit and Foote as well as the book I learned group theory from, *Groups and Symmetry* by Armstrong, avoid giving a notation for the conjugacy class of h. Wikipedia gives $\mathrm{Cl}(h)$, which is sensible, but I don't like it because it collides with the notation for the class group of a number field.

Exercise 4. Prove that a subgroup $H < G$ is normal iff it is a union of conjugacy classes.

The **center** of G, denoted $Z(G)$, is the set of elements $z \in G$ that commute with every other $g \in G$. Equivalently, the center consists of all $z \in G$ so that $[z] = \{z\}$.

Exercise 5. Prove that $Z(G)$ is a normal subgroup of G.

Exercise 6. (January 2018 Problem 4, modified) Let G be a finite group. Denote by $\mathrm{Aut}(G)$ the group of automorphisms of G, and by $Z(G) \subset G$ the center of G.

(a) Let $\mathrm{Inn}(G) \subset \mathrm{Aut}(G)$ be the subgroup consisting of automorphisms coming from conjugation, i.e. automorphisms of the form $x \mapsto gxg^{-1}$ for some $g \in G$. Prove that $\mathrm{Inn}(G)$ is isomorphic to $G/Z(G)$.

(b) Show that if $G/Z(G)$ is cyclic, then G is abelian.

(c) Show that if $\mathrm{Aut}(G)$ is cyclic, then G is abelian.

(d) Show that if G is abelian, then $x \mapsto x^{-1}$ is an automorphism of G. (**NB** in fact this is an if and only if.)

(e) Deduce that there is no group G such that $\mathrm{Aut}(G)$ is a nontrivial cyclic group of odd order.

Dihedral groups

Let us imagine a regular n-gon in the plane so that the vertices are on the unit circle at angles $2\pi k/n$, $0 \leq k \leq n-1$. One can show that the dihedral group D_n is generated by a rotation r, say counterclockwise by $2\pi/n$, and a reflection s, say about the horizontal axis. (I do not know why s is the letter chosen for the reflection other than that it is the letter after r. Some texts will instead call these ρ, σ because they're fancy.) The order of r is n, and the order of s is 2, and they have the relationship $srs = r^{-1}$.

We say that D_n has the **presentation** $\langle r, s : r^n, s^2, srsr\rangle$, where writing a string on the right of the presentation is the same as saying "*string* $= e$ in this group". The relationship $srs = r^{-1}$ means that s and r don't quite commute, but they come close enough that we know what changes we have to make when we try to commute them. So, even though this group is nonabelian, we can still write every element as $s^i r^k$ (or the other way around if you prefer) where $i \in \{0, 1\}$, $0 \leq k \leq n-1$. This idea of using group relations to write elements in a standardized way to make computations easier is common in group theory, and one often says that the standardized way of writing them is the **normal form**. E.g. in topology, the fundamental group of the Klein bottle is the nonabelian group with presentation $\langle a, b : aba^{-1}b\rangle$, and you can do the same kind of thing to write every element in a normal form like $a^m b^n$.

Exercise 7. (Familiarize yourself with D_n)

(a) Draw or cut out your favorite n-gon ($n \geq 4$ bc triangles are a little misleading sometimes) and visually convince yourself that $srs = r^{-1}$.

(b) Write down the Cayley table (multiplication table) for D_4. Yes, I'm serious.

(c) Find all the subgroups of D_4 and draw its lattice of subgroups. Do the same for D_5.

(d) Find all the distinct conjugacy classes of D_4 and D_5.

(e) Find all the normal subgroups of D_4 and D_5.

Symmetric and alternating groups

The symmetric group S_n consists of all bijections $\pi\colon \{1,\dots,n\} \to \{1,\dots,n\}$, with the group operation being function composition. (Using σ, π as the preferred letters for an arbitrary permutation is standard.) One can denote a permutation using "two-line" notation, by writing a two-row matrix with n in the top row and $\pi(n)$ in the bottom row, e.g.:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}.$$

Since the top row is always the same, you can also just write the second line, so the above could just be written as $\pi = 3412$.

However, both of these notations are clunky if we need to compose two permutations, e.g. what is $3412 \circ 3124$? For this reason, nobody actually uses either of those notations. Instead, the good notation is **cycle notation**. A cycle in a permutation is what you get by applying π iteratively to a single element: $n, \pi(n), \pi(\pi(n)), \dots$. Any given permutation splits up into a disjoint union of cycles, and so we can denote a permutation by its set of cycles. For example, the permutation I wrote as 3412 before has the cycle representation $(13)(24)$, meaning that 1 has been swapped with 3 and 2 with 4. The permutation 3124 has cycle representation (123). (We omit any cycles of length one.)

Cycles are composed right-to-left, like functions, so the composition I asked about in the previous paragraph becomes $(13)(24)(123)$. We can determine the cycle decomposition of the product by tracking each n:

(i) Starting with $n = 1$ and going right-to-left, we have $1 \mapsto 2 \mapsto 4$.

(ii) Since we ended on 4, now we start with 4 and have $4 \mapsto 2$.

(iii) We have $2 \mapsto 3 \mapsto 1$. This completes the cycle (142).

(iv) Only 3 is left, and must be mapped to itself. We can also see that going right-to-left we have $3 \mapsto 1 \mapsto 3$, as expected.

Thus, $(13)(24)(123) = (142)$.

Exercise 8. (Familiarize yourself with S_n)

(a) Compute $(1524)(14)(12)(34)$.

(b) Prove that S_n is generated by elements of the form (ij) (called **transpositions**). (Hint: prove that an arbitrary cycle can be written as a product of transpositions.)

(c) Prove that S_n is generated by elements of the form $(i, i+1)$ (adjacent transpositions).

(d) Prove that S_n is generated by elements of the form $(1i)$.

(e) Prove that S_n is generated by (12) and $(12\dots n)$.

(f) Prove that two elements $\pi_1, \pi_2 \in S_n$ are conjugate iff they have the same **cycle-type**. That is, iff the number and lengths of the cycles in the cycle decomposition of π_1 is the same as for π_2.

(g) Find all the subgroups of S_4 and draw its lattice of subgroups. Which subgroups are normal?

The symmetric group has an important subgroup, the alternating group A_n. By the previous exercise, every permutation can be written as a product of transpositions. The number of transpositions in a given representation is not an invariant of the permutation, but whether that number is even or odd actually is. The group A_n is the subgroup consisting of all even permutations, that is, all permutations whose representation as a product of transpositions has an even number of terms.

Equivalently, S_n can be embedded into $\mathrm{GL}_n(\mathbb{Z})$ as the permutation matrices, and then there is a homomorphism $\mathrm{GL}_n(\mathbb{Z}) \to \{\pm 1\}$ that takes a matrix to its determinant. (A matrix in $\mathrm{GL}_n(\mathbb{Z})$ always has determinant ± 1, because $1 = \det(AA^{-1}) = \det(A)\det(A^{-1})$, but $\det(A)$ and $\det(A^{-1})$ are both integers.) Composing these two maps gives a homomorphism $\epsilon\colon S_n \to \{\pm 1\}$ called the **sign homomorphism**, which is surjective because it takes any transposition to -1. The subgroup A_n is the kernel of this homomorphism, which moreover shows that it is a normal subgroup of S_n.

Exercise 9. (Familiarize yourself with A_n)

(a) Prove that a 3-cycle is even. More generally, any odd-length cycle is even.

(b) Prove that A_n is generated by 3-cycles, (ijk). (Hint: consider products of disjoint transpositions, and products of non-disjoint transpositions.)

(c) Find all the conjugacy classes in A_4.

(d) Give an example of two elements $\pi_1, \pi_2 \in A_4$ so that $[\pi_1] = [\pi_2]$ in S_4 but $[\pi_1] \neq [\pi_2]$ in A_4.

For $n \geq 5$, A_n is a simple group, i.e. it has no nontrivial normal subgroups. (I guess this is also true of A_3 but for a sorta silly reason.)

The quaternion group Q

Exercise 10. (Familiarize yourself with Q)

(a) Find all the conjugacy classes in Q.

(b) Find all the subgroups of Q, and draw its lattice of subgroups. Which subgroups are normal?

Free groups

The free group F_n has the following universal property: given any group G, and any function of sets $f\colon \{1, \ldots, n\} \to G$, there is a unique homomorphism $\widetilde{f}\colon F_n \to G$ such that $\widetilde{f}(g_i) = f(i)$. I don't have any really good questions to test your understanding of this one, if you come up with one let me know.

Isomorphism Theorems

Theorem (first isomorphism theorem)**.** Let $\phi\colon G \to H$ be a group homomorphism. Then $\ker\phi$ is a subgroup of G, $\operatorname{im}\phi$ is a subgroup of H, and the induced homomorphism $G/\ker\phi \to \operatorname{im}\phi$ is an isomorphism. In particular, if ϕ is surjective, then $H \cong G/\ker\phi$.

Theorem (second isomorphism theorem)**.** Let G be a group, $H < G$ a subgroup, and $N \triangleleft G$ a normal subgroup of G. Then $HN = \{hn : h \in H,\ n \in N\}$ is a subgroup of G, N is a subgroup of HN, $(HN)/N$ is a subgroup of G/N, $H \cap N$ is a normal subgroup of H, and the composition $H \hookrightarrow G \twoheadrightarrow G/N$ induces an isomorphism $H/(H \cap N) \xrightarrow{\sim} (HN)/N$.

Theorem (third isomorphism theorem)**.** Let G be a group, N a normal subgroup. Then if H is an subgroup of G such that $N < H < G$, then H/N is a subgroup of G/N. If, moreover, H is also normal, then H/N is normal in G/N, and the composition $G \twoheadrightarrow G/N \twoheadrightarrow (G/N)/(H/N)$ induces an isomorphism $G/H \xrightarrow{\sim} (G/N)/(H/N)$.

Theorem (lattice theorem)**.** The correspondence that takes a subgroup $H < G$ containing a normal subgroup N to the quotient H/N gives a bijection

$$\{\text{subgroups of } G \text{ containing } N\} \xrightarrow{\sim} \{\text{subgroups of } G/N.\}$$

Furthermore, this bijection preserves inclusions, products, and intersections.

Group Actions

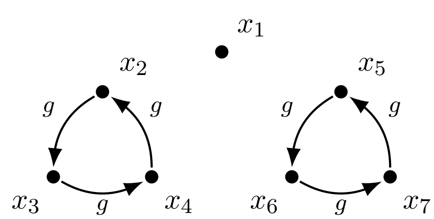
Group Actions

Group actions are the most important thing about groups. Their importance is witnessed by the fact that the groups D_n, S_n, GL_n, and SL_n are essentially defined as "the group that acts on the set X" for different sets X. This is part of the power of abstract algebra, we can take a group whose elements are really actions on some other set, and treat it as if those actions were small enough to hold in your hand.

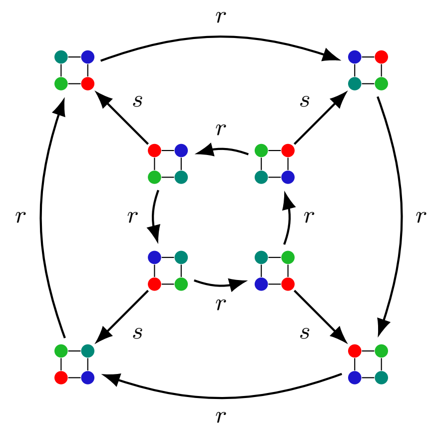
Formally, a group action is like a "scalar multiplication" by elements of a group on a set X. Because groups can be noncommutative, we need to specify whether we want our groups to act on the left or the right. Most of the time I will write actions on the left, but you can't get away from right actions sometimes. Then, for a group G to act on a set X on the left, often written ${}^{G\circlearrowleft}X$, we require that for each $x \in X$ and $g \in G$ we get an element $g \cdot x \in X$ such that $e \cdot x = x$ and $g_2 \cdot (g_1 \cdot x) = (g_2 g_1) \cdot x$. (For a right action, we write $X^{\circlearrowright G}$ and require that $(x \cdot g_1) \cdot g_2 = x \cdot (g_1 g_2)$.)

Equivalently, a group action is the same thing as, for each $g \in G$, specifying a function $\phi_g \colon X \to X$, where the function $\phi_g(x) = g \cdot x$. The two latter requirements above can be rephrased as $\phi_e = \mathrm{id}_X$, and $\phi_{g_1} \circ \phi_{g_2} = \phi_{g_1 g_2}$. Notice that because $g^{-1} \cdot (g \cdot x) = (g^{-1} g) \cdot x = e \cdot x = x$, the functions ϕ_g are in fact all bijections $X \to X$. Denoting the group of bijections $X \xrightarrow{\sim} X$ by S_X (by analogy to the symmetric group S_n), we can further rephrase a group action ${}^{G\circlearrowleft}X$ as a group homomorphism $G \to S_X$. Sometimes it is more convenient to view a group action in this way, and sometimes it is more convenient to view it more as a scalar multiplication.

If the set X is finite, you can visualize a group action by drawing a graph. For example, let C_6 be the (multiplicatively written) cyclic group of order 6 generated by g. Here is a possible action of C_6 on a set of size 7.



Below is a more complicated graph of D_4 acting on a square. Here, the generator r is rotation counterclockwise by 45°, and s is reflection about the diagonal going from upper right to lower left.



Here is some commonly-used terminology relating to group actions:

- Sometimes a set X with a group action ${}^{G}\circlearrowright X$ is called a **G-set**.

- Given an element $x \in X$, the set of elements of X that can be reached by applying some $g \in G$ is called the **orbit of** x, which I will denote $G \cdot x$, but some people prefer $\mathrm{Orb}_G(x)$. So, in the first example $G \cdot x_1 = \{x_1\}$ while $G \cdot x_2 = \{x_2, x_3, x_4\}$. In the second example, there is only one orbit, and we can get from any position of the square to any other position by applying an appropriate group element. An action where there is only one orbit is called **transitive**.

- An element $x \in X$ is a **fixed point** of the action if $G \cdot x = \{x\}$. The set of fixed points is denoted X^G. In the first example, $X^G = \{x_1\}$, while in the second example $X^G = \emptyset$. The standard notation for the subset of X fixed by a specific $g \in G$ is X^g, which I find really easy to get confused with X^G, so I'm going to write $\mathrm{Fix}(g)$ instead.

- Given an element $x \in X$, the set of $g \in G$ so that $g \cdot x = x$ is called the **stabilizer of** x. The stabilizer of a given $x \in X$ is always a subgroup of G. I will denote the stabilizer of x by $\mathrm{Stab}(x)$, but some people denote G_x. Again, I just find it confusing to try to keep track of the differences between $G \cdot x, X^G, X^g$, and G_x, and only the first two are common enough and sensible enough for me to keep straight. So, in the first example, $\mathrm{Stab}(x_1) = C_6$ (corresponding to being a fixed point) and $\mathrm{Stab}(x_2) = \{e, g^3\}$. In the second example, every stabilizer is equal to $\{e\}$. An action where every stabilizer is trivial is called **free**.

- A less-commonly used term, the **kernel** of an action is the set of $g \in G$ that fix every $x \in X$. When we think about an action as a homomorphism $G \to S_X$, the kernel of the action coincides with the kernel of the homomorphism. As a consequence, the kernel of an action is always a normal subgroup of G. In the first example, the kernel is $\{e, g^3\}$, while in the second example the kernel is $\{e\}$.

- An action that has trivial kernel is called a **faithful action**. So, the second example is faithful, while the first is not. To put it another way, an action is faithful iff $g \cdot x = h \cdot x$ for all $x \in X$ implies that $g = h$. The idea is that a faithful action "faithfully encodes" all the different elements of G as different functions on X, without collapsing any distinctions. A free action is always faithful, but not necessarily the other way around.

- Sometimes an action that is both transitive and free will be called **simply transitive**. The action of D_4 above is simply transitive.

Exercise 1.

(a) Give an example of a faithful action of C_6 on a set of size 5. (You can do this like I did above, by just drawing a graph of what the generator does to each of the 5 elements.)

(b) Is the action you gave free?

(c) If a finite group G acts freely on a finite set X, what can you conclude about $\#X$ relative to $\#G$?

The Orbit-Stabilizer Theorem

If group actions are the most important thing about groups, then the Orbit-Stabilizer Theorem is the most important thing about group actions.

Theorem (Orbit-Stabilizer)**.** Suppose G is a group, X is a set, and ${}^{G}\circlearrowright X$. For each $x \in X$, the map

$$\begin{aligned} G &\to (G \cdot x) \\ g &\mapsto g \cdot x \end{aligned}$$

is surjective and $|\mathrm{Stab}(x)|$-to-one. In particular, if G is finite, we obtain that

$$\frac{\#G}{\#\,\mathrm{Stab}(x)} = \#(G \cdot x).$$

Exercise 2. (Lagrange's theorem) Let G be a finite group, $H < G$, and denote by $[G : H]$ the number of cosets of H in G. Use the Orbit-Stabilizer Theorem to prove that $\#G = [G : H] \cdot (\#H)$, and in particular that both $\#H$ and $[G : H]$ divide $\#G$.

Exercise 3. (Class equation)

(a) Let G be a finite group, and let ${}^{G}\circlearrowright X$. Let $x_1, \ldots, x_n$ be representatives of each orbit in X. Prove that

$$\#X = \sum_{i=1}^{n} \#(G \cdot x_i).$$

(b) Let $g_1, \ldots, g_k \in G$ be representatives for the conjugacy classes of size greater than 1. Prove that

$$\#G = \#Z(G) + \sum_{i=1}^{k} \#[g_i].$$

(c) Use the Orbit-Stabilizer Theorem to prove that $\#[g_i] \mid \#G$.

Exercise 4. (Cauchy's theorem) Let G be a finite group, and let $p \mid \#G$ be a prime number. Consider the set $X \subset G^p$ consisting of all the tuples $(g_1, \ldots, g_p)$ satisfying $\prod_{i=1}^{p} g_i = e$.

(a) Let the cyclic group $C_p = \langle\sigma\rangle$ act on X by cyclic permutation: $\sigma \cdot (g_1, \ldots, g_p) = (g_2, \ldots, g_p, g_1)$. Prove that $\sigma \cdot (g_1, \ldots, g_p) \in X$, so this action is well-defined.

(b) Prove that $\#X = (\#G)^{p-1}$ by counting.

(c) Prove that the number of tuples in X fixed by σ is divisible by p. Prove also that this number is at least 1.

(d) Prove that there is an element $g \in G$ of exact order p.

I'm not going to state or prove the Sylow theorems, because they don't appear on the qual anymore, but the proofs are similar to the above proof of Cauchy's theorem, except you have to be both more careful and more clever. As Jordan Ellenberg once said, the Sylow theorems are hard to prove because you need to have not one but *two* good ideas.

Exercise 5. (January 2024 Problem 3) Let G be a group acting transitively on a set X. Let $H \subset G$ be a normal subgroup. Then H naturally acts on X, but the action need not be transitive. Prove that all orbits of H on X have the same cardinality. (You may assume G and X are finite if you like, but this is not necessary!)

Exercise 6. (August 2022 Problem 2) Let $\mathbb{F}_p$ be the finite field with p elements; here p is a prime number. Let $V = \mathbb{F}_p^2$, and recall that $G = \mathrm{GL}_2(\mathbb{F}_p)$ is the group of invertible linear transformations on V. G acts on V in the usual way (by multiplication).

(a) Describe the orbits of this action.

(b) Describe the stabilizer in G of the vector

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in V.$$

(c) Consider now the action of G on $V \times V$ (acting independently on each of the two vectors). How many orbits does this action have?

(d) What is the cardinality of G? Remember to justify your answer.

Exercise 7. (August 2023 Problem 5, edited) By S_n we mean the symmetric group on n elements.

(a) Let G be a subgroup of S_n. Suppose the action of G on $\{1, \ldots, n\}$ is transitive. (**NB** such a subgroup is called a "transitive" subgroup.) Prove that for any $x, y \in \{1, \ldots, n\}$, the stabilizer of x in G and the stabilizer of y in G are conjugate subgroups of G.

(b) Prove that if G is a transitive abelian subgroup of S_n, then $|G| = n$.

(c) Give an example of an integer n and a subgroup G of S_n such that G is transitive and abelian but not cyclic.

(d) For which integers n does S_n contain a transitive abelian subgroup G that is not cyclic? (Justify your answer.)

Group Theory 2

Semidirect products

An example

I'm going to begin describing semidirect products with an example you already know, dressed up in an unfamiliar garment. Often, we think about the group $\mathbb{Z}/n\mathbb{Z}$ acting on an n-gon by rotations. Imagine an n-gon made out of one continuous pipe, and within the pipe there is a fluid flowing around (maybe it's some modern art piece which nobody understands). Consider the group which acts on this object, where the possible symmetries are to rotate the n-gon *in the direction of the fluid flow*, or to reverse the direction the fluid is flowing.

We will call this group the *oriented cyclic group of order* n, OC_n, and concretely the elements of this group are pairs of the form (a, o), where $a \in \mathbb{Z}/n\mathbb{Z}$ is the *position* of the element (how much we're rotating), and $o \in \{\pm 1\}$ is the *orientation* (whether or not we're reversing the flow after the rotation). The group multiplication is defined so that the orientations multiply, so the composition of two negatively oriented elements is positively oriented. The positions add normally if they are both positively oriented, but if the first element is negatively oriented, instead the second position is subtracted from the first, because in that case the second rotation will happen in the opposite direction. In symbols, in all cases we have:

$$(a_1, o_1) \cdot (a_2, o_2) = (a_1 + o_1 a_2, o_1 o_2).$$

For example, if $n = 5$, here are a few computations in this group:

$$\begin{aligned} (3,1)\cdot(2,-1) &= (0,-1), & \qquad (1,-1)\cdot(3,-1) &= (1,1), \\ (2,-1)\cdot(3,1) &= (4,-1), & \qquad (3,-1)\cdot(1,-1) &= (2,1). \end{aligned}$$

Exercise 1.

(a) Verify that OC_n is a group. Is the multiplication associative? What is the identity? What is the inverse of (a, o)?

(b) What is the order of OC_n?

(c) Write out the multiplication table for OC_3. Yes, I'm serious.

(d) What is the more common name for this family of groups?

I have disguised these groups you already know in order to emphasize somewhat that this construction depended only on one thing: the fact that the groups $\mathbb{Z}/n\mathbb{Z}$ all have an action by a cyclic group of order 2, namely $a \mapsto -a$. This isn't just an action of $\{\pm 1\}$ on a *set* of size n that coincidentally happens to be a group: in addition, the map $a \mapsto -a$ is a group homomorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$. Inside the group OC_n (I'm not dropping the pretense), we can see how to get $-a$: first reverse the fluid flow, then rotate by a, then reverse the fluid flow back. In symbols: $(-a, 1) = (0, -1) \cdot (a, 1) \cdot (0, -1)^{-1}$. The negation action is built into the group as a conjugation!

The general case

Let N, Q be groups such that Q acts on N on the left. As mentioned, we are going to require that the action of Q on N "respects the group structure", so instead of Q acting on N by arbitrary functions, we want for Q to act on N by homomorphisms. This means that the action satisfies $(q \cdot n_1)(q \cdot n_2) = q \cdot (n_1 n_2)$, where $q \cdot n$ indicates the action of q on the element n. If we think about a general group action as a homomorphism $Q \to S_N$, here instead I'm asking for a homomorphsim $Q \to \mathrm{Aut}(N)$.

If we have such an action, we can construct the **semidirect product** $N \rtimes Q$. The semidirect product is a group whose underlying set is $N \times Q$, with multiplication defined by $(n_1, q_1)(n_2, q_2) = (n_1(q_1 \cdot n_2), q_1 q_2)$. You should think of this as *almost* being the direct product $N \times Q$, but the multiplication law has been "twisted" by the action of Q. It might be better to call this *one* semidirect product of N and Q, because we can get non-isomorphic products by choosing different actions of Q on N (always satisfying $(q \cdot n_1)(q \cdot n_2) = q \cdot (n_1 n_2)$).

With this definition, one can see that $N \rtimes Q$ has a normal subgroup isomorphic to N consisting of elements of the form (n, e_Q), and a subgroup isomorphic to Q consisting of elements of the form (e_N, q). The explanation for the cryptic multiplication is that it is defined so that conjugating an element of the normal subgroup (n, e_Q) by (e_N, q) corresponds to the action of q on n: $(e_N, q)(n, e_Q)(e_N, q^{-1}) = (q \cdot n, e_Q)$. Thus, our original group action has been "encoded" in the semidirect product as an action by conjugation. The subgroup isomorphic to Q need not be normal, as fixing an element $n \in N$, the set of elements $\{(n(q \cdot n^{-1}), q) : q \in Q\}$ forms a conjugate subgroup.

As noted above, which specific action you choose of Q on N can change the isomorphism type of the group $N \rtimes Q$. In situations where more than one semidirect product between the same two groups might come up, people will name the actions, e.g. $\phi, \psi \colon Q \to \mathrm{Aut}(N)$, and write $N \rtimes_\phi Q$ versus $N \rtimes_\psi Q$.

Notice that the direct product is a special case of the semidirect product: if ${}^{Q\circlearrowright}N$ trivially via $q \cdot n = n$ for all pairs $q \in Q$ and $n \in N$, then $N \rtimes Q = N \times Q$.

Exercise 2. How many nonisomorphic groups of the form $\mathbb{Z}/7\mathbb{Z} \rtimes C_6$ are there?

Semidirect products and splitting sequences

If N, Q are groups, and ${}^{Q\circlearrowright}N$ by homomorphisms, then there is a SES of groups

$$1 \to N \to N \rtimes Q \to Q \to 1.$$

Moreover, this exact sequence splits on the right, via the map $q \mapsto (e_N, q)$. More generally, whenever $N \triangleright G$, if the exact sequence

$$1 \to N \to G \to G/N \to 1$$

splits on the right, then ${}^{(G/N)}\circlearrowright N$ via conjugation, and $G \cong N \rtimes G/N$. In this case, letting H be the image of the splitting homomorphism, we say that G is the **internal semidirect product** of N and H.

With groups there is a real difference between a SES splitting on the right versus on the left. If the SES

$$1 \to N \to G \to G/N \to 1$$

splits on the left, then not only is $G \cong N \rtimes G/N$, but actually $G \cong N \times G/N$.

Exercise 3. Consider the group $G = \mathrm{SL}_2(\mathbb{F}_3)$.

(a) What is the order of G?

(b) Show that G has a subgroup H isomorphic to Q, the quaternion group of order 8.

(c) Show that H is normal.

(d) Prove that $G \cong Q \rtimes C_3$ by finding a splitting map $C_3 \cong G/H \to G$.

The semidirect product test

There's a standard test to show that G is the (internal) semidirect product of its subgroups N and Q. If:

- N is normal in G,
- $N \cap Q = \{e\}$ is the set containing only the identity element, and
- $G = NQ$ (that is, every element $g \in G$ can be written as nq for some $n \in N$ and $q \in Q$)

then $G = N \rtimes Q$ (where the action of Q on N is conjugation in G).

The commutator (derived) subgroup

An important subgroup of a group is the commutator subgroup, sometimes also called the derived subgroup. (I will not call it that because I think this terminology is misleading if you're familiar with other uses of the term "derived".) The **commutator subgroup** of G, denoted $[G, G]$, is the group generated by elements of the form $ghg^{-1}h^{-1}$, which are called **commutators**. This means a general element of $[G, G]$ will look like $g_1h_1g_1^{-1}h_1^{-1} \dots g_nh_ng_n^{-1}h_n^{-1}$. You will also (in the next section) see the notation $[H_1, H_2]$ for H_1, H_2 subgroups of a given group G. In this case, the notation means the subgroup generated by commutators $h_1h_2h_1^{-1}h_2^{-1}$ where $h_i \in H_i$, $i = 1, 2$.

The commutator subgroup will be trivial if and only if G is abelian. For any group G, $[G, G]$ is a normal subgroup of G. Here's a tricky way to see this, relying on the fact that commutators and conjugation are closely linked: suppose $x \in [G, G]$, and $g \in G$, then $gxg^{-1}x^{-1} \in [G, G]$ because it is a commutator, so call $y = gxg^{-1}x^{-1}$. Then we have $gxg^{-1} = yx$, and $yx \in [G, G]$ since both of $x, y \in [G, G]$.

Furthermore, $G/[G,G]$ is always an abelian group, since in the quotient every commutator is trivial. The group $G/[G,G]$ is called the *abelianization* of G, denoted G^{ab}. It has the universal property: if A is an abelian group, and $f\colon G \to A$ is a homomorphism, then there exists a unique homomorphism $\widetilde{f}\colon G^{\mathrm{ab}} \to A$ so that $f = \widetilde{f} \circ q$, where $q\colon G \to G^{\mathrm{ab}}$ is the quotient map. In diagram form:

$$\begin{tikzcd} G \arrow[d, "q"'] \arrow[dr, "f"] & \\ G^{\mathrm{ab}} \arrow[r, dashed, "\widetilde{f}"'] & A \end{tikzcd}$$

abelian groups

This is akin to the universal property of $S^{-1}M$, and analogous to my comment in that section, you should think of the abelianization as the "best abelian approximation" to G. When G is finite, the universal property has a payout in terms of sizes: G^{ab} is the largest abelian quotient of G.

Exercise 4. For each of the following groups G, compute $[G,G]$.

(a) D_4

(b) Q

(c) S_4

(d) A_4

(e) A_5

Sometimes, a group G might not admit *any* nontrivial maps to an abelian group, which would happen when $G^{\mathrm{ab}} = 1$, or to say it another way, when $[G,G] = G$. Such a group is called **perfect**.

Exercise 5. In this problem we will show that $G = \mathrm{SL}_2(\mathbb{F}_5)$ is a perfect group. Call a matrix $A \in \mathrm{SL}_2(\mathbb{F}_5)$ a *shearing matrix* if A is upper or lower triangular with 1s on the diagonal.

(a) Prove that $\mathrm{SL}_2(\mathbb{F}_5)$ is generated by the two matrices

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(b) Prove that T can be written as a product of shearing matrices. Conclude that the shearing matrices generate $\mathrm{SL}_2(\mathbb{F}_5)$.

(c) Prove that S and $S^\top$ lie in $[G,G]$. Conclude that $[G,G] = G$. (Hint: try conjugating S by matrices of the form D or TD, where D is diagonal.)

Observe that $\mathrm{SL}_2(\mathbb{F}_5)$ is not simple because it has a nontrivial center. It is the smallest perfect group that is not a simple group. Showing that matrix groups are perfect comes up because Tits' simplicity theorem lets you upgrade a group being perfect (plus some extra stuff) into that group being simple.

Nilpotent groups

A nilpotent group is a generalization of an abelian group. There are two equivalent definitions of nilpotent groups:

- G is nilpotent if the descending sequence of normal subgroups $G = G_0 \triangleright G_1 \triangleright G_2 \triangleright \dots$ eventually terminates in the trivial subgroup, where $G_{i+1} = [G_i, G]$.
- G is nilpotent if the ascending sequence of normal subgroups $\{e\} = Z_0 \triangleleft Z_1 \triangleleft Z_2 \triangleleft \dots$ eventually terminates in the full group G, where Z_{i+1} is defined so that $Z_{i+1}/Z_i = Z(G/Z_i)$. **Exercise 6.** Confirm that this gives a well-defined subgroup.

These are respectively called the **lower central series** and the **upper central series**. If these series terminate, their lengths are the same, and the length of either series is called the *nilpotency class* of G. The groups of nilpotency class 1 are exactly the abelian groups.

Exercise 7. (What's a central series?) There is a general definition of a "central series", although I think historically it was invented in retrospect to solidify the things that the lower and upper central series have in common. A **central series** is a sequence of normal subgroups

$$G = A_0 \triangleright A_1 \triangleright \cdots \triangleright A_n = \{e\}$$

that satisfy either of the two conditions

(i) $[G, A_i] < A_{i+1}$, or

(ii) $A_i/A_{i+1} < Z(G/A_{i+1})$.

I will write A^i to mean A_{n-i}.

(a) Prove that conditions (i) and (ii) above are equivalent.

(b) Given any central series A_i, prove that $A_i > G_i$.

(c) Given any central series A^i, prove that $A^{i+1}/A^i < Z(G/Z_i)$.

(d) Conclude that any central series has the same length.

Exercise 8. A finite group is called a p-**group** if $\#G = p^k$ for some k. Fix an arbitrary finite nontrivial p-group G.

(a) Show that $Z(G) \neq \{e\}$.

(b) Show that G is nilpotent.

Why are nilpotent groups called "nilpotent"? There doesn't seem to be anything particularly nilpotent-y about them. The term originally comes from the theory of Lie groups and Lie algebras. Here's one possible way to think about it: for each $g \in G$ we can define a function $f_g \colon G \to G$ by $f_g(x) = [g, x] = gxg^{-1}x^{-1}$. This function is not a homomorphism, it's just a function. A group is called nilpotent if there exists some n such that for all g, f_g^n is the trivial map $G \to \{e\}$, i.e. if each of the functions f_g is "nilpotent" in a more normal sense of the word.

Solvable groups

Another generalization of abelian groups are solvable groups. A group is **solvable** if there exists a sequence of subgroups $G = G_0 > G_1 > \cdots > G_n = \{e\}$ such that (i) $G_{i+1} \triangleleft G_i$ and (ii) the quotients G_i/G_{i+1} are all abelian. The terminology comes from Galois theory, as a polynomial can be solved by radicals if and only if its Galois group is a solvable group.

Given a group, it might be difficult to come up with such a sequence of subgroups just by thinking and messing around. Luckily, there's a standard sequence that you can always check. Given a group G, the **derived series** of G is the sequence of subgroups $G = G^{(0)} > G^{(1)} > G^{(2)} > \ldots$ where $G^{(i+1)}$ is the commutator subgroup of the preceding group $[G^{(i)}, G^{(i)}]$. (I would prefer to call this the "commutator series", but it seems like in actual practice nobody does that.)

Exercise 9. The derived series is usually not the same as the lower central series. The difference in the definition is $G^{(i+1)} = [G^{(i)}, G^{(i)}]$ versus $G_{i+1} = [G, G_i]$. Find an example of a group G demonstrating that these two series can be different.

Exercise 10. Prove that a finite group G is solvable if and only if the derived series eventually reaches the trivial group.

Exercise 11.

(a) Prove that any nilpotent group is solvable.

(b) Find an example of a group that is solvable but not nilpotent.

Exercise 12. Let $1 \to A \to B \to C \to 1$ be a short exact sequence of groups. Show that if A and C are solvable, then B is also solvable.

Exercise 13. (August 2019 Problem 5) Let G be the group of 2×2 invertible upper triangular matrices over the field $\mathbb{F}_p$.

(a) Prove that G has only one subgroup of order p, and that this subgroup is normal. (Hint: first show that an element of order p must have 1's on the diagonal.)

(b) Prove that G is solvable by exhibiting a homomorphism f from G to an abelian group A such that the kernel of f is also abelian.

(c) Prove that if $p \neq 2$, G is not nilpotent.

Exercise 14. (January 2022 Problem 1) On this problem, only the answer will be graded. Consider the symmetric group S_4.

(a) What is the order of S_4?

(b) What is the order of its center $Z(S_4)$?

(c) What is the order of its commutator subgroup $[S_4, S_4]$?

(d) Is S_4 a simple group?

(e) Is S_4 a solvable group?

(f) How many conjugacy classes does S_4 have?

Some other problems I like

Exercise 15. The additive group $\mathbb{Q}$ has the property that for any $q \in \mathbb{Q}$ and any $n \in \mathbb{Z}$, there exists an element $q' \in \mathbb{Q}$ such that $nq' = q$ (in particular $q' = q/n$). An abelian group with this property is called a *divisible* group.

(a) Show that any nontrivial divisible group is infinite.

(b) Show that if G is divisible, and N is a subgroup (normal since G is abelian), then G/N is divisible.

Exercise 16. (January 2017 Problem 1, (d) omitted) Give an example of each of the following.

(a) A group G with a normal subgroup N such that G is not a semidirect product $N \rtimes G/N$.

(b) A finite group G that is nilpotent but not abelian.

(c) A group G whose commutator subgroup $[G, G]$ is equal to G.

(e) A transitive action of S_3 on a set X of cardinality greater than 3.

Field and Galois Theory

Overview

Galois theory was created to understand the roots of polynomials in one variable. The key insight of Galois theory is that we can understand polynomials by understanding the possible symmetries of their roots. A motto to keep in mind about the Fundamental Theorem of Galois Theory (or any equivalence you meet for the rest of your life): what the equivalence gives you is flexibility. You can use that flexibility when solving problems, by seeing which equivalent formulation is best suited to information you have available. (Different parts of a problem might be better suited to different sides of the equivalence, so you may end up passing back and forth more than once!)

I think the most important things to know in these notes are: the examples given in Example 1, and the basic properties of cyclotomic extensions and finite fields given immediately before; the basic relations between the minimal polynomial of an element, the degree of that element, and the degree of a field extension; the composite and intersection of two fields; the definition of a separable extension, and examples of inseparable extensions; the Fundamental Theorem, and the Galois groups of all the examples in Example 1.

Important examples to keep in mind

The following examples of field extensions are in fact all Galois extensions, so whenever you read a fact about field extensions or about Galois theory you should start by testing what that fact says about these extensions.

Example 1.

- Starting with any field K, if $\mu \in K$ is not a square, then $K(\sqrt{\mu})$ is an extension of degree 2. In particular, this lets you think of lots of degree 2 extensions of $\mathbb{Q}$. **Exercise 1.** If n, m are different squarefree integers, show that $\mathbb{Q}(\sqrt{n})$ and $\mathbb{Q}(\sqrt{m})$ are different extensions of $\mathbb{Q}$.

- For every n, the extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ has degree $\varphi(n)$. (See **Cyclotomic extensions** below.)

- For any p, and any d, there is an extension of $\mathbb{F}_p$ of degree d. In fact, a non-transparent fact is that any two extensions of $\mathbb{F}_p$ of degree d are actually the same! So, we call this unique extension $\mathbb{F}_{p^d}$ *the* finite field with p^d elements. (Sometimes also called the "Galois field with p^d elements", and sometimes denoted $GF(p, d)$.) (See **Extensions of finite fields** below.)

- Combining some of the above examples, given an field K and an element $\mu \in K$ which is not a perfect nth power, then we can form the extension $K(\sqrt[n]{\mu}, \zeta_n)$, which is the field containing all the roots of the polynomial $x^n - \mu$ (see **Splitting fields and normal extensions** below). For example, $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$.

Quick Recap on Polynomials

(This first exercise is also in the ring theory notes.) Recall that there is a notion of "division with remainder" for polynomials: if $f(x)$ and $g(x)$ are polynomials over a field, then there are unique polynomials $q(x)$ and $r(x)$ so that $f(x) = g(x)q(x) + r(x)$ and either $\deg(r) < \deg(g)$ or $r(x) = 0$.

Exercise 2. Let K be a field, prove that $K[x]$ is a principal ideal domain.

Exercise 3.

(a) Let K be a field, $f \in K[x]$, and suppose there is $a \in K$ so that $f(a) = 0$. Prove that $(x - a) \mid f$.

(b) Conclude that a polynomial over a field cannot have more roots than its degree.

(c) Verify that the polynomial $x^2 - 1$ has 4 roots in $\mathbb{Z}/8\mathbb{Z}$.

Since we're going to be making statements about irreducible polynomials over various fields, you might reasonably wonder how to show that a given polynomial is irreducible. In general, it is pretty hard to tell directly. Here's a quick and dirty list of facts that can be helpful:

(1) Quadratic polynomials are easy to check (except in characteristic 2) because you can use the quadratic formula. In particular, a quadratic polynomial is irreducible if and only if its discriminant $b^2 - 4ac$ is not a square. (Things are more complicated in characteristic 2.)

(2) (Gauss' Lemma) A polynomial in $\mathbb{Q}[x]$ with coprime integer coefficients is irreducible if and only if it is irreducible in $\mathbb{Z}[x]$. Since we can always multiply through to clear denominators, this means that understanding irreducibility of polynomials over $\mathbb{Q}$ is the same as understanding irreducibility of polynomials over $\mathbb{Z}$. This is useful because...

(3) A polynomial in $\mathbb{Z}[x]$ with coprime coefficients is irreducible if it is irreducible over $\mathbb{Z}/m\mathbb{Z}$ for some integer m. This is usually most useful when the degree is 3 and m is a small prime number because, as mentioned below, in that case $\mathbb{Z}/p\mathbb{Z}$ is a field, so you can check irreducibility by just checking if any of the p elements of $\mathbb{Z}/p\mathbb{Z}$ are roots. (I want to state the fact for any m, even though on the qual you usually only use it for a prime number, because I think it's neat.)

(4) (Eisenstein's criterion) There's one more test which can only be used for polynomials in $\mathbb{Z}[x]$: suppose you write your polynomial $f = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$, and suppose there is a prime number p such that:

- the leading coefficient a_n is not divisible by p,
- every other coefficient *is* divisible by p, and
- the constant term a_0 is not divisible by p^2.

Then f is irreducible. (In fact, this can be modified to work for any PID, and indeed any integral domain. Eisenstein discovered it while working over the ring $\mathbb{Z}[i]$.)

(5) (Rational root theorem) Writing f as above, the rational root theorem says any rational root of f must be of the form $\pm d/e$, where $d \mid a_0$ and $e \mid a_n$. Again, for degree 2 or 3 polynomials this lets you compute if f is irreducible by ruling out all the possible rational roots.

- In particular, the rational root theorem implies that any rational root of a monic polynomial is in fact an integer root.

(6) (Galois theory) Finally, though not least, Galois theory gives us another tool for showing that a polynomial is irreducible. If we can, by hook or by crook, find the splitting field K of a given polynomial and compute the Galois group $G = \mathrm{Gal}(K/\mathbb{Q})$, we can check if the original polynomial is irreducible or not by checking whether or not G acts transitively on its roots. While this may sound far-fetched, in practice this is the most powerful item on this list.

Exercise 4. Some practice showing polynomials are irreducible using these techniques.

(a) Consider the polynomial $f(x) = x^2 - 2x - 4$. Show that $f(x)$ is irreducible over $\mathbb{Q}$ by reducing mod p for some prime p, and checking that it has no roots mod p. (If you have a number-theoretic instinct, you might wonder how to choose a good p in advance...)

(b) Consider the polynomial $x^3 + 2x^2 + 3x + 4$.

  (i) Use rational root theorem to check that this polynomial has no linear factors over $\mathbb{Q}$, and is therefore irreducible.

  (ii) Show that this polynomial is irreducible mod 3.

(c) Consider the polynomial $g(x) = x^4 - tx^2 - t$ over the field of rational functions $\mathbb{Q}(t)$. Show that this polynomial is irreducible by showing it is Eisenstein in the ring $\mathbb{Q}[t]$.

Fields

Definitions and basic examples

Example 2. You already are friends with the fields $\mathbb{Q}, \mathbb{R}, \mathbb{C}$. Here are some other examples that come up:

- If K is any field, I can form the field $K(x)$, the field of all rational functions in the variable x. This is the field of fractions of the ring of polynomials $K[x]$. I could equally well form $K(x, y)$, which is the field of rational functions in the variables x, y. These fields are good to keep in mind because a lot of counterexamples come from them.

- For any prime number p, the quotient ring $\mathbb{Z}/p\mathbb{Z}$ is a field. Often, when people are thinking about $\mathbb{Z}/p\mathbb{Z}$ as a field (rather than as one of the quotients $\mathbb{Z}/m\mathbb{Z}$ that just by chance happens to be a field), they will denote it as $\mathbb{F}_p$. If you have never done it before, **Exercise 5.** prove that $\mathbb{Z}/p\mathbb{Z}$ is a field (i.e. prove that you can "divide" by any nonzero element).

- A generalization of the previous example: if R is a ring, and m is a maximal ideal of R, then R/m is a field. If you have never done it before, **Exercise 6.** prove this (again, the sticking point is showing you can divide).

- A special case of this, which is most important to us, is the case when $R = K[x]$, and $m = (f)$ for an irreducible polynomial f. If you have never done it before, **Exercise 7.** prove that this is indeed a maximal ideal.

Comment. Notationally, writing square brackets $K[x]$ means "K adjoin x (as a ring)", while writing the parentheses $K(x)$ means "K adjoin x (as a field)". That is to say, $K[x]$ has all the elements of K, the element x, and anything that is has to have by adding, subtracting, and multiplying, while $K(x)$ also has everything it must have through division as well. This notation makes sense even if x is not an indeterminate variable, but an actual number. So, for example, one can write things like $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[1/2]$, $\mathbb{Q}[\sqrt{2}]$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\pi)$. If you have never done it before, **Exercise 8.** figure out what a general element of each of the preceding rings/fields looks like. Figure out why $\mathbb{Q}[\sqrt{2}] = \mathbb{Q}(\sqrt{2})$. Figure out what I mean when I write $\mathbb{C}(x + x^{-1})$.

A **field extension** of a field K is a field L such that $L \supset K$. (Sometimes it's better to say an extension is a field L and an injective homomorphism $f\colon K \to L$.) Often, we will write a field extension by just writing L/K, read aloud as "L over K". Unfortunately, this looks a lot like the notation for a quotient group, but that is *not* what this notation means. Rather, we imagine that the extension field lies physically "above" the base field, which we depict:

$$\begin{tikzcd} L \arrow[d, dash] \\ K \end{tikzcd}$$

A field extension is **algebraic** if it can be obtained by adjoining roots of polynomials to the base field. (Possibly infinitely many!)

Example 3.

- $\mathbb{C}$ is an algebraic field extension of $\mathbb{R}$. $\mathbb{R}$ is a non-algebraic field extension of both $\mathbb{Q}(\pi)$ and $\mathbb{Q}(\sqrt{2})$. $\mathbb{Q}(\pi)$ and $\mathbb{Q}(\sqrt{2})$ are both field extensions of $\mathbb{Q}$, but $\mathbb{Q}(\sqrt{2})$ is algebraic over $\mathbb{Q}$ while $\mathbb{Q}(\pi)$ is not.
- Given any field K, and any irreducible polynomial f, we can use our favorite construction to get an algebraic field extension of K that contains a root of f: $K[x]/(f)$. In the field $K[x]/(f)$, the equivalence class of x is a root of f, because we have modded out by f exactly with the goal of making $f(x) = 0$ in the quotient.
  - $\mathbb{C} \cong \mathbb{R}[x]/(x^2 + 1)$
  - $\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}[x]/(x^2 - 2)$.
  - Over $\mathbb{F}_3$, the polynomial $x^2 + 1$ is irreducible. So, I can form a field $\mathbb{F}_3[x]/(x^2 + 1)$, which it seems reasonable to call $\mathbb{F}_3(i)$. **Exercise 9.** List all the elements of this field.
  - Over $\mathbb{F}_5$, the polynomial $x^2 - 2$ is irreducible. So, I can form a field $\mathbb{F}_5[x]/(x^2 - 2)$, which it seems reasonable to call $\mathbb{F}_5(\sqrt{2})$. **Exercise 10.** How many elements does it have?

Given a field extension L/K, observe that L is a K-vector space. This modest observation is central to everything else.

Given a field extension L/K, we define the **degree** of the extension, written $[L : K]$, to be the dimension of L as a K-vector space. An extension is called "finite" if $[L : K]$ is finite, otherwise an extension is called "infinite". **Exercise 11.** find a $\mathbb{Q}$-basis of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and $\mathbb{Q}(\pi)$ as extensions of $\mathbb{Q}$. Do the numbers you find line up with your intuition?

Proposition (multiplicativity of degrees)**.** Suppose L is an extension of K, and M is an extension of L. Then $[M : K] = [M : L][L : K]$. In diagram form:

$$\begin{tikzcd}
M \arrow[d, dash, "d_2"'] \arrow[dd, dash, bend left, "d_1 d_2"] \\
L \arrow[d, dash, "d_1"'] \\
K
\end{tikzcd}$$

Here are several facts about some really fundamental examples.

Cyclotomic extensions

The complex numbers $\zeta_n^k = e^{2k\pi i/n}$ for $0 \le k < n$ are the n distinct roots of the polynomial $x^n - 1$. They are called "nth roots of unity" because they satisfy $x^n = 1$. If k and n share a common factor, then $\zeta_n^k = \zeta_d^\ell$, where $\ell = k/\gcd(n,k)$ and $d = n/\gcd(n,k)$, so the nth roots of unity include all the dth roots of unity for every d dividing n. The numbers ζ_n^k where k is coprime to n are the nth roots of unity that are not dth roots of unity for any smaller d, they are called the "primitive nth roots of unity". There are $\varphi(n)$ of them, where $\varphi(n)$ is Euler's φ function, which counts the number of positive integers up to n which are coprime to n.

If you are not already familiar with the function $\varphi(n)$, here are a few of its properties (which you may prove if you desire):

- $\varphi(p) = p - 1$ and $\varphi(p^k) = p^{k-1}(p-1)$ for any prime number p.
- $\varphi(n)$ is equal to the number of elements in $(\mathbb{Z}/n\mathbb{Z})^\times$, the group of units mod n.
- $\varphi(ab) = \varphi(a)\varphi(b)$ for any coprime integers a, b. (This follows from the previous bullet along with the Chinese Remainder Theorem.)

We define the **cyclotomic polynomial** $\Phi_n(x)$ as follows:

- $\Phi_1(x) = x - 1$,
- for $n > 1$, $\displaystyle\prod_{d|n} \Phi_d(x) = x^n - 1$,

where the notation means the product is taken over all positive integers d which divide n. The polynomial $\Phi_n(x)$ is monic, has coefficients in $\mathbb{Z}$, is irreducible over $\mathbb{Z}$, and its roots in $\mathbb{C}$ are exactly the primitive nth roots of unity. (The term "cyclotomic" means "circle cutting", and Gauss chose it because the nth roots of unity evenly divide the unit circle in the complex plane.)

Exercise 12. Compute $\Phi_n(x)$ for $n = 2, \ldots, 20$. Yes, I'm serious.

Exercise 13. Prove that $\Phi_p(x+1)$ is p-Eisenstein.

We can see that $\Phi_n(x)$ splits in the field $\mathbb{Q}(\zeta_n)$, since indeed $x^n - 1$ splits and Φ_n is a divisor. Later, we will see that $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is a Galois extension with Galois group of order $\varphi(n)$, which implies that Φ_n is irreducible over $\mathbb{Q}$ for general n. As far as I know, this is the best way to prove that Φ_n is irreducible. In Exercise 13, you proved that $\Phi_p(x)$ is irreducible by showing it is Eisenstein, and one can check the irreducibility of Φ_n directly (as Gauss did, and as Dummit and Foote do more briefly), but doing so is much more difficult than computing the Galois group of its splitting field.

Extensions of finite fields

Exercise 14. (Frobenius) Suppose K is a field of characteristic p. Show that the map $x \mapsto x^p$ is an injective field homomorphism $K \to K$. This map is known as the **Frobenius map**.

Exercise 15. Consider the finite field $\mathbb{F}_p$, and the polynomial $x^{p^n} - x$. As we will see later, this polynomial is separable, and so has distinct roots in its splitting field.

(a) Use 14 to show that the p^n roots of this polynomial form a field.

(b) Show that an extension of $\mathbb{F}_p$ has p^n elements if and only if it is a degree n extension.

(c) Let $L/\mathbb{F}_p$ be any extension of $\mathbb{F}_p$ of degree n. Show that every $\alpha \in L$ satisfies $\alpha^{p^n} = \alpha$, and hence that up to isomorphism there is a unique extension of $\mathbb{F}_p$ of degree n. This unique extension is denoted $\mathbb{F}_{p^n}$, or often people will write $\mathbb{F}_q$ and it is understood that $q = p^n$ for some p, n.

Exercise 16. (January 2023 Problem 4) Consider the finite field $\mathbb{F}_p$ where p is a prime number.

(a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p.)

(b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)

As a note, the multiplicative group of $\mathbb{F}_{p^n}$ is in fact *cyclic*, so there exists some element α which has exact order $p^n - 1$.

Minimal polynomial and degree of an element

Several of our examples indicate that individual *elements* of field extensions also have a notion of "degree" attached to them: if α is a root of an irreducible polynomial f of degree d, then α ought to be called "algebraic of degree d". We might be a little cautious, though, because it's not exactly clear how the degree of an individual element will relate to the degree of the field, because if I make a field extension like $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$, I not only get $\sqrt{2}$ and $\sqrt{3}$, but also $\sqrt{2} + \sqrt{3}$, $5/(\sqrt{6} - \sqrt{2})$, and a bunch of other stuff. One might guess that these elements are in fact algebraic numbers, and that their degrees are less than or equal to the degree of the given field extension, but it seems far from straightforward to prove that.

Here's where the linear algebra will help us! Suppose that $\alpha \in L$, then there is a map from $L \to L$ given by $x \mapsto \alpha x$, the "multiply by α" map. This map is *not* a ring homomorphism, but it is K-linear. If, moreover, $[L : K]$ is finite, then I can choose a finite K-basis for L and write the multiplication by α map as a matrix M_α. That matrix will have a minimal polynomial, which is actually independent of the particular basis chosen, and that minimal polynomial will have α as a root. (Note that since the matrix M_α has entries in K, its minimal polynomial will have coefficients in K. In particular, the minimal polynomial of α probably won't be just $x - \alpha$ unless $\alpha \in K$.)

Proposition. Let L/K be a finite extension, let $\alpha \in L$, and let $f(x)$ be the minimal polynomial of the matrix M_α. Then:

(1) $f(x)$ is irreducible over K, and

(2) $f(\alpha) = 0$.

The polynomial $f(x)$ is called the "minimal polynomial of α". (Another way to phrase this is: every element of a finite-degree field extension is algebraic.)

If α is algebraic of degree d over K, then $[K(\alpha) : K] = d$.

Exercise 17. The field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is degree 4 over $\mathbb{Q}$, and we can take a $\mathbb{Q}$-basis to be $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

(a) In this basis, write down the matrix associated to multiplication by $\sqrt{2} + \sqrt{3}$.

(b) Use that matrix to find the minimal polynomial of $\sqrt{2} + \sqrt{3}$.

Algebraic closure

A field K is **algebraically closed** if every polynomial with coefficients in K has a root in K. The **algebraic closure** of a field K, denoted $\overline{K}$, is an extension of K with the following two properties:

(1) $\overline{K}$ is algebraically closed, and

(2) no subfield of $\overline{K}$ containing K is algebraically closed.

Proposition. Every field is contained in an algebraically closed field. By Zorn's Lemma, this in particular implies every field has an algebraic closure.

Composite and intersection of two fields

Fix an algebraic closure $\overline{K}/K$, and let L_1, L_2 be extensions of K contained in $\overline{K}$. Then we can form other fields out of the extensions L_1, L_2:

- The intersection $L_1 \cap L_2$ is a field extension of K.
- The composite (or compositum) $L_1 L_2$ is defined to be the smallest extension of K that contains both L_1 and L_2 as subextensions. More concretely, for every pair of elements $\ell_1 \in L_1, \ell_2 \in L_2$, the product $\ell_1 \ell_2$ must lie in $L_1 L_2$ (hence the notation). But, then you also have to throw in all the finite sums of those products, and all the inverses of those sums, to make sure what you end up with is still a field.

Example 4.

- Let $K_1 = \mathbb{Q}(\sqrt{2})$, $K_2 = \mathbb{Q}(\sqrt{3})$. Then $K_1 \cap K_2 = \mathbb{Q}$, $K_1K_2 = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.
- Let $K_1 = \mathbb{Q}(\zeta_8)$, $K_2 = \mathbb{Q}(\sqrt{\sqrt{2}+1})$.

  Exercise 18. Prove that $K_1 \cap K_2 = \mathbb{Q}(\sqrt{2})$, $K_1K_2 = \mathbb{Q}(\zeta_{16})$. (It might be helpful to know that $\zeta_8 = \sqrt{i} = (1+i)/\sqrt{2}$.)

Separable polynomials and extensions

A polynomial $f \in K[x]$ is called **separable** if it has distinct roots in its splitting field, and an extension is called separable if it can be obtained by adjoining roots of separable polynomials. A trivial example of an inseparable polynomial is something like $(x-2)^2$ over $\mathbb{Q}$. In characteristic 0, every irreducible polynomial turns out to be separable, but this need not be the case in characteristic p. For example, in the field $\mathbb{F}_p(t)$, the polynomial $x^p - t$ is irreducible (by Eisenstein), but as soon as we adjoin a root, it will split as $(x - \sqrt[p]{t})^p$.

There is a criterion for determining if a polynomial is separable.

Exercise 19. Let $f \in K[x]$ be a polynomial, and let f' be its **formal derivative**. That is, f' is obtained from f by just applying the power rule to each of the terms in f. Prove that f is separable over K if and only if $\gcd(f, f') = 1$.

In particular, this shows that $x^{p^n} - x$ is a separable polynomial over $\mathbb{F}_p$, as mentioned previously.

A field where every finite extension is separable is called a **perfect field**, and we've already seen that all finite fields and fields of characteristic 0 are perfect. On the other hand, the above example shows that $\mathbb{F}_p(t)$ is not a perfect field, and inseparable extensions can arise when taking an irreducible polynomial of degree a power of p. (In fact, every inseparable irreducible polynomial is of the form $f(x^{p^k})$ for some irreducible polynomial f.)

Exercise 20. (August 2022 Problem 4) Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$).

(a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.

(b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.

(c) Show that K has a single degree p inseparable extension.

The primitive element theorem

Several of our examples have been of the form $K(\alpha)$ for some algebraic number α. However, we've also had examples like $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, where we adjoin two different algebraic numbers. A field where we've only adjoined one thing (called a **simple extension**) is sometimes easier to think about, so a reasonable question is: can every field extension be written as $K(\alpha)$ for some α? In this case, the element α is called a **primitive element** for the extension. Above, we computed that $\sqrt{2} + \sqrt{3}$ is a degree 4 element of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, so it must be a primitive element for that extension.

The following proposition gives a partial answer to when we can find a primitive element, and the answer is basically "in most of the cases where it matters".

Proposition (Primitive element theorem)**.** Let L/K be a finite extension of characteristic 0. Then there is an algebraic $\alpha \in L$ so that $L = K(\alpha)$. In particular, every finite extension of $\mathbb{Q}$ can be written as $\mathbb{Q}(\alpha)$ for some algebraic number.

There is a more precise, but also more abstract, form of the primitive element theorem, which allows us to obtain exact conditions under which a field extension is simple.

Proposition (Primitive element theorem, again)**.** If L/K is a separable extension (possibly of characteristic p), then we can find $\alpha \in L$ so that $L = K(\alpha)$. In maximum generality: there exists a primitive element $\alpha \in L$ if and only if there are only finitely many intermediate fields between L and K.

Exercise 21. Find an example of a field extension that does not have a primitive element.

Galois Theory

Splitting fields and normal extensions

We've talked about forming field extensions by adding on a root of a polynomial. However, you may have noticed, this does not always make the polynomial factor completely into linear terms! For example, in $\mathbb{Q}(\sqrt[3]{2})$, the polynomial $x^3 - 2$ factors into a linear term and an irreducible quadratic. (How do I know?) In order to use Galois theory to understand a polynomial, we need to have all of its roots living in the field.

We define the **splitting field** of a polynomial f (not necessarily irreducible) to be the smallest extension of K that contains all the roots of f. We might get the splitting field after adjoining only one root of f, or we might need to adjoin each root, one after the other.

Exercise 22. Suppose f is irreducible of degree n, and that L/K is the splitting field of f. Prove that $n \leq [L : K] \leq n!$. What do you think the most likely value of $[L : K]$ is for a "random" polynomial f? (This latter question is somewhat ill-defined, interpret it in a way that seems reasonable to you.)

A related notion is the notion of a normal extension. A **normal extension** is an extension L/K such that every polynomial which has *any one* root in L, actually has *all* of its roots in L. This is stronger than saying that L is the splitting field of some f: this is saying that L contains the splitting field for every f that has even a single root contained in L. That seems like it's infinitely more powerful than just being the splitting field of a single f.

... Well, it seems that way, but actually, the two things are exactly equivalent.

Proposition. A finite extension L/K is normal if and only if L is the splitting field of some polynomial over K.

Automorphism group of an extension

Okay, and we said we are going to study polynomials using group theory, so at some point we better introduce some groups. Given an extension L/K, we define the **automorphism group of the**

extension $\mathrm{Aut}(L/K)$ to be the group of all self-isomorphisms $\sigma\colon L \to L$ that fix K pointwise; that is, such that for every $x \in K$, $\sigma(x) = x$.

What exactly does that mean? The field L is both a K-vector space as well as being a field, so any automorphism must be both an invertible K-linear map, and must also respect the multiplication in L. Since an automorphism is a K-linear map, we can specify it on a basis. The most common situation is when $L = K(\alpha)$ for an algebraic element α, in which case a basis for L is $\{1, \alpha, \alpha^2, \ldots, \alpha^{n-1}\}$ when α is degree n. In order to be a ring homomorphism, we are forced to send $1 \mapsto 1$. However, once we have specified where to send α, we actually don't need to make any more choices, because if we send $\alpha \mapsto \beta$, we will be forced to send $\alpha^2 \mapsto \beta^2$, etc. Moreover, the next exercise says we do not have that many choices of where we can possibly send α.

Exercise 23. For every $\sigma \in \mathrm{Aut}(L/K)$, if $\alpha \in L$ is a root of an irreducible polynomial f, then $\sigma(\alpha)$ must also be a root of f.

Example 5.

Consider the extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$. We can write a basis for this as $\{1, \sqrt{2}\}$, as promised in the preceding paragraphs. Then any automorphism of this extension must fix 1, and must send $\sqrt{2}$ to $\pm\sqrt{2}$, so there are exactly 2 such automorphisms. Thus, the automorphism group is a group of order 2.

Consider the extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$. We can write a basis for this as $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$. However, to specify an automorphism, it's enough to specify what happens to $\sqrt[3]{2}$, because then the action on $\sqrt[3]{4}$ will be determined by multiplication. By the above, any automorphism must send $\sqrt[3]{2}$ to another root of $x^3 - 2$, but there is only one such root inside the field, so the only possible automorphism is the identity automorphism.

Consider the field $L = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$. **Exercise 24.** Think about it long enough to decide that $L/\mathbb{Q}$ has automorphism group S_3 (or D_3 if you are a fan of dihedral groups).

It is always the case that $|\mathrm{Aut}(L/K)| \leq [L : K]$. Equality is achieved if and only if L is the splitting field of a separable polynomial. (!!!) In such a case, we call $\mathrm{Aut}(L/K)$ the Galois group, and write it as $\mathrm{Gal}(L/K)$.

The Fundamental Theorem of Galois Theory

Now, we come to the big theorem. It basically says: all the data contained in a field is reflected in its Galois group.

Proposition (Fundamental Theorem of Galois Theory)**.** Suppose L/K is a Galois extension with Galois group G. There is a correspondence between the subgroups H of G and the intermediate fields $L \supset F \supset K$, given by taking the subfield of L fixed by H, denoted L^H. This correspondence has the following properties:

(a) It is a bijection, so every intermediate field F is the fixed field of some H. We would prove this by proving first that ...

(b) L is always Galois over any intermediate field F, and $\mathrm{Gal}(L/F)$ is a subgroup of G.

(c) The bijection is order-reversing: if $L \supset F_1 \supset F_2 \supset K$, the correspondence sends this to the chain $\{1\} \subset H_1 \subset H_2 \subset G$.

(d) The degree $[L : F]$ is equal to the size of the corresponding H.

(e) An intermediate field is Galois over K if and only if the corresponding subgroup H is normal in G. This is why they are called "normal extensions", they correspond to normal subgroups. In this case, $\mathrm{Gal}(F/K) = G/H$.

(f) If F_1, F_2 are intermediate extensions corresponding to the subgroups H_1, H_2, then the composite F_1F_2 corresponds to $H_1 \cap H_2$, and the intersection $F_1 \cap F_2$ corresponds to the subgroup generated by H_1 and H_2.

Exercise 25.

(a) Let $K = \mathbb{Q}(\zeta_n)$. What is $[K : \mathbb{Q}]$?

(b) Show that K is a Galois extension of $\mathbb{Q}$ by explicitly finding $[K : \mathbb{Q}]$-many automorphisms.

Exercise 26. Consider $\mathbb{F}_{p^n}/\mathbb{F}_p$. Let F be the Frobenius map. Prove that the powers of Frobenius give n distinct automorphisms of $\mathbb{F}_{p^n}$ fixing $\mathbb{F}_p$, and therefore that the Galois group is cyclic of order n.

Exercise 27.

(a) Show that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is Galois over $\mathbb{Q}$, and find its Galois group.

(b) Show that $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$ is Galois over $\mathbb{Q}$, and find its Galois group.

Exercise 28. (August 2016 Problem 5, modified) Consider the field $K = \mathbb{Q}(\zeta_7)$. According to your computation above, $\mathrm{Gal}(K/\mathbb{Q})$ is abelian and has even order, so has a normal subgroup of index 2. By the Fundamental Theorem of Galois Theory, this subgroup corresponds to a field $\mathbb{Q}(x) \subset K$ such that $[\mathbb{Q}(x) : \mathbb{Q}] = 2$, where x is some element of K. Compute an element x in terms of ζ_7 which generates this quadratic subfield.

Exercise 29. (Stolen from another school's algebra qual) Let $f \in \mathbb{Q}[x]$ be a degree 5 irreducible polynomial, let $K/\mathbb{Q}$ be its splitting field, and let $\alpha, \beta, \gamma, \delta, \epsilon \in K$ be the roots of f. Suppose that $\mathrm{Gal}(K/\mathbb{Q}) = S_5$, the symmetric group on 5 elements.

(a) Find the Galois group of $K/\mathbb{Q}(\alpha, \beta)$.

(b) Find the Galois group of $K/\mathbb{Q}(\alpha + \beta, \alpha\beta)$.

(c) The field $\mathbb{Q}(\alpha, \beta)$ contains the field $\mathbb{Q}(\alpha + \beta, \alpha\beta)$. Find the degree $[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}(\alpha + \beta, \alpha\beta)]$.

Exercise 30. (August 2017 Problem 4) Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K : \mathbb{Q}] = 4$, and $\sqrt{-m} \in K$ for some positive integer m. Show that $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Everything else

Here's a couple sentences about everything that appears on the Algebra qual topics list that doesn't appear in the rest of my notes. Of these, only division rings have appeared on a qual since 2017, and only in a problem 1.

Purely inseparable extensions and separability degree

We discussed briefly inseparable field extensions, which are algebraic field extensions that cannot be obtained by adjoining the root of a separable polynomial, they must be obtained by adjoining the root of inseparable polynomials. Examples include $x^p - t$ as a polynomial in the variable x with coefficients in $\mathbb{F}_p(t)$. One can quantify how "badly inseparable" a given polynomial/extension is:

- Any inseparable irreducible polynomial over a field of characteristic p (since there are no such over fields of characteristic 0) is of the form $f(x^{p^k})$, where $f(x)$ is a separable irreducible polynomial. The degree of $f(x)$ is the **separability degree** of the polynomial, and the exponent k appearing in the power x^{p^k} is the **inseparability degree**. A polynomial is called **purely inseparable** if its degree equals its inseparability degree, or equivalently if it is of the form $x^{p^k} - \lambda$ for some λ in the base field.

- Equivalently, any inseparable field extension M/K can be broken down into two extensions $M/L/K$, where L/K is the unique maximal subfield of M which is separable over K. If M is obtained by adjoining a root of $f(x^{p^k})$, L will be obtained by adjoining a root α of $f(x)$, and M will be obtained from L by adjoining $\sqrt[p^k]{\alpha}$. As with polynomials, we call $[L : K]$ the **separability degree** of M, and $[M : L]$ the **inseparability degree** of M, and M is **purely inseparable** if its degree equals its inseparability degree, or equivalently if $L = K$.

Division rings

A **division ring** is the noncommutative analog of a field. It's a ring R in which any nonzero $x \in R$ has an element $x^{-1} \in R$ which is a two-sided multiplicative inverse: $xx^{-1} = x^{-1}x = 1$. For example, the quaternion algebra $\mathbb{H}$ is a division ring. Every division ring is a simple ring, and a commutative ring is a division ring iff it is simple iff it is a field. However, for noncommutative rings there can be a difference, for example $M_2(\mathbb{C})$ is a simple ring that is not a division ring. In fact, matrix rings are the main source of simple rings that are not division rings, a notion made formal by the Wedderburn-Artin theorem.

Bimodules

Over a noncommutative ring R, a R**-bimodule** is an abelian group M with the structure of a left R-module as well as the structure of a right R-module, with the property that these two ring multiplications commute with one another: $r_1 \cdot (m \cdot r_2) = (r_1 \cdot m) \cdot r_2$. Every module over a commutative ring is automatically a bimodule, but you need to define bimodules to generalize some statements correctly to the noncommutative setting.

Support of a module

For a module M over a commutative ring R, the **support** of M, denoted $\operatorname{Supp} M$, is the set of prime ideals $\mathfrak{p} \subset R$ so that the localization is nonzero $M_\mathfrak{p} \neq 0$. For any $\mathfrak{p} \in \operatorname{Supp}(M)$, then $\mathfrak{p} \supset \operatorname{Ann}(M)$. If M is finitely generated, then $\mathfrak{p} \in \operatorname{Supp}(M)$ iff $\mathfrak{p} \supset \operatorname{Ann}(M)$, but this need not be the case for infinitely generated modules, for example $M = \bigoplus_{n \geq 2} \mathbb{Z}/n\mathbb{Z}$ has $\operatorname{Ann}(M) = 0$, but $0 \notin \operatorname{Supp}(M)$.

There's a closely related notion of an **associated prime** to a module. The prime ideal $\mathfrak{p} \subset R$ is **associated** to M if there is some $m \in M$ such that $\operatorname{Ann}(m) = \mathfrak{p}$. The set of all primes associated to M is denoted $\operatorname{Ass}_R(M)$, and that's pretty hilarious ngl. Every associated prime is in the support of M, but the support of M will also include primes containing associated primes. For example, $\mathbb{C}[x, y]/(x^2)$ is a $\mathbb{C}[x, y]$-module, and its only associated prime is (x). The prime ideal (x, y) is also in the support, but it is not an associated prime.

The terminology for support is similar to the terminology of "localization", in that it comes from a geometric notion, in this case the support of a function being the places x where $f(x) \neq 0$. In this case the prime $\mathfrak{p}$ is like the x, and the localization $M_\mathfrak{p}$ is like the $f(x)$.

Nilradical

The **nilradical** $\mathfrak{N}$ of a commutative ring R is the set of all nilpotent elements of R. The nilradical is always an ideal of R, and moreover it is the intersection of all the prime ideals of R: $\mathfrak{N} = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$. A ring whose nilradical is the 0 ideal (i.e. that has no nilpotents) is called **reduced**.

Nakayama's lemma

Nakayama's lemma is a name for several related facts important for understanding finitely-generated modules over local rings, and so is usually part of the process by which theorems can be proven about rings by proving them "everywhere locally". The version I remember best involves a local ring R with maximal ideal $\mathfrak{m}$, and a finitely-generated R-module M. The quotient $\kappa = R/\mathfrak{m}$ is a field, called the **residue field** of R, and $M \otimes_R R/\mathfrak{m} \cong M/\mathfrak{m}M$ is a finite-dimensional vector space over κ. Nakayama's theorem says that if $m_1, \ldots, m_n \in M$ form a basis for the quotient $M/\mathfrak{m}M$ as a vector space over κ, then in fact $m_1, \ldots, m_n$ form a minimal generating set for M as an R-module.

Here's an example application. Take $S = \mathbb{C}[x, y]$, $\mathfrak{m} = (x, y)$, and $R = S_\mathfrak{m}$. Let M be the module $\mathfrak{m}^2 = (x^2, xy, y^2)$. We can use Nakayama's lemma to prove quickly that M cannot be generated by fewer than 3 polynomials. To do this, we note that $R/\mathfrak{m} \cong \mathbb{C}$, and that we can think of $M/\mathfrak{m}M = \mathfrak{m}^2/\mathfrak{m}^3$ as the $\mathbb{C}$-vector space consisting of all degree 2 elements in $\mathfrak{m}$. Thus, $M/\mathfrak{m}M$ is clearly a 3-dimensional vector space spanned by x^2, xy, y^2, so these three must form a minimal generating set for M.

Symplectic matrices

A **symplectic matrix** is a matrix analogous to an orthogonal matrix. An orthogonal matrix can be defined as a matrix preserving an inner product, or more generally a nondegenerate symmetric bilinear form. Similarly, a symplectic matrix can be defined as a matrix preserving what is called a **symplectic form**, a non-degenerate *skew-symmetric* bilinear form. **Exercise 1.** A non-degenerate skew-symmetric form only exists if the dimension of the vector space is even, so every symplectic matrix is $2n \times 2n$ for some n.

The set of all symplectic matrices of a given size forms a group, called the **symplectic group**, denoted Sp_{2n}. Symplectic matrices are important in areas such as Hamiltonian mechanics in physics, in representation theory (where the symplectic group is one of the families in the "classical groups", along with SL_n and O_n), and in differential geometry.

Much of the time, working with symplectic matrices abstractly is similar to working with orthogonal matrices. For symmetric bilinear forms, the Gram-Schmidt process allows us to obtain a particularly nice basis, namely a basis $\{b_1, \ldots, b_n\}$ with the property that $\langle b_i, b_j \rangle = 0$ if $i \neq j$ and 1 if $i = j$. For symplectic forms, a very similar process gives what is called a *symplectic basis* $\{b_1, \ldots, b_n, c_1, \ldots, c_n\}$, which has the property that $\langle b_i, b_j \rangle = 0$, $\langle c_i, c_j \rangle = 0$, and $\langle b_i, c_j \rangle = 0$ if $i \neq j$ and 1 if $i = j$. Just as the columns of an orthogonal matrix form an orthogonal basis, the columns of a symplectic matrix form a symplectic basis.

As an example, consider the vector space $\mathbb{C}^4$ along with the symplectic form

$$\langle u, v \rangle = u^T \Omega v,$$

where Ω is the matrix

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

A matrix A is symplectic if $\langle Au, Av \rangle = \langle u, v \rangle$ for all vectors u, v, which means the same thing as

$A^T\Omega A = \Omega$. Some example symplectic matrices are the following:

$$\Omega \text{ itself}$$

$$\begin{pmatrix} 1 & 2 & 0 & 0 \\ 3 & 5 & 0 & 0 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & -3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Exercise 2.

(a) Suppose A is a 4×4 symplectic matrix for the bilinear form above. Since the vector space is complex, A has an eigenvector v, say of eigenvalue λ. Show that A also has an eigenvector of eigenvalue λ^{-1}.

(b) Conclude that $\det A = 1$, so $\mathrm{Sp}_4(\mathbb{C}) \subseteq \mathrm{SL}_4(\mathbb{C})$.

(c) Show that this subset is strict by finding a matrix in $\mathrm{SL}_4(\mathbb{C})$ which is *not* symplectic.